JOB MARKET PAPER

Of Fathers and Farms

How Intergenerational Persistence Shapes Structural Transformation

John Juneau

University of California, San Diego

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Abstract. The transmission of skills and preferences from parents to children generates intergenerational persistence in sector choice. How do these inherited connections alter the process of structural transformation? Using linked father-son records from historical U.S. census microdata, I document substantial differences in sector choices and earnings by parental background. I develop a model of structural change in which inherited skills and preferences shape workers' sector choices and the productive composition of subsequent generations. I estimate these mechanisms from observed sector choices and earnings distributions, accounting for endogenous selection into sectors. Counterfactual simulations imply that intergenerational persistence slowed U.S. industrialization between 1880 and 1940 by 23.2 percent. Inherited skills account for nearly all of this slowdown. Abstracting from these inherited connections also leads the model to substantially overestimate the growth in nonagricultural technology relative to agriculture needed to explain the historical transition.

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Introduction

Most people today know a doctor, a lawyer, or an academic who followed in their parents' footsteps. Occupational transmission is not limited to high-skilled professions; indeed, it is especially pronounced among farmers. This micro-level fact raises a macro-level question: how does intergenerational persistence in sector choice reshape the dynamics of structural transformation?

Research on structural change has typically focused on technological and demand forces that reallocate labor across sectors (Herrendorf, Rogerson and Valentinyi, 2014). Recent work has highlighted new generations of workers entering the labor force as a critical source of this reallocation (Hobijn, Schoellman and Vindas, 2018; Porzio, Rossi and Santangelo, 2022; Adão, Beraja and Pandalai-Nayar, 2024). Intuitively, a new worker may be more flexible to changing economic conditions than an incumbent who has formed strong ties through their experience in a given sector. Although new workers have no employment history of their own, they nevertheless enter the labor force connected to their parents' sectors through the skills and preferences they inherit. This paper studies how these inherited connections alter the reallocation of workers out of agriculture.

As a canonical episode of structural transformation, the United States' industrialization between 1880 and 1940 provides an ideal setting to quantify the paper's insights. Using linked father–son records from the full-count historical U.S. census microdata (Helgertz et al., 2024), I document large differences in sector choice between workers whose fathers worked in agriculture (i.e., "ag-born") and those whose fathers worked outside agriculture (i.e., "nonag-born"). These differences remain substantial after accounting for childhood school attendance and local labor markets: in 1880, ag-born workers were over 3.4 times as likely to work in agriculture as nonag-born workers. Moreover, ag-born and nonag-born workers have systematically different earnings within the same sector, suggesting that workers' productivity may also depend on their parents' sector.

Guided by these empirical patterns, I develop a two-sector macro model of structural transformation with intergenerational persistence. New workers choose between agriculture and nonagriculture based on the earnings and nonpecuniary benefits available in each sector. There are two sources of intergenerational persistence: higher productivity in their parents' sector (i.e., *inherited skills*) and nonpecuniary attachments to their parents' sector (i.e., *inherited preferences*). Inherited skills increase the earnings workers can obtain in their parents' sector, while inherited preferences make employment outside the parental sector less appealing even when it pays more. In the model, structural change

originates from exogenous changes in either sectoral production technologies or the non-pecuniary sectoral benefits common to all workers. I model consumer demand as in Comin, Lashkari and Mestieri (2021), so technological progress reallocates labor through changes in income and relative prices.

The model shows that intergenerational persistence alters long-run worker reallocation in two ways. First, inherited skills and inherited preferences can reduce the aggregate responsiveness of new workers to changes in relative wages. Consequently, a given technological improvement can induce less immediate industrialization in an economy with persistence than without. Second, intergenerational persistence also generates dynamics across generations. With inherited preferences, reallocation out of agriculture today induces further reallocation in the next generation. But with inherited skills, the next generation's labor productivity endogenously shifts in response to today's reallocation. This generates wage adjustments that can reverse the flow of workers in the next generation. Thus, ignoring intergenerational persistence can meaningfully alter the amount of industrialization we would expect from a given innovation as well as the innovation we would attribute to a given transition.

How do these theoretical insights materialize in the historical U.S. context? This requires quantifying the two sources of intergenerational persistence by revisiting the systematic gaps in sector choice and earnings between ag-born and nonag-born workers. Within-sector earnings differences can reflect both inherited skills and gains from self-selection (Lagakos and Waugh, 2013). For example, through inherited skills, ag-born workers may earn more in agriculture than nonag-born workers. However, through self-selection, ag-born workers may be less-positively selected into agriculture than nonag-born workers, which would reduce their earnings relative to nonag-born workers. A tractable benchmark provides closed-form expressions that clarify how earnings selection and inherited skills can be separated. I then jointly estimate the distribution of workers' sectoral skills, inherited skills and preferences, and relative sectoral payoffs using workers' sector choices and earnings.

Using the historical microdata, I estimate substantial inherited skills alongside a smaller inherited-preference wedge. In the preferred age-and-race specification, ag-born workers have an estimated agricultural skill advantage of 178 percent, while nonag-born workers have an estimated nonagricultural skill advantage of 42 percent. I also infer that workers value employment outside their parental sector approximately 8.5 percent less, holding potential earnings and common sectoral benefits fixed.

I conduct counterfactual simulations to quantify how U.S. industrialization would have unfolded without inherited skills or inherited preferences. For each counterfactual, I re-infer the initial fundamentals to match the same observed aggregate outcomes in 1880 and then expose the economy to the benchmark growth in sectoral technologies and common nonpecuniary benefits through 1940. Removing intergenerational persistence altogether increases industrialization by 8.9 percentage points (i.e., 23.2 percent of the observed reallocation of workers into nonagriculture).[1] Inherited skills account for nearly all of this effect. Without inherited preferences, industrialization increases by only 0.14 percentage points, while removing inherited skills alone increases it by 9.3 percentage points.

To understand why inherited skills account for most of the slowdown in industrialization, I distinguish between the *immediate* response of new workers to changing economic conditions and the *dynamic* effects transmitted through subsequent generations. Both inherited mechanisms reduce new workers' aggregate responsiveness to wages; in the linked 1880 U.S. sample, observed intergenerational persistence lowers this elasticity by 42 percent. After accounting for general equilibrium feedback, inherited preferences impose a slightly greater immediate constraint on reallocation than inherited skills. However, across generations, the two mechanisms have sharply different consequences. With inherited preferences, changes in parental composition lead to further reallocation out of agriculture, mostly offsetting the immediate slowdown. With inherited skills, the resulting shifts in sectoral labor productivity induce wage adjustments that instead amplify the slowdown. Thus, the immediate and dynamic effects of inherited skills reinforce each other, while those of inherited preferences largely offset each other.[2]

Finally, I show that accounting for intergenerational persistence changes what the model infers about the forces behind U.S. industrialization. As industrialization reduces the share of ag-born workers, inherited skills shift the supply of productive labor toward nonagriculture even without changes in sectoral technologies. Without accounting for this effect, matching the observed transition requires substantially more growth in nonagricultural relative to agricultural technology: 88.6 rather than 53.3 log points, an increase of 66 percent. Thus, parent sector dependence changes not only how workers respond to historical forces, but also the technological changes the model infers to explain the observed transition.

[1] For scale, Brandt, Hsieh and Zhu (2008) find that improvements in rural labor mobility contributed 8 (out of 38) percentage points to China's industrialization between 1978 and 2004.

[2] Under alternative calibrations with modest intergenerational persistence driven primarily by inherited preferences, the dynamic effects can more than offset the immediate dampening of reallocation. In such cases, inherited preferences modestly *accelerate* structural transformation.

Literature This paper bridges two vast yet distinct literatures: research on structural transformation and research on intergenerational occupational choice. I contribute to a growing strand emphasizing the critical role of new workers for labor reallocation as well as a nascent strand highlighting the macroeconomic consequences of micro-level occupational persistence. In addition, my work yields an insight broadly relevant for research in macroeconomics and international trade that uses quantitative models of labor allocation built on extreme-value discrete choice.

Relative to the recent work showcasing the importance of new generations of workers for structural change (Hobijn, Schoellman and Vindas, 2018; Porzio, Rossi and Santangelo, 2022; Adão, Beraja and Pandalai-Nayar, 2024), I show that individual workers' inherited connections to their parents' sectors meaningfully alter the aggregate process of reallocation.[3] Quantitative macroeconomic studies of structural change have almost exclusively abstracted from this mechanism. A prominent exception is Hayashi and Prescott (2008), who attribute Japan's depressed prewar output to an institutional barrier that kept agricultural employment roughly constant by forcing the designated heir to remain on the family farm. Relative to their aggregate restriction, my framework models persistence through worker-level sector choices and allows it to arise from either inherited skills or preferences, which I discipline using microdata. Crucially, I show that these two mechanisms have very different implications for structural transformation.

There is a very large literature measuring intergenerational persistence in occupation choice and evaluating specific determinants (Abramitzky, Boustan and Matiashvili, 2025). I contribute to a growing set of studies that focus on the macroeconomic consequences of this micro-level persistence.[4] Cassan, Keniston and Kleineberg (2021), Lo Bello and Morchio (2022), and Almgren, Kramer and Sigurdsson (2025) develop quantitative models to distinguish the sources of intergenerational occupational persistence and evaluate their implications for the allocation of talent and aggregate welfare. In contrast, I ask how persistence alters the long-run reallocation of workers in an economy experiencing major changes in economic fundamentals.

Finally, my analytical result speaks to a broader literature in quantitative macroeconomics and international trade that studies worker reallocation with extreme-value discrete choice models (Artuç, Chaudhuri and McLaren, 2010; Caliendo, Dvorkin and Parro, 2019). I show that even when different groups of workers have identical labor supply

[3] Kim and Topel (1995) and Lee and Wolpin (2006) also remark on new workers' role in labor reallocation.

[4] Several empirical papers focus on intergenerational persistence in agriculture and remark on the potential importance for structural change without explicitly studying it (Emran and Shilpi, 2011; Fernando, 2022; Haws, Just and Price, 2025; Liu, 2026).

elasticities, the aggregate labor supply elasticity can be substantially lower if these groups differ in their sector choices. In my setting, intergenerational persistence generates precisely these differences between ag-born and nonag-born workers. Moreover, because the composition of these groups changes across generations, so does the aggregate responsiveness of workers to wages.[5]

Roadmap Section 1 presents the motivating evidence on intergenerational persistence in the U.S. between 1880 and 1940. Section 2 develops a general equilibrium model of structural transformation with intergenerational persistence and characterizes its implications for worker reallocation within and across generations. Section 3 establishes how workers' sector choices and earnings identify inherited skills and preferences, and how aggregate economic outcomes identify the exogenous drivers of structural transformation. Section 4 estimates the model's labor supply parameters, calibrates consumer demand, and quantifies how intergenerational persistence shaped U.S. industrialization between 1880 and 1940. Section 5 concludes.

1. Motivating Facts

In this section, I document two patterns in the historical U.S. census that motivate the model developed in the next section. First, workers whose fathers were employed in agriculture are substantially more likely to work in agriculture themselves. Second, workers' earnings within the same sector differ systematically according to their fathers' sectors of employment. Both patterns hold even after accounting for differences across local labor markets. Together, these patterns suggest that parental background shapes both workers' sector choices and the earnings they obtain within sectors, motivating a framework that distinguishes inherited skills, inherited preferences, and endogenous selection.

To document these facts, I use the full-count U.S. decennial census microdata for 1880–1940 from IPUMS (Ruggles et al., 2024). I restrict attention to U.S.-born employed men aged 20–29 with observed industry, classifying workers as agricultural if their industry is coded as agriculture by IPUMS.[6] To identify their fathers' sectors, I link these workers to their childhood households using the IPUMS Multigenerational Longitudinal Panel

[5] Relatedly, Böhm, Etheridge and Irastorza-Fadrique (2025) examine how the mobility patterns of incumbent workers shape occupational labor supply elasticities, whereas my result emphasizes how differences in sector choice among groups of new workers shape their aggregate labor supply responsiveness.

[6] Agriculture comprises IND1950 codes 100–199.

(Helgertz et al., 2024).[7]

Sector Choices Figure 1 documents the evolution of young workers' agricultural employment shares. Panel A shows that the share of young workers employed in agriculture fell from 61 percent in 1880 to 23 percent in 1940. Despite this large reallocation, ag-born and nonag-born workers made very different sector choices: in 1880, over 80 percent of ag-born workers were employed in agriculture, compared with less than 20 percent of nonag-born workers. To summarize this persistence, I estimate the following linear probability model:

$$\mathbb{1}_{i,t}\{\text{Worker in Agriculture}\} = \alpha + \beta\, \mathbb{1}_{i,t-1}\{\text{Parent in Agriculture}\} + \boldsymbol{X}'_{i,t}\boldsymbol{\zeta} + \epsilon_{i,t}.$$

Panel B reports β, the difference in agricultural employment rates between ag-born and nonag-born workers, under three specifications: without controls; with age and race fixed effects; and with age and race fixed effects plus fully interacted childhood-by-adulthood commuting-zone fixed effects.

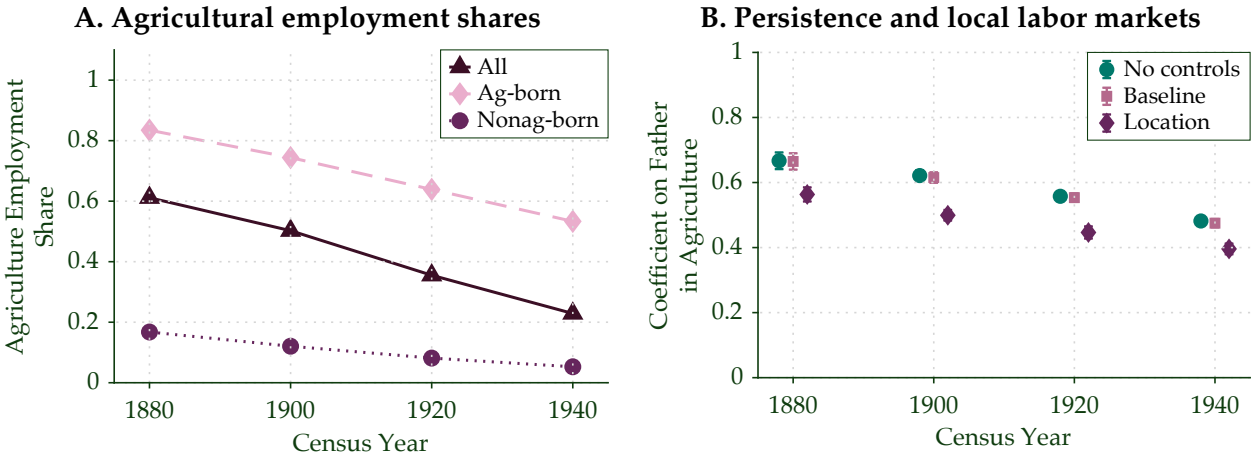


Figure 1: Agricultural Employment and Intergenerational Persistence, 1880–1940

Note: Panel A plots agricultural employment shares among linked U.S.-born employed men aged 20–29, separately for all workers, workers whose fathers were employed in agriculture (ag-born), and workers whose fathers were employed outside agriculture (nonag-born). Panel B plots coefficients on an indicator for an agricultural father from separate linear probability models of agricultural employment. The no-controls specification includes no covariates; the baseline specification includes age and race fixed effects; and the location specification additionally assigns a separate fixed effect to each observed childhood-to-adulthood commuting-zone pair. Whiskers in Panel B denote 95 percent confidence intervals from standard errors clustered by adulthood commuting zone. Agriculture comprises IND1950 codes 100–199.

[7] I first link each worker to his childhood record in an earlier census and then use within-household relationships to identify his father and his father's industry. Appendix A.1.2 establishes robustness to alternative historical census links from the Census Tree (Price, Buckles and Clement, 2021; Buckles et al., 2025) and the Census Linking Project (Abramitzky et al., 2025).

The unconditional agricultural employment gap between parental groups was 66.7 percentage points in 1880 and remained 48.2 percentage points in 1940. One concern is that this gap may exclusively reflect differences across local labor markets. Put simply, if agricultural jobs are concentrated in the rural hinterland while nonagricultural jobs are concentrated in industrial cities, then ag-born workers may be more likely to work in agriculture simply because they are more likely to live in rural areas. However, Panel B shows that the gap remains large after accounting for age and race and after comparing workers within the same childhood-to-adulthood commuting-zone pair.

Earnings Parental sector predicts not only where workers are employed, but also how much they earn within the same sector. Because 1940 is the first census to report individual earnings, I focus on that year and further restrict the sample to workers with positive wage and salary income, positive weeks worked, and positive usual weekly hours. Within each destination sector, I estimate

$$\log \text{Earnings}_{i,t} = \alpha + \beta \, \mathbb{1}_{i,t-1}\{\text{Parent in Agriculture}\} + \boldsymbol{X}_{i,t}'\boldsymbol{\zeta} + \epsilon_{i,t}.$$

Figure 2 plots the estimated β coefficients using log hourly earnings. It reports the same three specifications as the sector-choice analysis: no controls; age and race fixed effects; and age and race fixed effects plus fully interacted childhood-by-adulthood commuting-zone fixed effects.

Without controls, ag-born workers earn approximately 48 percent less than nonag-born workers in agriculture and 26 percent less in nonagriculture. Adding age and race fixed effects yields gaps of approximately 47 and 25 percent, respectively; additionally comparing workers within the same childhood-to-adulthood commuting-zone pair narrows the gaps to approximately 35 and 14 percent, but both remain substantial. At first glance, it may seem surprising that ag-born workers earn less than nonag-born workers in agriculture, despite being much more likely to work there. On its face, this pattern may suggest that nonag-born workers are ubiquitously more productive than ag-born workers, and that ag-born workers disproportionately enter agriculture only because they are constrained or otherwise prefer to do so. However, this overlooks a potentially important selection effect: with relatively few nonag-born workers entering agriculture, they may be especially positively selected on their agricultural skills. Hence, these earnings differences could reflect both workers inheriting productive skill advantages in their parents' sectors and potent selection effects.

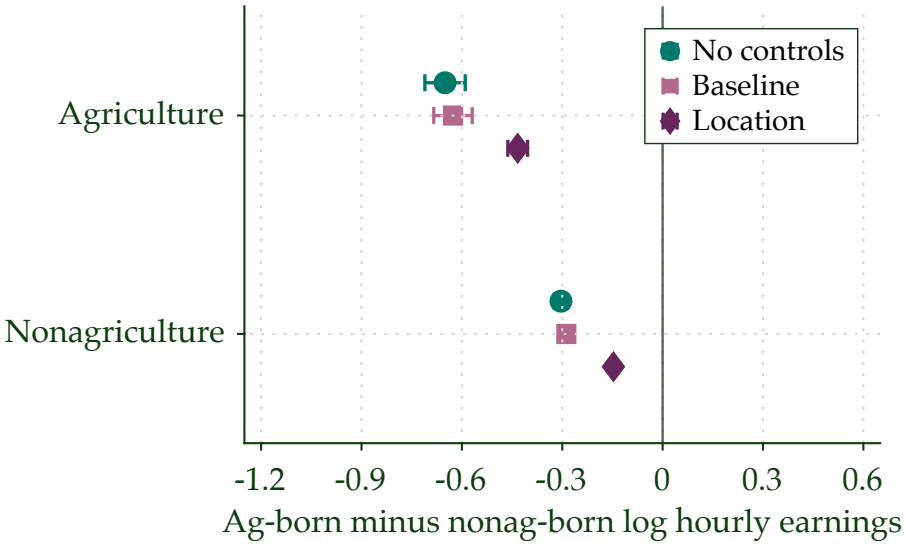


Figure 2: Earnings Differences by Parental Sector, 1940

Note: Each marker reports the coefficient on an indicator for whether a worker's father was employed in agriculture from a separate regression of log hourly wage and salary income within the indicated destination sector; negative values therefore indicate lower earnings for ag-born workers than for nonag-born workers. The no-controls specification includes no covariates; the baseline specification includes age and race fixed effects; and the location specification additionally assigns a separate fixed effect to each observed childhood-to-adulthood commuting-zone pair. None of the specifications controls for childhood schooling; Appendix Table 12 reports the corresponding schooling sensitivity. The sample consists of linked U.S.-born employed men aged 20–29 with observed own and paternal industries, positive wage and salary income, positive weeks worked, and positive usual weekly hours. Whiskers denote 95 percent confidence intervals from standard errors clustered by adulthood commuting zone. Appendix Table 11 reports the estimates and regression diagnostics.

Together, these findings establish substantial intergenerational persistence in sector choice and systematic earnings differences by parental sector. On their own, these facts do not establish the exact sources of persistence, especially whether they reflect inherited skills or inherited preferences. The next section develops a model that distinguishes these mechanisms and traces their implications for structural transformation.

2. A Model of Structural Change with Intergenerational Persistence

I develop a two-sector general equilibrium model of structural transformation in which worker allocations are linked across generations. In each period, a new generation of workers replaces the previous one. Workers are classified according to whether their parents worked in agriculture (ag-born) or nonagriculture (nonag-born), so the composition of the labor force depends on the previous generation's sectoral allocation.

The labor supply block builds on a Roy model of comparative advantage and self-

selection (Roy, 1951; Lagakos and Waugh, 2013). Workers choose between agriculture and nonagriculture based on their potential earnings and the nonpecuniary benefits of employment in each sector. I introduce two sources of intergenerational persistence: inherited skills, which raise workers' productivity in their parents' sectors, and inherited preferences, which make leaving a parent's sector less attractive. I combine this labor supply framework with sectoral production and nonhomothetic consumer demand to determine equilibrium wages and allocations. Finally, I characterize how intergenerational persistence affects workers' aggregate responsiveness to wages and the immediate and dynamic effects of technological change. For analytical transparency, I use independent sectoral skills with common-shape Fréchet marginals to obtain closed-form allocations and characterize the immediate and dynamic responses to technological change. The aggregation result for labor supply elasticities below is more general and also applies when sectoral skills are correlated. In the preferred quantitative specification, Section 4.1 allows sector-specific Fréchet shapes and dependence through a Frank copula and evaluates the corresponding choice and effective-labor integrals numerically.

2.1. Labor Supply

There is a unit mass of households indexed by i and two sectors of production indexed by $j \in \{A, N\}$, where A and N denote agriculture and nonagriculture, respectively. Households are dynastic and workers live for a single period: the worker inhabiting household i in period $t-1$ is the parent of the worker inhabiting household i in period t. Let $k \in \{A, N\}$ denote a worker's parental sector, and refer to workers whose parents were employed in sector k as type-k workers.

The Worker's Problem Worker i with a parent in sector k chooses the sector that maximizes indirect utility:

$$s_{i,t} \quad = \quad \arg\max_{j \in \{A,N\}} \left\{ V_{i,j,t}(k) \equiv B_{i,j,t}(k) \cdot y_{i,j,t}(k) \right\},$$

Utility depends on labor income, $y_{i,j,t}(k)$, and the nonpecuniary benefits of working in the sector, $B_{i,j,t}(k)$.

I introduce the two inherited mechanisms through these two components:

$$B_{i,j,t}(k) = B_{j,t}(1-\tau_k)^{\mathbb{1}(k \neq j)}, \qquad y_{i,j,t}(k) = w_{j,t}\left[\epsilon_{i,j,t}\kappa_k^{\mathbb{1}(k=j)}\right].$$

Here, $B_{j,t} > 0$ is the common nonpecuniary benefit of sector j, and $\tau_k \in [0, 1)$ captures the inherited preference for remaining in parental sector k by reducing the value of employment outside it. The wage per efficiency unit in sector j is $w_{j,t} > 0$. In the analytical benchmark, workers draw independent sector-specific skills, $\epsilon_{i,j,t}$, from the common Fréchet distribution $F(\epsilon) = \exp(-\epsilon^{-\gamma})$, where $\gamma > 1$ and larger values of γ imply less skill dispersion. Finally, $\kappa_k \geq 1$ is the inherited skill advantage in parental sector k.

These two mechanisms have distinct economic interpretations. Inherited preferences affect utility without directly affecting earnings, whereas inherited skills increase both earnings and effective labor in the parental sector. For example, $\kappa_A = 1.5$ means that an ag-born worker can supply 50 percent more effective labor to agriculture than an otherwise identical nonag-born worker. Likewise, $\tau_k = 0.5$ means that a type-k worker requires earnings outside the parental sector to be twice as high as earnings within it to be willing to leave, holding common sectoral nonpecuniary benefits and idiosyncratic skills fixed.

Sector Allocations of Ag-born and Nonag-born Workers Let $\phi_{A,t}(k)$ denote the fraction of type-k workers who choose agriculture. The optimal sector choices imply

$$\begin{aligned} \phi_{A,t}(A) &\equiv \mathbb{P}_i\left[s_{i,t} = A \mid k = A\right] = \frac{1}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \frac{(1-\tau_A)}{\kappa_A}\right]^{\gamma}}, \\ \phi_{A,t}(N) &\equiv \mathbb{P}_i\left[s_{i,t} = A \mid k = N\right] = \frac{1}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \frac{\kappa_N}{1-\tau_N}\right]^{\gamma}}. \end{aligned} \tag{1}$$

Fewer ag-born and nonag-born workers choose agriculture when the relative wage or nonpecuniary benefit of nonagriculture is higher. Stronger inherited agricultural skills increase the fraction of ag-born workers choosing agriculture, while stronger inherited nonagricultural skills reduce the fraction of nonag-born workers choosing agriculture. Stronger inherited preferences widen this difference from both sides by making each group less likely to leave its parental sector. These inherited mechanisms therefore generate differences in sector choice between workers of different parental backgrounds, even when they face the same wages and common nonpecuniary benefits. Expressions analogous to (1) hold for the fractions of ag-born and nonag-born workers in nonagriculture, which also satisfy $\phi_{N,t}(k) = 1 - \phi_{A,t}(k)$.

Effective Labor Supplied by Ag-born and Nonag-born Workers Let $h_{j,t}(k)$ denote the average effective labor supplied by type-k workers who choose sector j. The Fréchet

assumption implies

$$h_{j,t}(k) \equiv \mathbb{E}_i\left[\epsilon_{i,j,t}\kappa_k^{\mathbb{1}(k=j)} \mid s_{i,t} = j, k\right] \quad = \quad \bar{\gamma} \cdot \kappa_k^{\mathbb{1}(k=j)} \cdot \phi_{j,t}(k)^{-1/\gamma}, \tag{2}$$

where $\bar{\gamma} \equiv \Gamma(1 - 1/\gamma)$ and $\Gamma(\cdot)$ is the Gamma function. Holding sector choice probabilities fixed, inherited skill increases workers' average effective labor in their parents' sector. However, workers who are less likely to enter a sector are more positively selected, as captured by $\phi_{j,t}(k)^{-1/\gamma}$. Thus, differences in average effective labor between parental groups reflect both inherited skill and selection, which can work in opposite directions.

Aggregation and Intergenerational Dynamics The allocation of workers across sectors in period $t-1$ is a state variable for the economy in period t because it determines the composition of ag-born versus nonag-born workers in the new generation. As such, the aggregate allocation of workers across sectors evolves according to the following law of motion:

$$\Phi_{A,t} \quad = \quad \Phi_{A,t-1} \cdot \phi_{A,t}(A) \quad + \quad (1 - \Phi_{A,t-1}) \cdot \phi_{A,t}(N), \tag{3}$$

where $\Phi_{A,t}$ denotes the fraction of all workers in agriculture in period t and $\Phi_{A,t-1}$ denotes the fraction of all workers in agriculture in period $t-1$. Aggregate agricultural employment is the weighted average of the agricultural employment shares of ag-born and nonag-born workers, with weights determined by the previous generation's sectoral allocation. Even though workers make static sector choices, current reallocation affects future aggregate employment by changing the composition of subsequent generations. There is an analogous expression for the fraction of workers in nonagriculture, which moreover satisfies: $\Phi_{N,t} = 1 - \Phi_{A,t}$.

Aggregate effective labor supply in agriculture is given by:

$$H_{A,t} \quad = \quad \Phi_{A,t-1} \cdot \phi_{A,t}(A) \cdot h_{A,t}(A) \quad + \quad (1 - \Phi_{A,t-1}) \cdot \phi_{A,t}(N) \cdot h_{A,t}(N), \tag{4}$$

where $H_{A,t}$ denotes the total effective labor supply in agriculture in period t, and an analogous expression holds for $H_{N,t}$. Aggregate effective labor supply in agriculture equals the sum of the effective labor supplied by ag-born and nonag-born workers, accounting for both the number of workers in each group and their average effective labor. Consequently, changes in parental composition affect not only the number of workers in each sector but also the effective labor they supply, which feeds back into equilibrium wages.

Finally, the total labor income in the economy is given by:

$$Y_t \quad = \quad w_{A,t} \cdot H_{A,t} \quad + \quad w_{N,t} \cdot H_{N,t}. \tag{5}$$

Aggregate income equals the sum of wage payments across the two sectors. I next introduce sectoral production and nonhomothetic consumer demand, under which sectoral expenditure shares depend on real income and relative prices.

2.2. Characterizing Aggregate Sectoral Labor Supply Elasticities

Before closing the model, I characterize how workers respond to changes in relative wages and common nonpecuniary benefits, holding the composition of parental groups fixed. The following aggregation result does not depend on the Fréchet assumption: it holds for any differentiable two-sector sector-choice probabilities, including those generated by correlated skills. Define the within-group relative sectoral labor supply elasticity for type-k workers as

$$\mathcal{E}_t(k) \;\equiv\; \frac{\partial \log(\phi_{N,t}(k)/\phi_{A,t}(k))}{\partial \log(w_{N,t}/w_{A,t})} \;=\; \frac{\partial \log(\phi_{N,t}(k)/\phi_{A,t}(k))}{\partial \log(B_{N,t}/B_{A,t})}.$$

The equality of the wage and common-benefit elasticities follows because both enter workers' choices through their product. Under independent, common-shape Fréchet skills, $\mathcal{E}_t(k) = \gamma$ for both parental groups; with a more general joint skill distribution, these elasticities need not be equal.

Define the aggregate relative sectoral labor supply elasticity as

$$\mathcal{E}_t \;\equiv\; \frac{\partial \log(\Phi_{N,t}/\Phi_{A,t})}{\partial \log(w_{N,t}/w_{A,t})} \;=\; \frac{\partial \log(\Phi_{N,t}/\Phi_{A,t})}{\partial \log(B_{N,t}/B_{A,t})},$$

and define a weighted average of the within-group elasticities:

$$\overline{\mathcal{E}}_t \;\equiv\; \frac{\displaystyle\sum_{k\in\{A,N\}} \Phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\mathcal{E}_t(k)}{\displaystyle\sum_{k\in\{A,N\}} \Phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)}.$$

The weights depend on each parental group's size and its allocation across the two sectors.

Proposition 1. *Suppose both parental groups are present and have differentiable, interior sector-choice probabilities with positive within-group elasticities. Holding parental composition fixed, the aggregate relative sectoral labor supply elasticity satisfies*

$$\mathcal{E}_t \;=\; \overline{\mathcal{E}}_t \left[1 - \frac{\mathrm{Var}(\mathbb{E}[D_{i,t} \mid k])}{\mathrm{Var}(D_{i,t})}\right], \qquad 0 < \mathcal{E}_t \leq \overline{\mathcal{E}}_t,$$

where $D_{i,t} \equiv \mathbb{1}(s_{i,t} = A)$. Equality holds if and only if agricultural employment probabilities are identical across parental groups.

Proof. See Appendix B.1.1. □

Proposition 1 shows that systematic differences in sector choices across parental groups attenuate the aggregate response relative to a weighted average of their within-group responses. The attenuation term can be expressed entirely in terms of observable allocations:

$$\frac{\mathrm{Var}(\mathbb{E}[D_{i,t} \mid k])}{\mathrm{Var}(D_{i,t})} \;=\; \frac{\Phi_{A,t-1}(1-\Phi_{A,t-1})[\phi_{A,t}(A) - \phi_{A,t}(N)]^2}{\Phi_{A,t}(1-\Phi_{A,t})}.$$

It is therefore not necessary to estimate the underlying distribution of skills to measure this source of aggregation attenuation. For a stationary economy ($\Phi_{A,t} = \Phi_{A,t-1}$), the attenuation term simplifies to $[\phi_{A,t}(A) - \phi_{A,t}(N)]^2$.

This result applies beyond the independent Fréchet benchmark, even when the two parental groups have different within-group elasticities. Under independent, common-shape Fréchet skills, $\overline{\mathcal{E}}_t = \gamma$, so the proposition reduces to $\mathcal{E}_t = \gamma[1 - \mathrm{Var}(\mathbb{E}[D_{i,t} \mid k])/\,\mathrm{Var}(D_{i,t})]$. In more flexible models, the weighted within-group elasticity must also be computed, but the share of responsiveness lost through aggregation depends only on sector-choice differences. The result compares the aggregate elasticity with the weighted within-group benchmark, not necessarily with the elasticity of each parental group separately.

After closing the model by specifying production and consumer demand, I show how this attenuation shapes the immediate response to changes in fundamentals under the tractable independent Fréchet benchmark, and how inherited skills and preferences transmit their effects across generations.

2.3. Production, Demand, and Equilibrium

I now close the model by introducing sectoral production and consumer demand, and defining the competitive equilibrium.

Production and Labor Demand Each sector has a representative firm that produces output using effective labor as its only input:

$$Q_{j,t} = Z_{j,t} H_{j,t}, \qquad j \in \{A, N\}, \tag{6}$$

where $Q_{j,t}$ denotes sectoral output, $H_{j,t}$ is the total effective labor supplied to the sector, and $Z_{j,t} > 0$ denotes its production technology. Firms take the price of their output, $p_{j,t}$, and the wage per efficiency unit of labor, $w_{j,t}$, as given. With linear production, positive output in competitive equilibrium requires

$$w_{j,t} = p_{j,t} Z_{j,t}, \qquad j \in \{A, N\}. \tag{7}$$

Thus, wages depend on both sectoral technologies and goods prices, with the latter determined endogenously through consumer demand and goods-market clearing.

Consumer Demand I represent aggregate consumer demand using the nonhomothetic CES expenditure system of Comin, Lashkari and Mestieri (2021), evaluated at aggregate labor income. This specification abstracts from the effects of household income heterogeneity on sectoral demand.

$$\sum_{j \in \{A,N\}} \Omega_j^{1/\sigma} \left(\frac{X_{j,t}}{Q_t^{\vartheta_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \tag{8}$$

where $X_{j,t}$ is consumption of sector j goods, Q_t is aggregate consumption, $\Omega_j > 0$ is a sectoral demand weight, $\sigma \in (0, 1)$ governs substitutability between goods, and $\vartheta_N \geq \vartheta_A = 1$ governs differences in their income elasticities. Because $\sigma < 1$, agricultural and nonagricultural goods are complements, so an increase in a good's relative price raises its expenditure share, holding aggregate consumption fixed. Meanwhile, $\vartheta_N > \vartheta_A$ implies that rising aggregate consumption shifts expenditure toward nonagriculture, holding relative prices fixed. When $\vartheta_A = \vartheta_N = 1$, the aggregator reduces to the standard homothetic CES specification, in which changes in aggregate income do not shift relative demand across sectors, holding relative prices fixed.

Define the sectoral expenditure share as $\omega_{j,t} \equiv p_{j,t} X_{j,t} / Y_t$ and the expenditure per unit of aggregate consumption as $P_t \equiv Y_t / Q_t$. Expenditure minimization subject to (8) implies

$$\frac{\omega_{A,t}}{\omega_{N,t}} = \frac{\Omega_A}{\Omega_N} \cdot \left(\frac{p_{N,t}}{p_{A,t}} \right)^{\sigma-1} \cdot {Q_t}^{(\vartheta_A - \vartheta_N)(1-\sigma)}. \tag{9}$$

This expression summarizes the relative price and income effects on sectoral expenditure shares: a higher relative price of agricultural goods increases agriculture's expenditure share, while higher aggregate consumption shifts expenditure toward nonagriculture.

Competitive Equilibrium Normalize the agricultural goods price to $p_{A,t} = 1$ and define the relative price of nonagricultural goods as $p_t \equiv p_{N,t}/p_{A,t}$. Given an initial agricultural employment share $\Phi_{A,0}$, the model parameters, and exogenous paths of sectoral technologies and nonpecuniary benefits, a competitive equilibrium consists of sequences of worker allocations, effective labor supplies, production, consumption, and prices satisfying the following conditions in every period:

1. Workers optimally choose sectors, generating the group-specific employment shares in (1) and average effective labor supplies in (2).
2. Aggregate employment shares and effective labor supplies satisfy (3) and (4), respectively, with analogous expressions for nonagriculture.
3. Competitive firms produce according to (6) and pay wages according to (7). Aggregate labor income satisfies (5).
4. Aggregate consumer demand satisfies the relative expenditure-share relationship in (9), subject to the budget constraint

$$p_{A,t}X_{A,t} + p_{N,t}X_{N,t} = Y_t.$$

   Aggregate consumption is determined by (8), with expenditure per unit of consumption given by $P_t = Y_t/Q_t$.
5. Sectoral goods markets clear:

$$X_{j,t} = Q_{j,t}, \qquad j \in \{A, N\}. \tag{10}$$

2.4. Implications for Structural Transformation

I now characterize how inherited skills and preferences affect the economy's response to exogenous changes in economic fundamentals. I focus on a permanent 1-percent improvement in agricultural technology, $Z_{A,t}$, holding all other fundamentals fixed. I first examine the *immediate* response of the generation entering the labor market when the shock occurs, and then the *dynamic* effects transmitted to subsequent generations.

For this analysis, I compare three stylized economies: one without intergenerational persistence, one with inherited skills, and one with inherited preferences. To obtain sharp analytical results, I impose the following simplifying assumptions:

Assumption 1. *The three stylized economies satisfy:*

1. *The no-IGP economy has $\kappa_A = \kappa_N = 1$ and $\tau_A = \tau_N = 0$; the κ-IGP economy has $\kappa_A = \kappa_N > 1$ and $\tau_A = \tau_N = 0$; and the τ-IGP economy has $\kappa_A = \kappa_N = 1$ and $\tau_A = \tau_N > 0$.*
2. *Each economy begins in a stationary equilibrium with the same aggregate agricultural employment share, $\bar{\Phi}_A$.*
3. *The κ-IGP and τ-IGP economies have the same positive gap between ag-born and nonag-born workers' agricultural employment shares.*
4. *Aggregate consumer demand is homothetic: $\vartheta_A = \vartheta_N = 1$.*

The common stationary employment share and sector-choice gap ensure that the stylized economies are comparable. All three begin with the same aggregate worker allocation and experience the same technological improvement. Moreover, the common sector-choice gap, together with stationarity, implies that the κ-IGP and τ-IGP economies have identical worker allocations within each parental group. Because both economies have the same within-group elasticity γ under the independent Fréchet benchmark, Proposition 1 implies that they also have the same aggregate labor supply elasticity. Thus, differences in their immediate responses to technological change reflect differences in general equilibrium wage feedback rather than workers' underlying responsiveness to wages.

The parameter restrictions isolate each inherited mechanism and impose symmetry within the two persistence economies, while homothetic demand eliminates income-driven shifts in relative sectoral demand. Relative prices still respond to changes in effective labor supplies, allowing me to isolate how inherited skills and preferences generate different wage adjustments. The quantitative analysis later allows inherited skills to differ across sectors and restores nonhomothetic consumer demand. This comparison allows me to distinguish how the source of intergenerational persistence affects both the immediate response to technological change and the subsequent transmission of worker reallocation across generations.

Immediate Effects An improvement in agricultural technology increases the relative abundance of agricultural goods. Because sectoral goods are complements ($\sigma < 1$), the relative price of nonagricultural goods rises sufficiently to increase the relative wage of

nonagriculture, holding effective labor supplies fixed. This induces workers in the newest generation to leave agriculture. Under the independent Fréchet benchmark, Proposition 1 implies that, for a given increase in the relative wage, worker reallocation is smaller in the two stylized economies with intergenerational persistence. Moreover, because the two IGP economies have the same aggregate labor supply elasticity, they exhibit the same initial responsiveness to relative wages.

The full general equilibrium response also depends on how worker reallocation feeds back into relative wages. As workers leave agriculture, relative effective labor supply shifts toward nonagriculture, reducing its relative price and partially offsetting the initial increase in its relative wage. Importantly, the magnitude of this feedback depends on the source of intergenerational persistence. Although the two IGP economies have identical initial worker allocations, they differ in the effective labor supplied by workers in each sector. With inherited preferences, a given reallocation of workers produces a larger increase in relative nonagricultural effective labor supply than with inherited skills. Consequently, the relative wage increases less in the τ-IGP economy, inducing less reallocation out of agriculture.

Proposition 2. *Under Assumption 1, an improvement in agricultural technology induces workers to leave agriculture in all three economies, but the magnitude of the immediate response satisfies*

$$\left.\frac{d\Phi_{A,t}}{d\log Z_{A,t}}\right|_{\text{no-IGP}} < \left.\frac{d\Phi_{A,t}}{d\log Z_{A,t}}\right|_{\kappa\text{-IGP}} < \left.\frac{d\Phi_{A,t}}{d\log Z_{A,t}}\right|_{\tau\text{-IGP}} < 0.$$

Thus, inherited preferences dampen the immediate reallocation out of agriculture more strongly than inherited skills.

Proof. See Appendix B.1.2. □

Thus, even when inherited skills and preferences generate the same aggregate labor supply elasticity, they can have different immediate effects on structural transformation because of their different general equilibrium wage feedbacks. In particular, inherited preferences dampen the immediate reallocation out of agriculture more strongly than inherited skills.

Dynamic Effects Worker reallocation in the current generation also affects the sector choices of future generations by changing the composition of ag-born and nonag-born workers. To isolate this intergenerational transmission, I examine how the agricultural

employment share in period t affects the next generation's allocation, holding the path of economic fundamentals fixed.

There are two channels through which this transmission operates. First, when workers leave agriculture in period t, fewer workers in period $t+1$ are ag-born. Because ag-born workers are more likely to work in agriculture, this compositional change directly reduces agricultural employment in the next generation. Holding next-generation wages fixed, equation (3) implies

$$\left.\frac{\partial \Phi_{A,t+1}}{\partial \Phi_{A,t}}\right|_{w_{N,t+1}/w_{A,t+1}} = \phi_{A,t+1}(A) - \phi_{A,t+1}(N) > 0. \tag{11}$$

Thus, the direct dynamic effect propagates reallocation out of agriculture across generations. Moreover, this effect is identical in the κ-IGP and τ-IGP economies at their matched initial allocations.

Second, changes in parental composition also affect the effective labor supplied to each sector, inducing relative wage adjustments. As the fraction of ag-born workers declines, relative effective labor supply shifts toward nonagriculture, reducing its relative wage and opposing the direct dynamic effect. With inherited preferences, this wage adjustment is insufficient to reverse the direct effect, so reallocation out of agriculture continues into the next generation. But with inherited skills, fewer ag-born workers also means fewer workers inheriting agricultural skills and more inheriting nonagricultural skills. This generates a larger shift in relative effective labor supply and a stronger wage adjustment, which can reverse the direct effect and induce workers in the next generation to return to agriculture.

Proposition 3. *Under Assumption 1, the equilibrium response of next-generation agricultural employment to the current generation's allocation satisfies*

$$\left.\frac{d\Phi_{A,t+1}}{d\Phi_{A,t}}\right|_{\textit{no-IGP}} = 0, \qquad \left.\frac{d\Phi_{A,t+1}}{d\Phi_{A,t}}\right|_{\tau\textit{-IGP}} > 0.$$

In the κ-IGP economy, the response can be positive or negative depending on whether the direct effect of changing parental composition or the induced relative wage adjustment dominates.

Proof. See Appendix B.1.3. □

Thus, inherited preferences propagate reallocation in the same direction across generations, whereas inherited skills can reverse its direction.

Cumulative Effects Propositions 2 and 3 establish how intergenerational persistence affects the immediate response to technological change and its transmission to subsequent generations. However, these results do not establish whether intergenerational persistence ultimately accelerates or slows structural transformation. For sufficiently weak intergenerational persistence, I can characterize the cumulative effects analytically.

Proposition 4. *Under Assumption 1, for a sufficiently small permanent improvement in agricultural technology, sufficiently weak inherited preferences increase reallocation out of agriculture relative to an economy without intergenerational persistence, whereas sufficiently weak inherited skills reduce it. These results hold in every generation after the immediate response and in the long run.*

Proof. See Appendix B.1.4. □

When intergenerational persistence is weak, its immediate dampening of reallocation is second order in the gap between ag-born and nonag-born workers' sector choices, whereas its dynamic effects are first order. Consequently, inherited preferences can accelerate structural transformation even though they initially slow worker reallocation. However, this result need not hold when intergenerational persistence is stronger.

To examine how these results extend to stronger intergenerational persistence, Figure 3 compares the responses to a permanent improvement in agricultural technology across successive generations. I vary the strength of inherited skills and preferences, holding the initial aggregate agricultural employment share fixed across economies. Each response is expressed relative to the no-IGP economy, so positive values indicate greater reallocation out of agriculture.

Figure 3 confirms that both inherited mechanisms dampen immediate reallocation, with inherited preferences generating the larger slowdown for a given degree of persistence. Across subsequent generations, however, their effects diverge. With inherited skills, the dynamic effects reinforce the initial slowdown for the two smaller degrees of persistence, but partially offset it for the largest. Nevertheless, reallocation remains below the no-IGP benchmark in all three cases shown. With inherited preferences, the dynamic effects offset the initial slowdown for all three degrees of persistence. For the smallest degree of persistence, these effects are sufficiently large that the economy eventually reallocates more workers out of agriculture than the no-IGP economy. For stronger persistence, the dynamic effects are insufficient to overcome the initial slowdown. Appendix

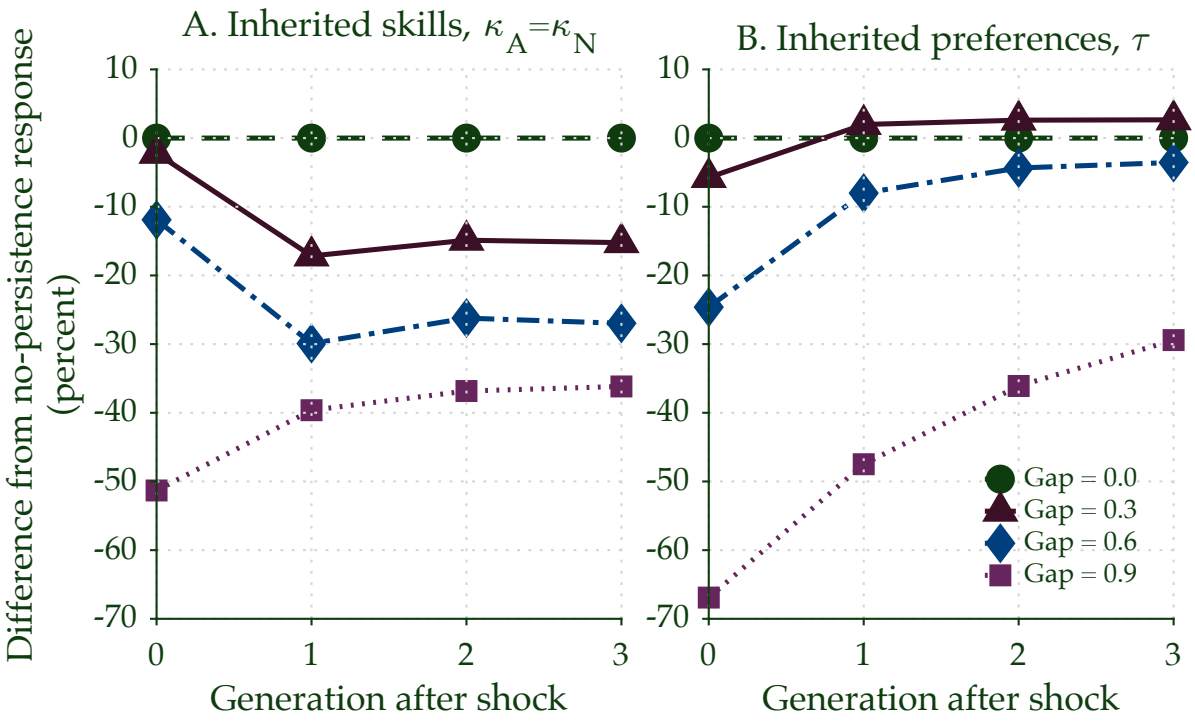


Figure 3: Numerical Responses Relative to No Intergenerational Persistence

Note: The figure reports responses to a permanent 0.01-log-unit increase in agricultural technology, Z_A. Responses are measured relative to the no-IGP benchmark as $100 \times$ (IGP response/no-IGP response $- 1$); positive values indicate faster reallocation, and generation zero is the first exposed to the shock. For each parameterization, fundamentals are separately re-anchored so that the unshocked economy remains at the common stationary agricultural employment share $\bar{\Phi}_A = 0.6$. The legend reports the resulting parent-sector employment gap, $\phi_A(A) - \phi_A(N)$. Panel A generates the positive gaps using symmetric inherited skills, $\kappa_A = \kappa_N \in \{1.373, 2.019, 4.403\}$ and $\tau_A = \tau_N = 0$; Panel B uses symmetric inherited preferences, $\tau_A = \tau_N \in \{0.271, 0.505, 0.772\}$ and $\kappa_A = \kappa_N = 1$. The remaining parameters are $\gamma = 2$, $\sigma = 0.3$, $\vartheta_A = \vartheta_N = 1$, and $\Omega_A = \Omega_N = 1$. This is an illustrative numerical exercise rather than a quantitative calibration to a particular economy.

Figures 14–17 repeat the exercise under alternative stationary allocations, nonhomothetic demand, Fréchet shape parameters, and elasticities of substitution across goods.

Thus, inherited preferences can either accelerate or slow structural transformation, depending on the strength of intergenerational persistence. Which mechanism dominates, and by how much, in the historical U.S. economy is a quantitative question that I address in the subsequent analysis.

3. Identification

Now I establish that workers' sector choices and earnings identify the inherited skills and inherited preferences parameters. With latent skills drawn independently from a com-

mon Fréchet distribution, the identification argument is especially transparent. However, the same economic logic applies to more flexible joint skill distributions, which I use in the quantitative application. Finally, I characterize the identification of sectoral technologies and common nonpecuniary benefits that drive structural transformation.

3.1. Identifying Inherited Skills and Preferences

The central distinction underlying identification is that inherited skills affect both potential earnings and sector choice, whereas inherited preferences affect sector choice without directly changing potential earnings. However, observed earnings also reflect workers' endogenous selection into sectors. I first establish the identifying relationships for a general joint distribution of sectoral skills, then show how independent Fréchet skills yield particularly transparent closed-form expressions.

General Identifying Relationships Let workers' idiosyncratic skills (ϵ_A, ϵ_N) follow a common joint distribution F across parental groups, with finite means in each sector. Define log comparative advantage as $D \equiv \log \epsilon_A - \log \epsilon_N$, with strictly increasing CDF G implied by F. Write $R_t \equiv (B_{N,t}/B_{A,t}) \cdot (w_{N,t}/w_{A,t})$ for the relative sectoral payoff common to both parental groups. Ag-born and nonag-born workers choose agriculture when, respectively,

$$D \geq \log R_t + \log(1-\tau_A) - \log \kappa_A, \qquad D \geq \log R_t + \log \kappa_N - \log(1-\tau_N).$$

Given F, each parental group's agricultural employment share identifies its sector choice threshold, $q_k \equiv G^{-1}(1-\phi_{A,t}(k))$.

These thresholds determine how workers from each parental group select into sectors. Let $m_{j,t}(k) \equiv \mathbb{E}_F[\epsilon_{i,j,t} \mid s_{i,t} = j, \text{ type } k]$ denote the average idiosyncratic skill of type-k workers who choose sector j. Conditional on F, these selected skill means are determined by the corresponding choice thresholds. Let $y_{j,t}(k)$ denote their mean earnings. Assuming both parental groups face the same wage per efficiency unit within each sector, their earnings ratios satisfy

$$\underbrace{\frac{y_{j,t}(j)}{y_{j,t}(\bar{j})}}_{\text{Earnings ratio}} = \underbrace{\kappa_j}_{\substack{\text{Inherited} \\ \text{skills}}} \cdot \underbrace{\frac{m_{j,t}(j)}{m_{j,t}(\bar{j})}}_{\substack{\text{Differential} \\ \text{selection}}}, \tag{ID-1}$$

where $\bar{j}$ denotes the sector other than j.

This decomposition illustrates why observed earnings differences need not reveal underlying inherited skills. For example, nonag-born workers are much less likely to enter agriculture than ag-born workers. Those who do enter may therefore be especially positively selected on their agricultural skills. Inherited agricultural skills then have two competing effects on the relative earnings of ag-born workers: they directly raise their agricultural earnings, but can also weaken their positive selection by inducing more workers with relatively low agricultural skills to enter the sector. Consequently, ag-born workers may earn less in agriculture despite possessing an inherited agricultural skill advantage.

Having identified inherited skills from selection-adjusted earnings differences, the remaining persistence in sector choice identifies inherited preferences. Subtracting the two parental groups' choice thresholds gives

$$q_N - q_A = \log \kappa_A + \log \kappa_N - \log(1 - \tau_A) - \log(1 - \tau_N). \tag{ID-2}$$

The common relative sectoral payoff cancels, leaving a relationship between observed sector choices and the inherited mechanisms. Under the restriction $\tau_A = \tau_N \equiv \tau$, this relationship identifies inherited preferences conditional on the joint skill distribution and inherited skills.

Together, equations (ID-1) and (ID-2) summarize the identifying relationships conditional on the joint skill distribution. When that distribution is unknown, its parameters must also be disciplined by the earnings data.

Closed-Form Identification under Common Fréchet Skills The common Fréchet specification makes these identifying relationships particularly transparent. It delivers closed-form expressions for the distribution of earnings, the selection corrections, and the sector choice thresholds.

First, the dispersion of log earnings identifies the Fréchet shape parameter γ. Within each destination-by-parental-sector cell, maximum indirect utility is Fréchet distributed and independent of which sector is chosen. Because log earnings differ from log maximum utility only by constants within each cell,

$$\gamma \quad = \quad \frac{\pi}{\sqrt{6 \operatorname{Var}[\log y_{i,j,t}(k) \mid s_{i,t} = j, \text{ type } k]}}. \tag{ID-3}$$

Thus, within-cell log-earnings variance identifies γ independently of sectoral wages and

the inherited mechanisms. This specification also implies that log-earnings variance must be identical across all four destination-by-parental-sector cells. Appendix C.1.1 provides the derivation.

Second, the Fréchet assumption provides a simple correction for workers' endogenous selection into sectors:

$$m_{j,t}(k) \quad = \quad \Gamma\left(\frac{\gamma-1}{\gamma}\right) \quad \cdot \quad \phi_{j,t}(k)^{-1/\gamma}.$$

Workers who are less likely to choose a sector have higher average idiosyncratic skills conditional on entering it. Consequently, the relative selection terms in equation (ID-1) depend only on γ and the parental groups' sector-choice probabilities. Inherited skills are therefore identified by

$$\kappa_A = \underbrace{\frac{y_{A,t}(A)}{y_{A,t}(N)}}_{\text{Earnings ratio}} \Bigg/ \underbrace{\left(\frac{\phi_{A,t}(A)}{\phi_{A,t}(N)}\right)^{-1/\gamma}}_{\text{Selection effect}}, \tag{ID-4}$$

and, analogously,

$$\kappa_N = \underbrace{\frac{y_{N,t}(N)}{y_{N,t}(A)}}_{\text{Earnings ratio}} \Bigg/ \underbrace{\left(\frac{\phi_{N,t}(N)}{\phi_{N,t}(A)}\right)^{-1/\gamma}}_{\text{Selection effect}}. \tag{ID-5}$$

Finally, given inherited skills, the remaining persistence in sector choice identifies inherited preferences. Define the intergenerational agricultural employment odds ratio as

$$\mathcal{O} \equiv \frac{\phi_{A,t}(A)/\phi_{N,t}(A)}{\phi_{A,t}(N)/\phi_{N,t}(N)}.$$

Under independent Fréchet skills, the general choice relationship in equation (ID-2) reduces to

$$\mathcal{O}^{1/\gamma} = \frac{\kappa_A \kappa_N}{(1-\tau)^2}.$$

Hence,

$$\tau = 1 - \sqrt{\frac{\kappa_A \kappa_N}{\mathcal{O}^{1/\gamma}}}. \tag{ID-6}$$

The independent Fréchet specification therefore delivers a sequential identification argument: earnings dispersion identifies skill heterogeneity, sector-choice probabilities determine the selection corrections needed to identify inherited skills, and residual persistence in sector choice identifies inherited preferences.

Quantitative Implementation The closed-form identification relies on independent sectoral skills with a common Fréchet shape parameter. In particular, this assumption restricts all four destination-by-parental-sector groups to have identical log-earnings variances, contrary to the substantial differences observed in the historical data.

In the quantitative application, I relax these restrictions by allowing sector-specific Fréchet shape parameters and dependence across sectoral skills through a Frank copula. The identification logic remains unchanged, but the selection corrections now depend on the joint skill distribution and must be computed numerically. I therefore estimate the skill distribution, inherited skills, inherited preferences, and the relative sectoral payoff jointly by simulated method of moments. The estimator targets two parental-group sector-choice shares, two within-sector earnings ratios, and four cell-specific log-earnings variances. Alternative skill distributions are considered in the appendix.

Identification continues to rely on the assumption of common sectoral wages per efficiency unit across parental groups. It also requires that observed wage-and-salary earnings adequately represent the relevant latent skill distributions conditional on sector and parental background. In particular, the model accounts for selection into sectors but not selection into the sample with observed earnings, which is important because the historical census excludes self-employment income, including that of many farmers.

3.2. Exogenous Drivers of Structural Transformation

I next show how aggregate economic outcomes identify sectoral technologies, $Z_{A,t}$ and $Z_{N,t}$, and the relative common nonpecuniary benefit, $B_{N,t}/B_{A,t}$. Only the ratio of common benefits is identified because scaling both by the same factor does not change choices. The argument applies to the independent Fréchet model and to the flexible joint distribution used in the quantitative analysis.

Proposition 5 (Identification of Fundamentals)**.** *Fix the joint skill distribution, inherited-skill and inherited-preference parameters, and consumer-demand parameters. Assume that the CDF of log comparative advantage is continuous and strictly increasing, with support on the real line, and that the relevant conditional mean skills are finite. Normalize $p_{A,t} = 1$ and impose $\tau_A = \tau_N \equiv \tau$. Given $\Phi_{A,t-1} \in (0,1)$, the observed outcomes $\Phi_{A,t} \in (0,1)$, $p_t > 0$, and $Q_t > 0$ uniquely identify $Z_{A,t}$, $Z_{N,t}$, and $B_{N,t}/B_{A,t}$, provided the outcomes are consistent with the model.*

Proof. See Appendix C.1.2. □

First, conditional on the inherited composition $\Phi_{A,t-1}$, aggregate agricultural employment is a continuous, strictly decreasing function of the common relative sectoral payoff R_t. Thus, the observed $\Phi_{A,t}$ identifies R_t uniquely. Given R_t and the known joint distribution, the model determines the sector choices of both parental groups and their conditional mean skills, hence aggregate effective labor $H_{A,t}$ and $H_{N,t}$. For the flexible distribution, these quantities are evaluated numerically rather than using the independent-Fréchet closed forms.

Second, relative demand and goods-market clearing determine relative sectoral technology:

$$\frac{Z_{N,t}}{Z_{A,t}} = \frac{H_{A,t}}{H_{N,t}} \cdot \frac{\Omega_N}{\Omega_A} \cdot p_t^{-\sigma} \cdot Q_t^{(\vartheta_N - \vartheta_A)(1-\sigma)}. \tag{ID-7}$$

Third, nonhomothetic demand and market clearing identify the level of agricultural technology:

$$Z_{A,t} = \frac{\Omega_A Q_t^{\vartheta_A}}{H_{A,t}} \left[\Omega_A + \Omega_N \left(p_t Q_t^{\vartheta_N - \vartheta_A} \right)^{1-\sigma} \right]^{\sigma/(1-\sigma)}. \tag{ID-8}$$

The level of nonagricultural technology then follows from the identified ratio. Finally, the competitive-wage condition implies $w_{N,t}/w_{A,t} = p_t Z_{N,t}/Z_{A,t}$. The relative common nonpecuniary benefit is therefore

$$\frac{B_{N,t}}{B_{A,t}} = R_t \frac{Z_{A,t}}{p_t Z_{N,t}}. \tag{ID-9}$$

These relationships identify the three fundamentals period by period. The inversion is nevertheless recursive because $\Phi_{A,t}$ determines the parental composition entering the following period. The consumer-demand parameters are treated as given here and calibrated separately in the quantitative application.

4. Quantitative Application: U.S. Industrialization

Here I quantify how intergenerational persistence shaped U.S. industrialization between 1880 and 1940. I estimate the inherited mechanisms and the joint distribution of sectoral skills using linked historical census records, calibrate consumer demand using aggregate data, and infer the fundamentals reproducing the observed transition. I then simulate counterfactual economies to examine how inherited skills and inherited preferences affected worker reallocation within and across generations.

4.1. Estimation of Labor Supply Parameters

Data and Empirical Moments I estimate labor supply parameters using linked father–son records from the 1940 census, the first U.S. census to report individual wage and salary earnings. I focus on employed, U.S.-born men aged 20–29 whose own and fathers' sectors can be identified. The estimated parameters are held fixed in the subsequent analysis of the 1880–1940 transition. Throughout, inherited preferences are symmetric across parental sectors, $\tau_A = \tau_N \equiv \tau$.

The estimator matches eight moments: the agricultural employment probabilities of ag-born and nonag-born workers, the two within-sector ratios of mean hourly earnings by parental sector, and the four variances of log hourly earnings conditional on destination and parental sector. These moments provide complementary information. Sector choices discipline the relative sectoral choice thresholds, earnings ratios discipline inherited skills after accounting for differential selection, and the four earnings variances help identify the dispersion and dependence of sectoral skills that govern those selection corrections. I use the full linked sample for sector-choice moments, workers with positive reported wage-and-salary income and valid weeks and hours for mean earnings, and the further full-time/full-year sample (at least 35 usual weekly hours and 40 weeks worked) for log-earnings variances. Mean earnings ratios use arithmetic means of hourly earnings, while the dispersion moments use variances of log hourly earnings. The resulting samples contain 5,022,389 sector choices, 3,119,595 observations for mean earnings, and 2,077,253 full-time/full-year observations for earnings variances. I retain the recorded 1940 top code of \$5,000 in constructing hourly earnings so that the baseline matches the adjusted historical series used throughout the paper.

I report two specifications based on the same model and eight-moment estimator. The *baseline* adjusts the empirical moments for age and race. The *schooling-and-location* specification additionally adjusts for childhood school attendance and adult commuting-zone fixed effects. The latter helps assess sensitivity to observed human capital and local labor-market differences, although schooling, migration, and adult location may themselves be related to inherited skills and preferences.

Preferred Joint Skill Distribution I retain Fréchet marginal skill distributions but allow skill dispersion to differ across sectors and the two skill draws to be correlated. Specifi-

cally,

$$F_j(\epsilon_j) = \exp\left(-\epsilon_j^{-\gamma_j}\right), \qquad j \in \{A, N\}, \qquad \Pr(\epsilon_A \leq a, \epsilon_N \leq n) = C_{\theta_F}(F_A(a), F_N(n)),$$

where C_{θ_F} is the Frank copula. The distribution therefore has two shape parameters, $\gamma_A, \gamma_N > 1$, and a dependence parameter, θ_F. I jointly estimate these parameters with inherited skills κ_A and κ_N, inherited preferences τ, and the relative payoff R governing sector choice in 1940. Relative skill-scale parameters are normalized, with R absorbing the relevant common relative-payoff scale.

I estimate the seven parameters by simulated method of moments (SMM) using the eight empirical targets described above. For each candidate parameter vector, the joint skill distribution determines parental-group sector choices, mean skills conditional on the chosen sector, and within-cell log-earnings variances. The estimator minimizes discrepancies between the corresponding simulated and empirical moments. Unlike the independent-Fréchet formulas in Section 3.1, the selection corrections here account explicitly for the estimated joint distribution and are computed numerically. I feature this specification because it provides the best fit to the eight moments used to discipline sector choice, selection, and inherited skills among the distributions considered. This criterion does not imply that it best describes every feature of the observed earnings distributions; I assess that distinction and alternative joint skill distributions in the appendix. Appendix A.2.4 provides the simulation, weighting, optimization, and distributional details.

Fit of Targeted Moments Table 1 compares the eight empirical targets with their fitted model counterparts. The model closely matches the choice probabilities and within-sector mean-earnings ratios in both specifications, while the larger remaining discrepancies are concentrated in the nonagricultural earnings variances.

Table 1: Empirical and Model-Implied SMM Moments

	Baseline		Schooling and location	
Moment	Data	Model	Data	Model
$\Pr(A \mid A\text{-born})$	0.529	0.531	0.469	0.471
$\Pr(A \mid N\text{-born})$	0.055	0.055	0.090	0.090
Mean ratio, agriculture	0.534	0.535	0.638	0.638
Mean ratio, nonagriculture	1.264	1.265	1.121	1.122
Variance, agriculture/A-born	0.767	0.773	0.691	0.682
Variance, agriculture/N-born	0.749	0.739	0.652	0.661
Variance, nonagriculture/A-born	0.396	0.351	0.337	0.301
Variance, nonagriculture/N-born	0.259	0.294	0.235	0.263
Eight-moment SMM objective	–	0.008	–	0.007

Note: The baseline moments adjust for age and race; the schooling-and-location moments additionally adjust for childhood school attendance and adult commuting zone. Mean ratios compare ag-born with nonag-born workers in agriculture and nonag-born with ag-born workers in nonagriculture, matching the estimator's target definitions. Variances are variances of log hourly earnings by destination and parental sector. The objective applies the estimator's transformations and weights to the unrounded moments, so it cannot be reconstructed exactly from the displayed entries.

Estimates of Inherited Skills and Preferences Table 2 reports estimates under the baseline and schooling-and-location specifications. The inheritance parameters have the same interpretation in both: κ_A and κ_N multiply efficiency units in the parental sector, while τ reduces the nonpecuniary value of leaving that sector. The copula and sector-specific shape parameters affect how strongly observed earnings gaps reflect selection rather than inherited skills.

The baseline estimates imply inherited skill multipliers of 2.776 in agriculture and 1.422 in nonagriculture, together with an inherited-preference wedge of 0.085. The corresponding schooling-and-location estimates are 1.992, 1.220, and 0.088. Thus, adding schooling and location controls attenuates the estimated agricultural skill advantage but leaves the inherited-preference wedge similar; the sector-specific dispersion and dependence parameters adjust simultaneously because all seven parameters are estimated jointly.

Inherited Skills and Selection in Observed Earnings The central identifying challenge is that average earnings within a destination sector reflect both inherited skills and the selection of workers with different parental backgrounds. Figure 4 decomposes each model-implied log earnings ratio (ag-born minus nonag-born) into the direct contribution of in-

Table 2: Fréchet–Frank Estimates of the Joint Skill Distribution and Inherited Mechanisms

	Baseline	Schooling and location
Skill distribution		
Agricultural Fréchet shape, γ_A	1.401	1.488
Nonagricultural Fréchet shape, γ_N	2.385	2.535
Frank dependence parameter, θ_F	3.574	3.379
Inherited mechanisms and common payoff		
Inherited agricultural skills, κ_A	2.776	1.992
Inherited nonagricultural skills, κ_N	1.422	1.220
Inherited preferences, τ	0.085	0.088
Relative sectoral payoff, R	3.113	2.458

Note: Both columns use a Fréchet–Frank joint skill distribution and target two parental-group choice shares, two within-sector arithmetic-mean hourly-earnings ratios, and four destination-by-parental-sector log-hourly-earnings variances. Baseline moments adjust for age and race; the second column additionally adjusts for childhood school attendance and adult commuting zone. Entries are point estimates. The choice, mean-earnings, and earnings-variance samples are defined in the text and need not have the same number of observations.

herited skills and the relative selection of workers into the sector. The inherited-skill contributions are $\log \kappa_A$ in agriculture and $-\log \kappa_N$ in nonagriculture. The selection contributions are the corresponding log ratios of conditional mean idiosyncratic skills computed under the fitted Fréchet–Frank distribution, not the closed-form $-1/\gamma$ expressions from the independent-Fréchet benchmark. The figure reports the baseline and schooling-and-location specifications separately and compares the fitted total with the corresponding empirical ratio of arithmetic mean hourly earnings.

Fit and Data Limitations A distinctive feature of the preferred estimator is that it disciplines the joint skill distribution with all four destination-by-parental-sector earnings variances, rather than imposing a common variance as independent Fréchet does. The remaining targeted moments and additional distributional validation checks are reported in Appendix A.2.4. That appendix also compares the preferred specification with independent Fréchet and bivariate lognormal skills; the alternative distributions are not used for the headline quantitative results.

The model corrects for endogenous *sector choice*, but does not model selection into the sample with observed wage-and-salary income. The 1940 census omits self-employment earnings, which is especially consequential for agricultural workers. Using the observed

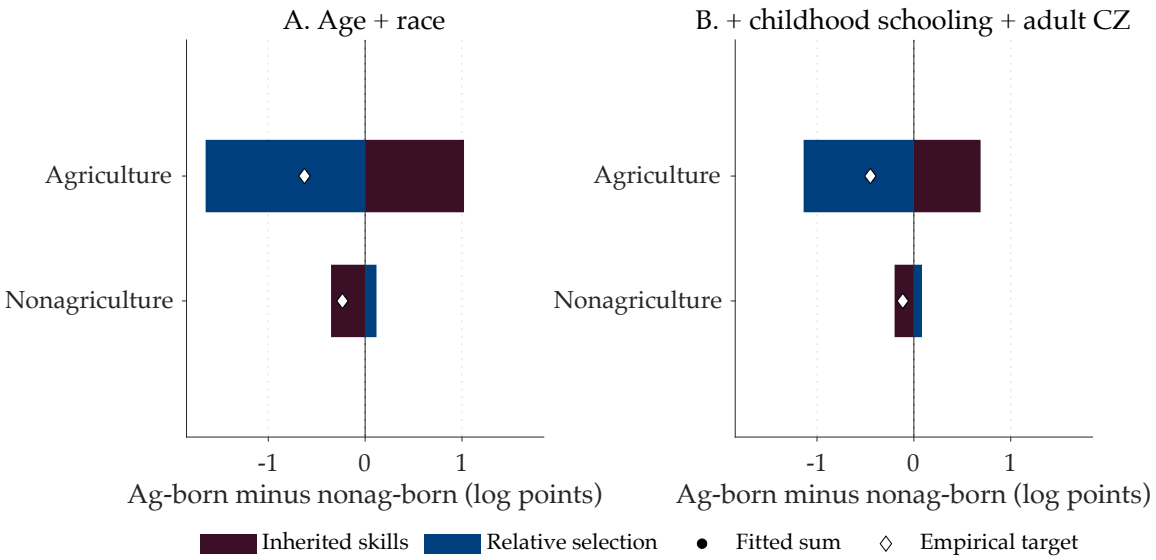


Figure 4: Inherited Skills, Selection, and Within-Sector Earnings Gaps

Note: In both panels, quantities are log ratios of arithmetic mean hourly earnings, ag-born minus nonag-born, for workers in the indicated destination sector. In agriculture, inherited skills contribute $\log \widehat{\kappa}_A$ and relative selection contributes $\log[\widehat{m}_A(A)/\widehat{m}_A(N)]$. In nonagriculture, the corresponding terms are $-\log \widehat{\kappa}_N$ and $\log[\widehat{m}_N(A)/\widehat{m}_N(N)]$. Model-implied components add exactly to the *fitted* log ratio of mean earnings. Empirical markers show the adjusted log ratio of arithmetic means targeted by SMM; because estimation is overidentified, the fitted and empirical markers need not coincide. The panels correspond to the age-and-race baseline and the additional schooling-and-location controls.

earnings moments to estimate the underlying skill distribution requires assuming that any remaining selection into observed earnings is ignorable conditional on destination sector, parental sector, and the relevant controls. A separate maintained assumption is that workers of different parental backgrounds face the same wage per efficiency unit within a destination sector, after accounting for observed characteristics. The schooling-and-location specification probes sensitivity to some observable differences but does not remove these limitations; moreover, location and schooling may be endogenous to family background.

4.2. Consumer Demand Calibration

I set $\sigma = 0.3$, consistent with the estimates in Comin, Lashkari and Mestieri (2021). I normalize $\Omega_N = 1$ and $\vartheta_A = 1$, calibrate Ω_A from the initial agricultural output share, and choose $\vartheta_N > \vartheta_A$ by minimum distance. I observe the agricultural output share, $\omega_{A,t}$, from Dennis and İşcan (2009), relative prices $p_t \equiv p_{N,t}/p_{A,t}$ from Alvarez-Cuadrado and Poschke (2011), and real GDP per capita Y_t/P_t from Maddison (1995) for each $t \in \{1880, 1900, 1920, 1940\}$. I index the last two series to one in 1880. With $\Omega_N = 1$, the 1880

agricultural output share identifies

$$\Omega_A = \frac{\omega_{A,1880}}{1 - \omega_{A,1880}}.$$

I then estimate ϑ_N by minimizing the squared distance between observed and fitted agricultural output shares in 1900, 1920, and 1940:

$$\widehat{\vartheta}_N = \arg\min_{\vartheta_N > 1} \sum_{t \in \{1900,1920,1940\}} [\omega_{A,t} - \widehat{\omega}_{A,t}(\vartheta_N; \sigma = 0.3)]^2,$$

where

$$\frac{\widehat{\omega}_{A,t}}{1 - \widehat{\omega}_{A,t}} = \Omega_A p_t^{\sigma-1} Q_t^{(1-\vartheta_N)(1-\sigma)}.$$

The resulting calibration is $\Omega_A = 0.449$ and $\widehat{\vartheta}_N = 4.630$. Appendix Figure 18 compares fitted and observed agricultural output shares. The calibrated consumer-demand parameters are held fixed across the two labor-supply control specifications. Table 3 summarizes the benchmark parameters and their empirical discipline.

4.3. Inferring Drivers of U.S. Structural Transformation

I infer the historical paths of sectoral technologies and the relative common nonpecuniary benefit by matching the observed agricultural employment share, relative goods price, and real GDP per capita in each census year. Proposition 5 establishes the inversion conditional on the joint skill distribution and inherited parameters. For each labor-supply specification, I recover parental-group choices and effective labor using the estimated Fréchet–Frank distribution; the inversion does not substitute the independent-Fréchet choice or selection formulas. The observed employment share from the previous census determines the parental composition of the current generation. I use the age-and-race estimates as the preferred benchmark for the main figures, and repeat the full inversion using the schooling-and-location estimates as a sensitivity exercise in the appendix. Appendix D.1 describes the numerical implementation and fit to the targeted aggregate outcomes.

Table 3: Benchmark Model Parameters

Parameter	Interpretation	Value	Method	Empirical discipline
Panel A: Labor supply and joint skill distribution				
γ_A	Agricultural Fréchet shape	1.401	SMM estimate	Destination-by-parental-sector log-earnings variances and selection into agriculture.
γ_N	Nonagricultural Fréchet shape	2.385	SMM estimate	Destination-by-parental-sector log-earnings variances and selection into nonagriculture.
θ_F	Frank dependence parameter	3.574	SMM estimate	Joint fit of sector choices, within-sector earnings ratios, and earnings variances.
κ_A	Inherited agricultural skills	2.776	SMM estimate	Agricultural earnings differences by parental sector, accounting for selection.
κ_N	Inherited nonagricultural skills	1.422	SMM estimate	Nonagricultural earnings differences by parental sector, accounting for selection.
τ	Inherited preferences	0.085	SMM estimate	Intergenerational sector-choice persistence net of inherited skills.
R	Relative sectoral payoff in 1940	3.113	SMM estimate	Levels of the two parental groups' sector-choice probabilities.
Panel B: Consumer demand				
Ω_A	Agricultural demand weight	0.449	Calibrated	Agricultural output share in 1880.
Ω_N	Nonagricultural demand weight	1	Normalized	Relative demand-weight normalization.
σ	Elasticity of substitution	0.300	Externally set	Comin, Lashkari and Mestieri (2021).
ϑ_A	Agricultural income parameter	1	Normalized	Reference income-exponent normalization.
ϑ_N	Nonagricultural income parameter	4.630	Calibrated	Minimum-distance fit to agricultural output shares in 1900, 1920, and 1940.
Panel C: Initial condition				
$\Phi_{A,0}$	Initial agricultural employment share	0.665	Observed	Agricultural employment share in 1860, determining parental composition in 1880.

Note: Panel A reports the preferred age-and-race Fréchet–Frank estimates from Table 2; all seven parameters are estimated jointly using the eight moments described in the text. Consumer-demand parameters use historical agricultural output shares, relative prices, and real GDP per capita. The 1860 agricultural employment share determines the parental composition of the generation observed in 1880.

The inversion exactly targets aggregate employment, prices, and real GDP per capita, but it does not target the agricultural employment shares of ag-born and nonag-born workers separately in the earlier years. The resulting subgroup predictions are therefore an informative check on the maintained assumption that the inherited parameters estimated in 1940 also apply in 1880–1920. Without comparable individual earnings before 1940, identifying historical changes in inherited skills and preferences would require additional assumptions.

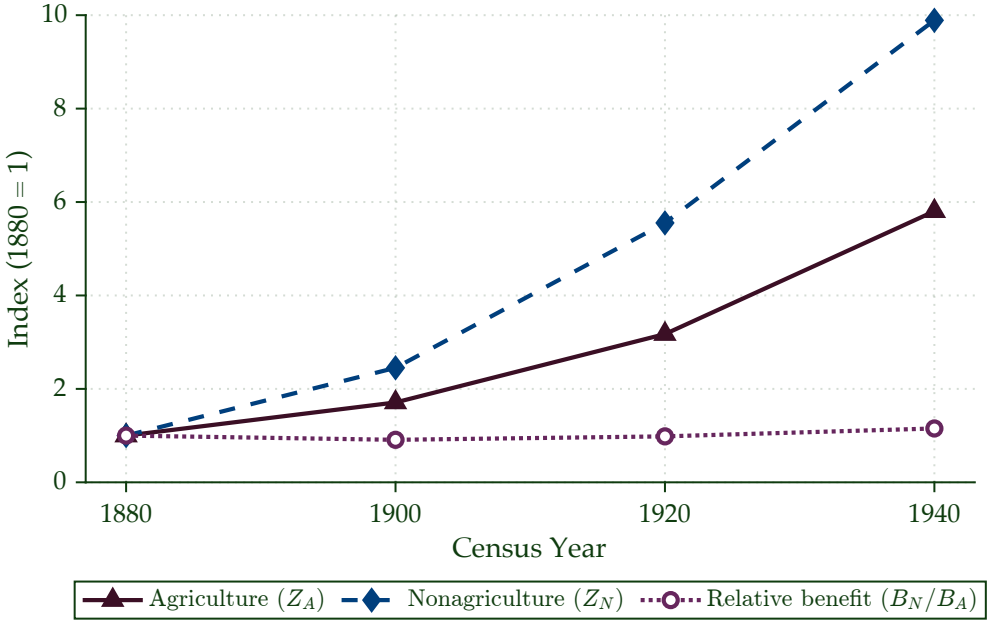


Figure 5: Inferred Drivers of U.S. Structural Transformation, 1880–1940

Note: The figure plots $Z_{A,t}$, $Z_{N,t}$, and $B_{N,t}/B_{A,t}$ identified from the observed aggregate agricultural employment share, relative price, and real GDP per capita using the baseline age-and-race Fréchet–Frank estimates. Each series is indexed to one in 1880. Appendix figures repeat the exercise using the schooling-and-location estimates.

In the benchmark inversion, agricultural technology grows by a factor of 5.80 and nonagricultural technology by a factor of 9.89 between 1880 and 1940. The relative common nonpecuniary benefit changes by a factor of 1.15. These are the fundamentals required to account for the observed transition *conditional on* the inherited mechanisms and joint skill distribution estimated from the 1940 data.

Because sectoral effective labor depends on workers' inherited skills and parental composition, the technological changes inferred from observed prices and employment also depend on whether intergenerational persistence is included. Figure 6 compares the preferred benchmark inversion with one that removes inherited skills and inherited preferences while retaining the estimated joint skill distribution and consumer-demand param-

eters. The inferred 1880–1940 change in $Z_{N,t}/Z_{A,t}$ is 53.3 log points with persistence and 88.6 log points without it. The corresponding changes in $B_{N,t}/B_{A,t}$ are 14.2 and -5.4 log points. The comparison isolates the effect of accounting for inherited connections on the fundamentals needed to rationalize the *same* observed historical outcomes.

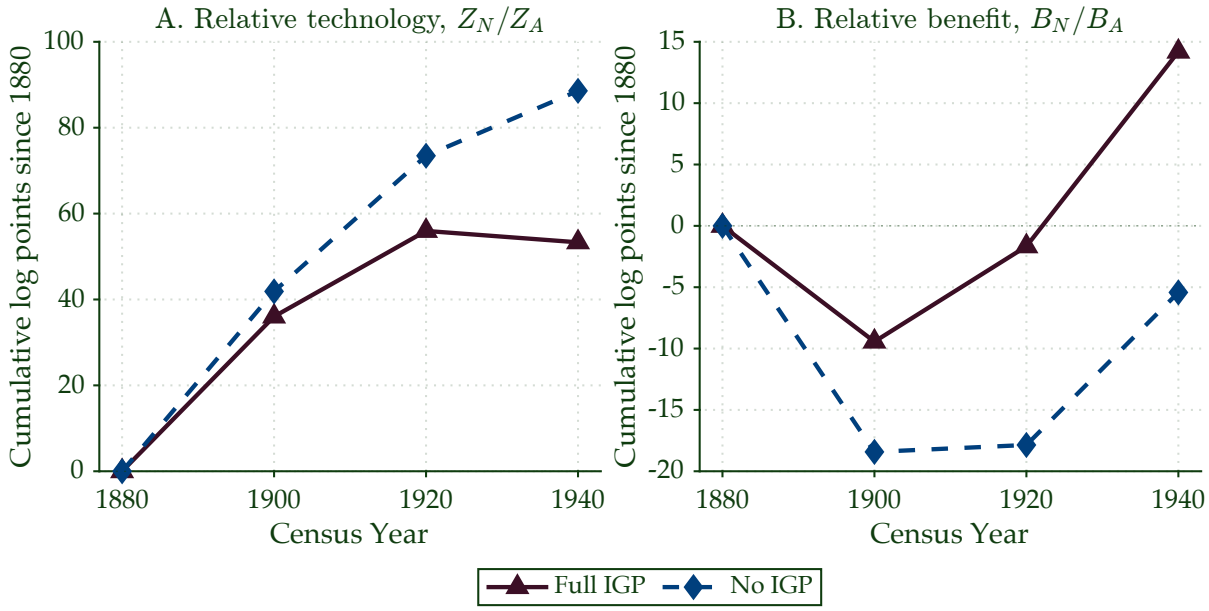


Figure 6: Inferred Relative Drivers With and Without Intergenerational Persistence

Note: Panel A plots cumulative log growth in relative technology, $Z_{N,t}/Z_{A,t}$; Panel B plots cumulative log growth in the relative common nonpecuniary benefit, $B_{N,t}/B_{A,t}$. Each model series is normalized to zero in 1880. Both inversions use the same preferred Fréchet–Frank joint skill distribution, consumer demand, and observed aggregate targets; the no-persistence inversion sets $\kappa_A = \kappa_N = 1$ and $\tau = 0$. The appendix reports the schooling-and-location counterpart together with an external historical relative-TFP comparison.

4.4. Counterfactual U.S. Structural Transformation

I quantify the effect of intergenerational persistence on structural transformation by removing inherited skills, inherited preferences, and both mechanisms together. Removing inherited skills sets $\kappa_A = \kappa_N = 1$; removing inherited preferences sets $\tau = 0$. In each counterfactual, the estimated joint skill distribution is held fixed, including its two Fréchet shape parameters and Frank dependence parameter. This is important: removing inheritance is not the same as replacing the joint skill distribution with independent Fréchet draws.

For each labor-supply specification and counterfactual regime, I re-infer the 1880 levels of sectoral technologies and common nonpecuniary benefits so that all economies match the same initial agricultural employment share, relative price, and real GDP per capita.

I then expose the counterfactual economy to the *growth rates* in fundamentals inferred under the full-persistence benchmark for the corresponding control specification. The parental composition of each new generation is determined recursively by the previous generation's simulated allocation. Thus, counterfactual differences after 1880 reflect the removed inherited mechanisms under common benchmark growth in fundamentals, not different initial aggregate conditions.

Table 4 reports counterfactual changes in the agricultural employment share between 1880 and 1940, first for the baseline and then for the schooling-and-location estimates. Under the preferred baseline specification, removing all intergenerational persistence produces an additional 8.87 percentage points of reallocation out of agriculture, equal to 23.2 percent of the observed 1880–1940 decline in agricultural employment. The corresponding effect using schooling-and-location controls is 6.11 percentage points (16.0 percent of the observed decline).

Table 4: Cumulative Effects on U.S. Worker Reallocation, 1880–1940

Counterfactual	$\Delta\Phi_A$ (pp)	Effect (pp)	Effect (%)
Panel A: Baseline (age and race)			
Benchmark	-38.3	–	–
No inherited skills	-47.58	-9.32	24.3
No inherited preferences	-38.40	-0.14	0.4
No intergenerational persistence	-47.13	-8.87	23.2
Panel B: Schooling and location			
Benchmark	-38.3	–	–
No inherited skills	-44.83	-6.57	17.2
No inherited preferences	-38.36	-0.10	0.3
No intergenerational persistence	-44.37	-6.11	16.0

Note: $\Delta\Phi_A \equiv 100(\Phi_{A,1940} - \Phi_{A,1880})$ is negative for reallocation out of agriculture. Effects are counterfactual minus full-persistence benchmark in percentage points; positive percentage effects indicate *more* reallocation than the benchmark, with numerator and denominator measured under the signed convention. Each counterfactual is re-anchored to common observed 1880 aggregate targets, retains the specification's estimated joint skill distribution, and faces that specification's full-persistence growth in fundamentals thereafter.

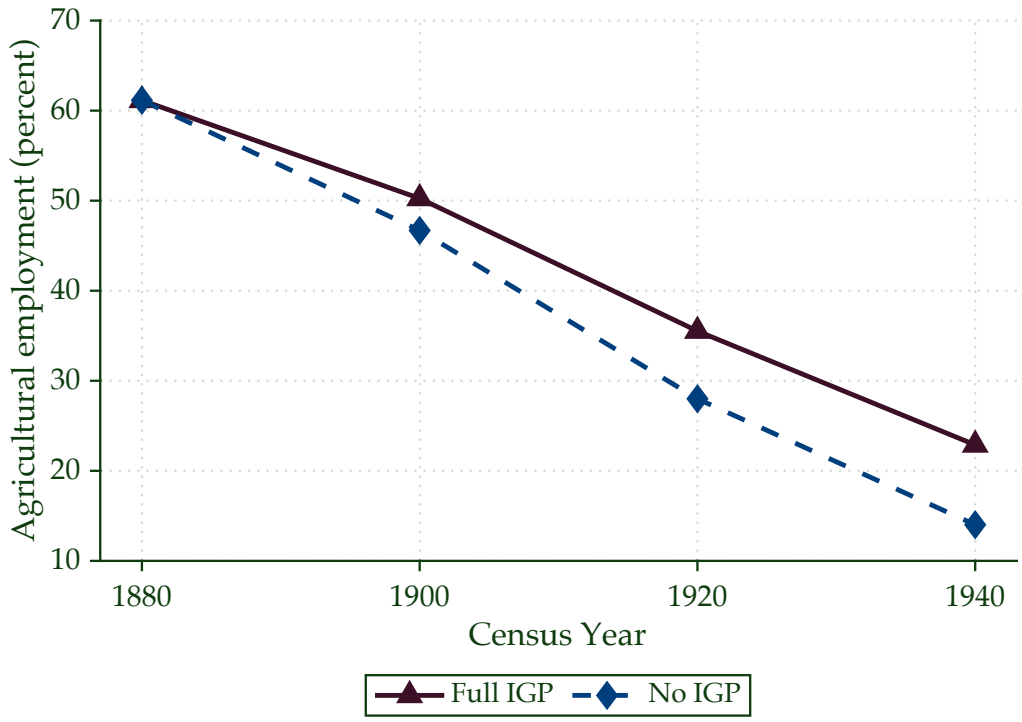


Figure 7: Counterfactual Agricultural Employment Shares, 1880–1940

Note: The figure plots agricultural employment under full intergenerational persistence (Full IGP) and without inherited skills or preferences (No IGP), using the preferred age-and-race Fréchet–Frank estimates. Both paths match the same observed 1880 employment share; the No IGP path evolves under Full IGP growth in inferred fundamentals but sets $\kappa_A = \kappa_N = 1$ and $\tau = 0$. The corresponding figure using schooling-and-location estimates is reported in the appendix.

Aggregate Labor Supply Responsiveness Figure 8 applies Proposition 1 along the preferred counterfactual paths. Aggregation substantially reduces responsiveness under full intergenerational persistence. In 1880, the weighted average of within-group elasticities is 2.572, while the aggregate elasticity is 1.328, an attenuation of 48.35 percent. In 1940, the corresponding elasticities are 2.238 and 1.554, so attenuation remains substantial at 30.58 percent. These are model-implied attenuation rates based on the fitted subgroup allocations; the 1880 value is distinct from the 41.69 percent attenuation calculated from observed allocations in Section 2.2.

Aggregation attenuation is distinct from the entire gap between the Full IGP and No IGP elasticities. Under Fréchet–Frank, within-group responsiveness varies with the prevailing sector-choice threshold, so the Full IGP weighted within-group elasticity need not equal the No IGP elasticity. In 1880, the two are nearly identical (2.572 versus 2.604), and aggregation accounts for 97.46 percent of the overall elasticity gap. By 1940, the Full IGP weighted within-group elasticity, 2.238, exceeds the No IGP elasticity, 1.770. Nevertheless, the Full IGP aggregate elasticity remains lower because the 0.684 aggregation con-

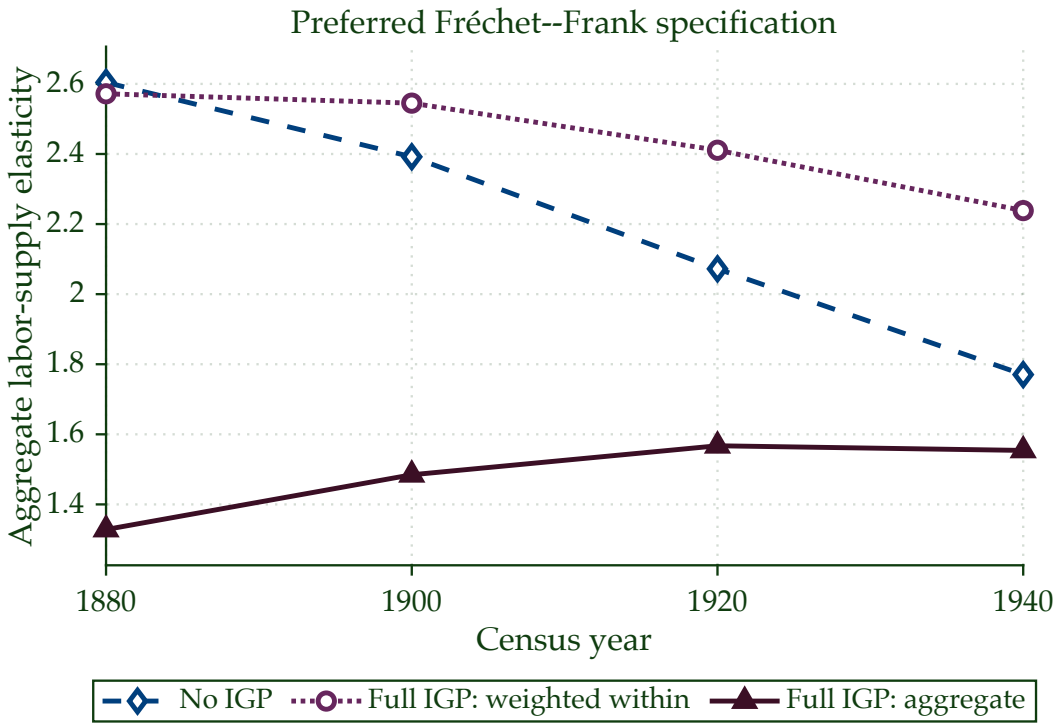


Figure 8: Aggregate Labor Supply Responsiveness With and Without Intergenerational Persistence

Note: All series are partial-equilibrium elasticities with respect to relative sectoral wages, evaluated at the prevailing equilibrium of the indicated economy. The derivatives hold parental composition, common non-pecuniary benefits, inherited parameters, and the joint skill distribution fixed. The weighted within-group elasticity is defined in Proposition 1; its difference from the Full IGP aggregate elasticity isolates aggregation attenuation. The No IGP elasticity is evaluated at the counterfactual economy's own equilibrium along its reanchored path, so its difference from the Full IGP aggregate elasticity also reflects differences in within-group responsiveness at the two economies' prevailing choice thresholds. The figure uses the preferred age-and-race Fréchet–Frank estimates.

tribution more than offsets the -0.468 difference in within-group responsiveness. This is an exact accounting of elasticities evaluated at different prevailing equilibria, not a causal decomposition holding allocations fixed.

These elasticities characterize workers' responsiveness to relative wages at a given point in the transition. The cumulative effects of intergenerational persistence also depend on how changes in parental composition are transmitted across generations. To distinguish contemporaneous adjustment from intergenerational propagation, I decompose each cumulative counterfactual effect into an immediate component from changes in fundamentals at a fixed parental composition and a dynamic component from changes in parental composition at fixed fundamentals. These effects interact, so I use a two-component Shapley decomposition that averages over the two possible orders of changing fundamentals and parental composition in each historical interval. I sum the contribu-

tions over 1880–1940 and subtract the corresponding contributions under full persistence. Table 5 reports the decomposition under both control specifications.

Table 5: Immediate and Dynamic Components of Cumulative Effects

Counterfactual	Immediate (pp)	Dynamic (pp)	Total (pp)
Panel A: Baseline (age and race)			
No inherited skills	-0.66	-8.66	-9.32
No inherited preferences	-1.35	1.21	-0.14
No intergenerational persistence	-1.10	-7.77	-8.87
Panel B: Schooling and location			
No inherited skills	0.02	-6.59	-6.57
No inherited preferences	-1.23	1.13	-0.10
No intergenerational persistence	-0.38	-5.73	-6.11

Note: The decomposition changes fundamentals and inherited parental composition in both possible orders and averages the resulting marginal contributions. Components are differences relative to the corresponding full-persistence benchmark, using the signed agricultural-share convention (negative indicates additional reallocation out of agriculture). Immediate and dynamic components must add to the cumulative percentage-point effect in Table 4 within numerical tolerance.

In the baseline no-persistence counterfactual, the immediate component is -1.10 percentage points and the dynamic component is -7.77 percentage points. The latter captures how inherited connections change the productive composition of subsequent generations and, through general-equilibrium wages, their sectoral employment decisions. The importance and sign of each component depend on whether persistence operates through inherited skills or inherited preferences; the counterfactual comparisons distinguish these mechanisms rather than imposing an additive decomposition of the inherited-skill and inherited-preference treatments. Removing inherited skills accounts for nearly all of the cumulative baseline effect: its immediate and dynamic components are both negative and sum to -9.32 percentage points. For inherited preferences, the immediate effect of -1.35 percentage points is nearly offset by the dynamic effect of 1.21 percentage points, leaving a net effect of only -0.14 percentage points. Adding schooling and location controls attenuates the no-persistence effect from 8.87 to 6.11 percentage points, but preserves the same qualitative contrast between inherited skills and preferences.

5. Conclusion

New generations entering the labor force may be more flexible than incumbent workers, but their sector choices are not independent of their parents' sectors. Using linked historical U.S. census records and a macroeconomic model of structural transformation, I find that removing intergenerational persistence would have increased the decline in agricultural employment between 1880 and 1940 by approximately 8.9 percentage points, or 23.2 percent of the observed decline. Most of this effect comes from inherited skills, which dampen the immediate reallocation out of agriculture and reinforce this slowdown across subsequent generations through changes in effective labor supply and relative wages. Inherited preferences also dampen immediate reallocation, but their dynamic effects largely offset the initial slowdown. Accounting for these inherited mechanisms also changes the sectoral technological progress the model infers from the observed transition.

These findings have implications beyond U.S. industrialization. Intergenerational mobility is often studied for its implications for inequality and economic opportunity (OECD, 2018; Narayan et al., 2018), but inherited connections also shape how an economy responds to economic change. The analytical benchmark shows that systematic differences in sector choices across parental groups can reduce aggregate labor supply responsiveness even when the groups have identical within-group elasticities. Models that abstract from these differences may therefore overstate how strongly wage changes translate into worker reallocation. Moreover, distinguishing inherited skills from inherited preferences matters because their effects on reallocation differ immediately and across generations.

One direction for future work is to apply this framework to contemporary developing economies, where agricultural occupations are often transmitted across generations and labor reallocation can lag agricultural productivity growth (Emran and Shilpi, 2011; Krishna and Meenakshi, 2022). Linked data on parents' sectors, workers' earnings, and sector choices could help establish whether inherited skills or preferences contribute to slower structural transformation.

Another direction is to incorporate intergenerational persistence into models of adjustment to shorter-run sectoral shocks, including trade shocks. Recent evidence suggests that the eventual recovery of U.S. labor markets exposed to import competition came disproportionately from workers who were below working age when the shock began (Autor et al., 2025). Allowing these workers' sector choices to depend on their parents' sectors could reveal whether the arrival of new generations dampens or propagates the effects of such shocks.

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A. Appendix: Empirical Results

A.1. Data Construction

This appendix documents linked-census sample construction and reports additional robustness results for the intergenerational-persistence estimates.

A.1.1. MLP Sample Selection

Appendix Table 6 documents how the MLP linkage procedure changes the analysis sample. Panel A begins with U.S.-born employed men ages 20–29 with a valid current industry and reports the number linked to a childhood census record, the number for whom a father is identified in the childhood household, and the number with a valid paternal industry. The final sample ranges from 49 percent of eligible workers in 1880 to 68 percent in 1940. Panel B compares the full eligible population with the final linked sample in 1940. Despite the reduction in sample size, the two samples are similar along the observed dimensions reported in the table. The linked sample is modestly more agricultural, white, and educated and modestly less urban and Southern, while its sector-specific earnings moments remain close to those in the full sample. The standardized differences in the final column are small, with the largest equal to 0.103 standard deviation for age.

Table 6: MLP Linkage and Sample Selection

Panel A: MLP sample construction

Year	Eligible	Childhood link	Father identified	Final sample	Final share
1880	1,756,491	1,152,891	1,047,863	864,166	49.2%
1900	2,619,056	1,644,774	1,565,404	1,394,395	53.2%
1920	4,786,523	3,638,704	3,343,024	2,971,796	62.1%
1940	7,393,949	5,903,317	5,491,663	5,022,389	67.9%

Panel B: Full versus linked sample, 1940

Statistic	Full eligible sample	Final MLP sample	Difference	Std. difference
Agricultural employment share	0.215	0.229	0.014	0.034
Mean age	24.556	24.264	-0.291	-0.103
White share	0.923	0.941	0.018	0.067
Urban share	0.536	0.518	-0.017	-0.035
High-school graduate share	0.381	0.408	0.027	0.055
South share	0.290	0.273	-0.016	-0.036
Valid hourly-earnings share	0.634	0.621	-0.013	-0.027
Mean log hourly earnings: Agriculture	-1.931	-1.944	-0.013	-0.014
Variance of log hourly earnings: Agriculture	0.899	0.913	0.015	—
Mean log hourly earnings: Nonagriculture	-0.778	-0.783	-0.005	-0.008
Variance of log hourly earnings: Nonagriculture	0.434	0.429	-0.005	—

Note: Panel A begins with U.S.-born employed men ages 20–29 with a valid current industry. "Childhood link" requires an MLP link to the relevant preceding census, using 1880 as the childhood census for workers observed in 1900 because the 1890 census was destroyed. "Father identified" additionally requires a household-head father in the childhood record, and the final sample additionally requires a valid paternal industry. The final share is relative to the eligible population in the first column. Panel B compares the full eligible population with the final MLP sample in 1940; the Difference column reports the linked-sample mean minus the full-sample mean. The standardized difference divides this difference by the full-eligible-sample standard deviation; earnings means are standardized within sector, and standardized differences are not reported for variance statistics. The hourly-earnings sample requires positive valid wage income, weeks worked, and usual weekly hours; recorded topcoded income of $5,001 is set to its $5,000 censoring threshold.

Appendix Table 7 documents the additional sample selection induced by the earnings restrictions. Although the final linked sample remains close to the eligible population on observed characteristics, requiring observed wage and salary earnings selects substantially away from agriculture, consistent with the absence of self-employment income in the 1940 census.

Table 7: Nested Analysis Samples and Composition, 1940

Panel A: Sample sizes

Sample	Observations	Percent of eligible	Percent of linked
Eligible workers	7,393,949	100.0	—
Full linked sample	5,022,389	67.9	100.0
Positive hourly earnings	3,119,595	42.2	62.1
Earnings moments sample	2,077,253	28.1	41.4

Panel B: Sample composition

Characteristic	Eligible	Full linked	Positive earnings	Earnings moments
Agricultural employment share	0.215	0.229	0.116	0.114
Mean age	24.556	24.264	24.392	24.641
White share	0.923	0.941	0.949	0.953
Urban share	0.536	0.518	0.608	0.636
High-school graduate share	0.381	0.408	0.455	0.492
South share	0.290	0.273	0.250	0.247

Note: Eligible workers are U.S.-born employed men ages 20–29 with a valid industry. The full linked sample additionally requires an identified father with a valid industry. Positive hourly earnings require positive wage and salary income, weeks worked, and usual weekly hours. The earnings moments sample further requires at least 40 weeks worked and at least 35 usual weekly hours. Columns are nested from left to right.

A.1.2. Alternative Historical Linkages

My baseline analysis uses the Multigenerational Longitudinal Panel (MLP) links distributed by IPUMS. As an alternative, I use the Census Tree crosswalks to link each adult observation to his childhood census record and then identify his father as the household head in that census. I also repeat the analysis using the automated linking procedure of Abramitzky et al. (2021), hereafter ABE. My primary ABE results use exact-name conservative links, which require a unique match within a two-year age window, and I additionally report the less restrictive exact-name standard links. I apply the same age, sex, birthplace, employment, industry, earnings, schooling, and geographic restrictions under all linkage procedures.

Appendix Tables 8 and 9 report estimates across linkages. The first compares reduced-form sector-choice and earnings gaps, including the age-and-race specification used for the baseline empirical figures. The alternative links produce somewhat smaller sector-choice gaps, particularly in 1880 and 1900, but the differences narrow substantially by 1920 and 1940. The primary ABE estimates are especially close to the Census Tree estimates, while the larger ABE standard sample produces somewhat smaller gaps. The sec-

ond re-estimates the preferred seven-parameter, eight-moment SMM model using each 1940 linkage sample. The inherited-skill and inherited-preference estimates are similar across linkage procedures in both control specifications. The estimated dependence parameter varies somewhat more, but all four samples imply positive skill dependence and the same substantive role for inherited skills and preferences.

Table 8: Reduced-Form Robustness to Alternative Historical Linkages

Outcome	Specification	MLP	Census Tree	ABE exact cons.	ABE exact std.
Panel A: Sector choice					
Sector choice, 1880	Unadjusted	0.667	0.594	0.583	0.537
Sector choice, 1880	Age + race	0.665	0.591	0.579	0.533
Sector choice, 1880	School + child/adult CZ FE	0.556	0.466	0.468	0.412
Sector choice, 1900	Unadjusted	0.621	0.563	0.545	0.511
Sector choice, 1900	Age + race	0.615	0.556	0.538	0.502
Sector choice, 1900	School + child/adult CZ FE	0.499	0.421	0.422	0.375
Sector choice, 1920	Unadjusted	0.558	0.532	0.521	0.488
Sector choice, 1920	Age + race	0.554	0.528	0.520	0.486
Sector choice, 1920	School + child/adult CZ FE	0.445	0.416	0.417	0.379
Sector choice, 1940	Unadjusted	0.482	0.457	0.445	0.424
Sector choice, 1940	Age + race	0.475	0.451	0.441	0.419
Sector choice, 1940	School + child/adult CZ FE	0.396	0.373	0.367	0.345
Panel B: Log hourly earnings					
Log hourly earnings, Agricultural workers	Unadjusted	-0.650	-0.590	-0.584	-0.520
Log hourly earnings, Nonagricultural workers	Unadjusted	-0.304	-0.299	-0.274	-0.275
Log hourly earnings, Agricultural workers	Age + race	-0.627	-0.569	-0.576	-0.515
Log hourly earnings, Nonagricultural workers	Age + race	-0.288	-0.277	-0.262	-0.254
Log hourly earnings, Agricultural workers	School + child/adult CZ FE	-0.422	-0.380	-0.393	-0.345
Log hourly earnings, Nonagricultural workers	School + child/adult CZ FE	-0.150	-0.145	-0.141	-0.135

Note: The table compares estimates obtained from IPUMS MLP links with estimates obtained from Census Tree and ABE links for the 1870–1880, 1880–1900, 1910–1920, and 1930–1940 census pairs. ABE conservative uses exact-name links that are unique within a two-year age window, while ABE standard uses the less restrictive exact-name procedure. Panel A reports linear-probability coefficients on an indicator for a father employed in agriculture, where the outcome is an indicator for the son working in agriculture. Its age-and-race rows reproduce the specification used in the paper's baseline sector-choice figure; the fully adjusted rows absorb childhood-schooling, childhood-commuting-zone, and adulthood-commuting-zone fixed effects. Panel B reports coefficients on father employed in agriculture from regressions of log hourly earnings estimated separately for agricultural and nonagricultural workers in 1940. Its age-and-race rows reproduce the specification used in the paper's baseline earnings figure; the fully adjusted rows use the same three sets of fixed effects as Panel A. The Panel B sample has positive hourly earnings and the full-time, full-year restrictions used for the earnings-dispersion moments.

Table 9: Preferred SMM Estimates across Historical Linkages

Parameter	MLP	Census Tree	ABE exact cons.	ABE exact std.
Panel A: Age and race				
$\widehat{\gamma}_A$	1.401	1.403	1.379	1.364
$\widehat{\gamma}_N$	2.385	2.363	2.381	2.381
$\widehat{\theta}_F$	3.574	3.227	3.108	2.775
$\widehat{\kappa}_A$	2.776	2.712	2.781	2.697
$\widehat{\kappa}_N$	1.422	1.418	1.379	1.382
$\widehat{\tau}$	0.085	0.078	0.096	0.088
$\widehat{R}$	3.113	3.112	3.327	3.301
SMM objective	0.008	0.006	0.007	0.007
Panel B: Schooling and location				
$\widehat{\gamma}_A$	1.488	1.476	1.454	1.470
$\widehat{\gamma}_N$	2.535	2.508	2.528	2.521
$\widehat{\theta}_F$	3.379	2.357	2.419	2.901
$\widehat{\kappa}_A$	1.992	1.974	2.022	1.922
$\widehat{\kappa}_N$	1.220	1.236	1.222	1.196
$\widehat{\tau}$	0.088	0.099	0.103	0.075
$\widehat{R}$	2.458	2.575	2.702	2.551
SMM objective	0.007	0.005	0.006	0.005

Note: The table estimates the preferred seven-parameter joint-skill model separately using IPUMS MLP, Census Tree, ABE exact-name conservative, and ABE exact-name standard links. Each column targets the same two parental-group choice shares, two within-sector arithmetic-mean hourly-earnings ratios, and four destination-by-parental-sector log-hourly-earnings variances, with the weighting matrix used for the main estimates. Panel A adjusts the empirical moments for age and race; Panel B additionally adjusts for childhood school attendance and adult commuting-zone fixed effects. The MLP columns reproduce the main estimates, which use one million simulation draws. The alternative-linkage estimates use 200,000 simulation draws and are independently re-evaluated on a second set of 200,000 draws; all optimizations converged and no estimate lies on a parameter boundary. Objectives are calculated using unrounded moments and parameters.

Appendix Table 10 compares sample retention directly across the four linkage procedures. The Census Tree retains a share of eligible workers similar to the MLP, whereas the exact-name ABE procedures trade substantially smaller samples for more conservative links.

Table 10: Sample Retention Across Historical Linkage Procedures

Panel A: Sample flow in 1940

Procedure	Eligible (millions)	Percent of eligible			Final (millions)
		Linked	Father identified	Final	
MLP	7.39	79.8	74.3	67.9	5.02
Census Tree	9.51	75.6	69.0	63.1	6.00
ABE exact conservative	9.51	23.8	21.8	19.9	1.90
ABE exact standard	9.51	32.0	29.1	26.6	2.53

Panel B: Final sample as a percent of eligible

Procedure	1880	1900	1920	1940
MLP	49.2	53.2	62.1	67.9
Census Tree	44.6	56.1	57.2	63.1
ABE exact conservative	10.3	13.8	17.8	19.9
ABE exact standard	16.1	21.0	24.5	26.6

Note: MLP retention rates use the eligible universe in the IPUMS MLP files; Census Tree and ABE rates use the full-count eligible adult universe. "Linked" denotes a link to the relevant childhood census, "Father identified" additionally requires an identified father in the childhood household, and the final sample additionally requires a valid paternal industry.

A.1.3. Persistence Across Schooling, Geography, and Sectors

Figure 9 reports additional schooling and geographic sensitivity specifications complementary to Panel B of Figure 1. The schooling adjustment leaves the parent-sector gap nearly unchanged, while either childhood or adulthood commuting-zone fixed effects attenuate the gap but leave it large throughout 1880–1940.

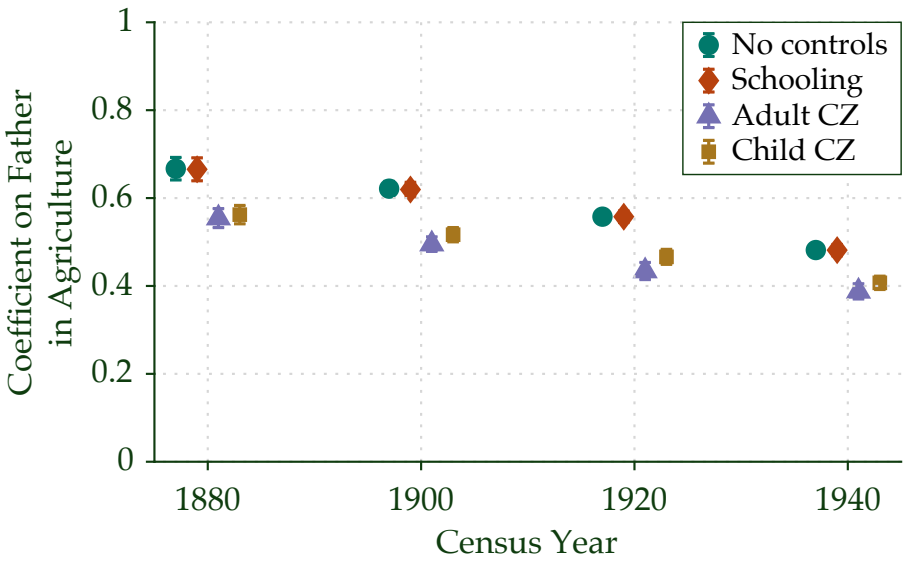


Figure 9: Schooling and Geographic Sensitivity of Agricultural Persistence

Note: Each marker reports the coefficient on an indicator for whether a worker's father was employed in agriculture from a separate linear probability model whose dependent variable indicates whether the worker was employed in agriculture. The specifications include no additional controls, childhood-school-attendance fixed effects, adulthood commuting-zone fixed effects, or childhood commuting-zone fixed effects. The sample consists of linked U.S.-born employed men aged 20–29 with observed own and paternal industries; agriculture comprises IND1950 codes 100–199. Whiskers denote 95 percent confidence intervals from standard errors clustered by adulthood commuting zone.

Figure 10 compares persistence across sectors. Agricultural employment remains strongly associated with paternal agricultural employment whether the comparison group consists of workers with fathers in manufacturing or services, whereas persistence between manufacturing and services is substantially smaller.

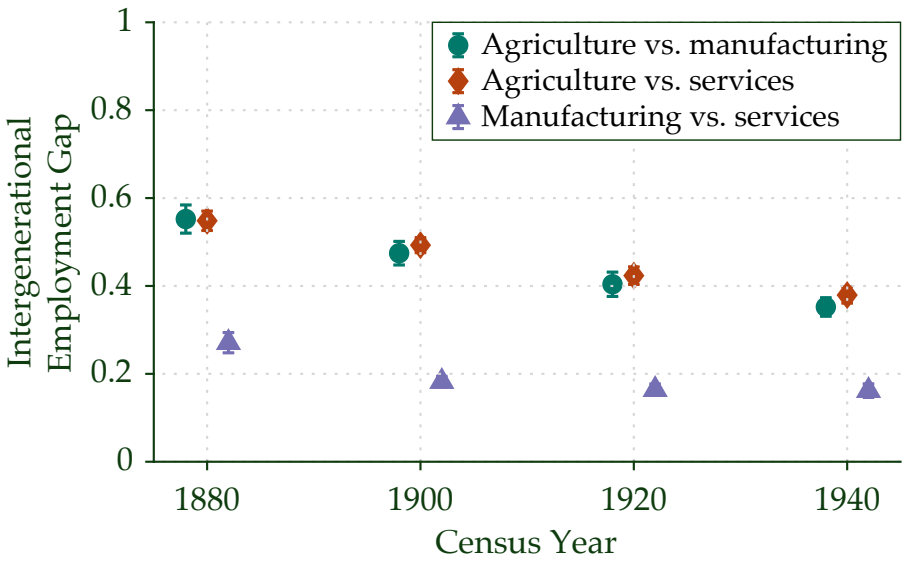


Figure 10: Intergenerational Persistence Across Sectors, 1880–1940

Note: Each marker reports a coefficient from a separate linear probability model estimated in the indicated census year. The agriculture-versus-manufacturing and agriculture-versus-services series report the coefficient on an agricultural-father indicator, restricting the comparison group as indicated; the outcome is agricultural employment. The manufacturing-versus-services series restricts the sample to workers with nonagricultural fathers and reports the coefficient on a manufacturing-father indicator; the outcome is manufacturing employment. Manufacturing comprises IND1950 codes 300–499, and services comprise the remaining nonagricultural industries. All specifications absorb childhood-school-attendance-by-commuting-zone fixed effects; whiskers denote 95 percent confidence intervals from standard errors clustered by adulthood commuting zone.

A.1.4. Detailed Earnings Regressions

Appendix Table 11 reports the estimates and regression diagnostics underlying Figure 2.

Table 11: Earnings Differences by Parental Sector, 1940

	No controls		Baseline		Location	
	Agricultural workers	Nonagricultural workers	Agricultural workers	Nonagricultural workers	Agricultural workers	Nonagricultural workers
Father in agriculture	-0.650***	-0.304***	-0.627***	-0.288***	-0.433***	-0.147***
	(0.0307)	(0.0101)	(0.0292)	(0.0089)	(0.0150)	(0.0050)
Age fixed effects	No	No	Yes	Yes	Yes	Yes
Race fixed effects	No	No	Yes	Yes	Yes	Yes
Adult × child CZ fixed effects	No	No	No	No	Yes	Yes
Observations	363,143	2,756,452	363,143	2,756,452	350,312	2,722,683
R^2	0.074	0.036	0.098	0.118	0.237	0.217

Note: The dependent variable is log hourly wage and salary income, and the reported coefficient corresponds to an indicator for the worker's father being employed in agriculture. Regressions are estimated separately within each destination sector. The middle two columns include age and race fixed effects. The final two columns additionally include fully interacted childhood-by-adulthood commuting-zone fixed effects, assigning a separate fixed effect to each observed commuting-zone pair. None of the specifications controls for childhood schooling; Appendix Table 12 reports the corresponding schooling sensitivity. The sample consists of linked U.S.-born employed men aged 20–29 with observed own and paternal industries, positive wage and salary income, positive weeks worked, and positive usual weekly hours. Standard errors clustered by adulthood commuting zone are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.1.5. Schooling Controls in the Earnings Analysis

Appendix Table 12 examines sensitivity to childhood school attendance. Both specifications include childhood-by-adult commuting-zone-pair fixed effects; the second adds school-attendance fixed effects. Appendix Figure 11 broadens the exercise using 1940 indicators for completing some high school and graduating from high school.

Table 12: Schooling Sensitivity of Earnings Differences by Parental Sector

	No schooling control		With schooling control	
	Agricultural workers	Nonagricultural workers	Agricultural workers	Nonagricultural workers
Father in agriculture	-0.431***	-0.143***	-0.433***	-0.151***
	(0.0151)	(0.0056)	(0.0150)	(0.0055)
Childhood-schooling fixed effects	No	No	Yes	Yes
Adult × child CZ fixed effects	Yes	Yes	Yes	Yes
Observations	350,312	2,722,683	350,312	2,722,683
R^2	0.231	0.156	0.231	0.162

Note: The dependent variable is log hourly wage and salary income, and the reported coefficient corresponds to an indicator for the worker's father being employed in agriculture. Regressions are estimated separately within each destination sector and include a separate fixed effect for each observed childhood-to-adulthood commuting-zone pair. The final two columns additionally include childhood-school-attendance fixed effects. All four columns use the common sample retained by the schooling-controlled specification. The sample consists of linked U.S.-born employed men aged 20–29 with observed own and paternal industries, positive wage and salary income, positive weeks worked, and positive usual weekly hours. Standard errors clustered by adulthood commuting zone are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

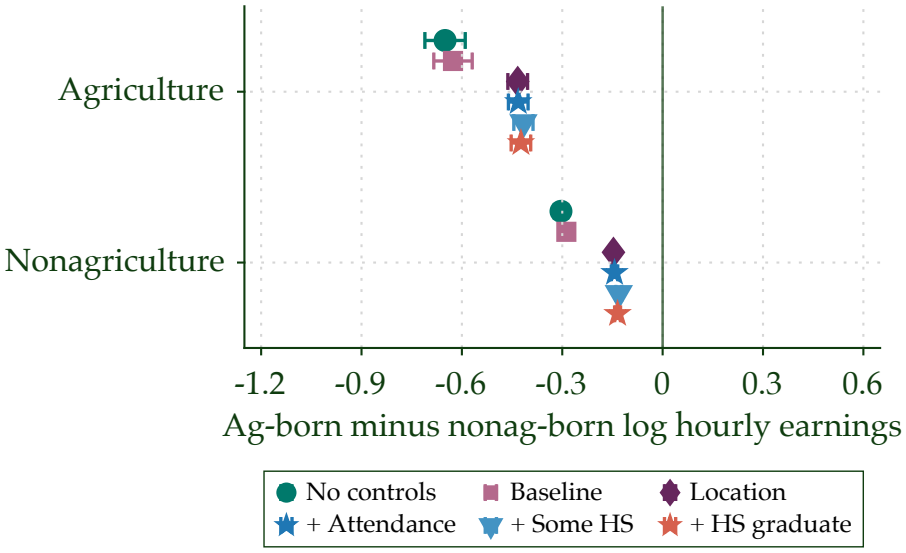


Figure 11: Schooling and Educational-Attainment Sensitivity of Earnings Differences

Note: Each marker reports the coefficient on an indicator for whether a worker's father was employed in agriculture from a separate regression of log hourly earnings, estimated within the indicated destination sector. The baseline includes age and race fixed effects; Location additionally includes adult commuting-zone fixed effects; and Attendance additionally controls for childhood school attendance. The final two specifications add indicators for completing some high school or graduating from high school to the Location specification and use a common sample with valid 1940 educational attainment. Whiskers denote 95 percent confidence intervals from standard errors clustered by adult commuting zone.

Occupation fixed effects. Appendix Table 13 assesses whether the earnings gaps remain after accounting for workers' occupations. Both specifications include fixed effects for observed pairs of childhood and adult commuting zones; the second additionally absorbs worker-occupation fixed effects. The implied earnings discounts attenuate to approximately 29 percent in agriculture and 13 percent in nonagriculture, but remain substantial and statistically significant.

Table 13: Occupation Sensitivity of Earnings Differences by Parental Sector

	No occupation FE		With occupation FE	
	Agricultural workers	Nonagricultural workers	Agricultural workers	Nonagricultural workers
Father in agriculture	-0.431***	-0.143***	-0.353***	-0.130***
	(0.0151)	(0.0056)	(0.0137)	(0.0045)
Occupation fixed effects	No	No	Yes	Yes
Adult × child CZ fixed effects	Yes	Yes	Yes	Yes
Observations	350,277	2,722,452	350,277	2,722,452
R^2	0.231	0.156	0.271	0.252

Note: The dependent variable is log hourly wage and salary income, and the reported coefficient corresponds to an indicator for the worker's father being employed in agriculture. Regressions are estimated separately within each destination sector and include a separate fixed effect for each observed childhood-to-adulthood commuting-zone pair. The final two columns additionally include worker-occupation fixed effects based on OCC1950 codes. All four columns use the common sample retained by the occupation-controlled specification. The sample otherwise follows Table 12. Standard errors clustered by adulthood commuting zone are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.2. Labor Supply Parameter Estimates

This appendix provides additional details and robustness checks for the labor-supply parameter estimates in Section 4.1. The complete conditional log-income variance derivation used to identify γ appears in Appendix C.1.1. The first exercises use the independent common-shape Fréchet model to illustrate the paper's closed-form identification logic; they are not the preferred quantitative specification. I then report a like-for-like eight-moment comparison of alternative joint skill distributions and conclude with robustness checks based on the analytical benchmark.

A.2.1. Decomposing Log-Odds Persistence

This subsection reports the decomposition implied by the independent common-shape analytical benchmark, not by the preferred seven-parameter SMM specification. Equation (ID-6) decomposes the log intergenerational odds ratio additively as

$$\log \mathcal{O} = \underbrace{\gamma \left(\log \kappa_A + \log \kappa_N\right)}_{\text{inherited skills}} + \underbrace{-2\gamma \log(1-\tau)}_{\text{inherited preferences}}.$$

I define each mechanism's contribution to persistence as its component divided by $\log \mathcal{O}$. Using $(\widehat{\gamma}, \widehat{\kappa}_A, \widehat{\kappa}_N, \widehat{\tau}) = (2.195, 1.497, 1.736, 0.178)$ and $\widehat{\mathcal{O}} = 19.270$ gives

$$\widehat{s}_\kappa = \frac{\widehat{\gamma}(\log \widehat{\kappa}_A + \log \widehat{\kappa}_N)}{\log \widehat{\mathcal{O}}} = 0.709, \qquad \widehat{s}_\tau = \frac{-2\widehat{\gamma} \log(1-\widehat{\tau})}{\log \widehat{\mathcal{O}}} = 0.291.$$

Thus inherited skills account for 70.9 percent of observed log-odds persistence and inherited preferences account for the remaining 29.1 percent.

A.2.2. Location and Schooling Controls

This exercise likewise retains the analytical benchmark and varies only the controls used to construct its closed-form moments. Table 14 carries the childhood-school-attendance control through the estimation of all four labor-supply parameters. Adding schooling barely changes the location-adjusted estimates, indicating that the attenuation relative to the baseline comes from conditioning on adult location rather than observed childhood school attendance.

Table 14: Analytical-Benchmark Estimates with Location and Schooling Controls

	Baseline	Location	Location + school attendance
Fréchet shape, $\widehat{\gamma}$	2.195	2.319	2.322
	(0.033)	(0.033)	(0.033)
Inherited agricultural skills, $\widehat{\kappa}_A$	1.497	1.306	1.301
	(0.033)	(0.020)	(0.020)
Inherited nonagricultural skills, $\widehat{\kappa}_N$	1.736	1.422	1.414
	(0.027)	(0.022)	(0.021)
Inherited preferences, $\widehat{\tau}$	0.178	0.154	0.154
	(0.012)	(0.007)	(0.007)

Note: All columns control for age and race. Location adds adult commuting-zone fixed effects. The final column also controls for school attendance. Parentheses report adult-CZ cluster-bootstrap standard errors using 499 resamples; each resample jointly re-estimates the earnings and sector-choice moments. The common inherited-preference parameter imposes $\tau_A = \tau_N \equiv \tau$.

A.2.3. Inherited Skills and Preferences among Movers

The analytical-benchmark estimates may partly reflect systematic differences in the local labor markets occupied by ag-born and nonag-born workers or advantages conferred by local family networks. To assess this concern, I repeat the location procedure among workers whose adult commuting zone differs from their childhood commuting zone. The Movers column of Table 15 uses mover-specific hourly-earnings moments and sector-choice probabilities, while retaining the location estimate of the Fréchet shape parameter. The inherited-skill and inherited-preference estimates become smaller, but remain economically meaningful among movers.

Table 15: Analytical-Benchmark Estimates among Movers

	Baseline	Location	Movers
Panel A: Estimated Intergenerational Persistence			
Inherited Skills, $\widehat{\kappa}_A$	1.497	1.306	1.176
Inherited Skills, $\widehat{\kappa}_N$	1.736	1.422	1.263
Inherited Preferences, $\widehat{\tau}$	0.178	0.154	0.077
Panel B: Mean hourly earnings			
Agriculture, ag-born: $\widehat{y}_A(A)$	0.19	0.20	0.22
Agriculture, nonag-born: $\widehat{y}_A(N)$	0.36	0.32	0.29
Nonagriculture, ag-born: $\widehat{y}_N(A)$	0.48	0.52	0.54
Nonagriculture, nonag-born: $\widehat{y}_N(N)$	0.60	0.59	0.61
Panel C: Intergenerational persistence			
Ag. employment share, ag-born	0.529	0.471	0.321
Ag. employment share, nonag-born	0.055	0.089	0.116
Sector-choice odds ratio	19.270	9.122	3.626

Note: The Baseline and Location columns reproduce the corresponding analytical-benchmark estimates in Table 14. The Movers column applies the Location procedure to workers whose adult and childhood commuting zones differ, using mover-specific earnings and sector-choice moments and the Location estimate of γ. All columns adjust the earnings means and sector-choice probabilities for age and race fixed effects; Location and Movers additionally include adult commuting-zone fixed effects. Earnings moments use workers with valid positive hourly earnings; sector-choice moments entering κ_j, τ, and Panel C use the full linked sample for the relevant column.

A.2.4. Alternative Skill Distributions

Table 16 compares independent common-shape Fréchet skills, bivariate lognormal skills with a Gaussian copula, and the preferred sector-specific Fréchet marginals joined by a Frank copula. In every specification, the bivariate object is the pair of agricultural and nonagricultural skills, (ϵ_A, ϵ_N), rather than skills indexed by parental origin. Ag-born and nonag-born workers draw from the same joint distribution; parental origin affects sectoral payoffs only through κ_A, κ_N, and τ.

The baseline specification assumes independent unit-scale Fréchet skills with a common shape parameter:

$$F_j(\epsilon) \equiv \Pr(\epsilon_j \leq \epsilon) = \exp\left(-\epsilon^{-\gamma}\right), \qquad \epsilon > 0, \quad j \in \{A, N\}, \qquad \epsilon_A \perp \epsilon_N. \tag{12}$$

The bivariate lognormal specification instead assumes

$$\begin{pmatrix} \log \epsilon_A \\ \log \epsilon_N \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_A^2 & \rho\sigma_A\sigma_N \\ \rho\sigma_A\sigma_N & \sigma_N^2 \end{pmatrix} \right]. \tag{13}$$

Finally, the Fréchet–Frank specification allows sector-specific Fréchet marginals,

$$F_j(\epsilon) = \exp\left(-\epsilon^{-\gamma_j}\right), \qquad j \in \{A, N\}, \tag{14}$$

and joins them using the Frank copula

$$\begin{aligned} \Pr(\epsilon_A \leq a, \epsilon_N \leq n) &= C_{\theta_F}(F_A(a), F_N(n)), \\ C_\theta(u, v) &= -\frac{1}{\theta} \log\left[1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right]. \end{aligned} \tag{15}$$

for $\theta \neq 0$, with $C_0(u, v) = uv$ denoting independence. All marginal location or scale parameters are normalized because the relative sector payoff R absorbs their relative level.

I estimate every distribution against exactly the same eight targets: two parental-group agricultural choice probabilities, two within-sector ratios of arithmetic mean hourly earnings, and four destination-by-parental-sector variances of log hourly earnings. The baseline moments adjust for age and race; the second panel additionally adjusts every target family for childhood school attendance and adult commuting-zone fixed effects. All models use the same transformed errors and weights within a panel. The independent Fréchet model has five parameters, while each dependent model has seven, so all three are overidentified by the eight moments.

Table 16: SMM Estimates under Alternative Skill Distributions

	Independent Fréchet	Bivariate lognormal	Fréchet–Frank
Panel A: Baseline (age and race)			
Common shape, γ	1.827	–	–
Agricultural shape, γ_A	–	–	1.401
Nonagricultural shape, γ_N	–	–	2.385
Agricultural log-skill SD, σ_A	–	1.363	–
Nonagricultural log-skill SD, σ_N	–	0.584	–
Dependence, ρ or θ_F	0	0.423	3.574
Inherited agricultural skills, κ_A	1.831	2.014	2.776
Inherited nonagricultural skills, κ_N	1.848	1.278	1.422
Inherited preferences, τ	0.178	0.421	0.085
Relative payoff, R	2.092	3.232	3.113
Eight-moment objective	0.20974	0.05410	0.00763
Panel B: Schooling and location			
Common shape, γ	1.946	–	–
Agricultural shape, γ_A	–	–	1.488
Nonagricultural shape, γ_N	–	–	2.535
Agricultural log-skill SD, σ_A	–	1.332	–
Nonagricultural log-skill SD, σ_N	–	0.537	–
Dependence, ρ or θ_F	0	0.411	3.379
Inherited agricultural skills, κ_A	1.496	1.699	1.992
Inherited nonagricultural skills, κ_N	1.480	1.112	1.220
Inherited preferences, τ	0.154	0.354	0.088
Relative payoff, R	1.884	2.929	2.458
Eight-moment objective	0.20765	0.03236	0.00655

Note: All specifications minimize the same weighted quadratic form in the eight transformed moment errors using fixed common random numbers. Choice errors use log odds, earnings ratios and variances use logs, and the weights are $1/2$ for each choice and mean-ratio moment and $1/4$ for each cell variance. The Fréchet column imposes a common shape parameter and independence across sectoral skills. The bivariate lognormal allows sector-specific log-skill dispersions and correlation ρ; the Fréchet–Frank specification allows sector-specific shape parameters and dependence governed by the Frank copula parameter θ_F. The relative sector payoff R is a nuisance parameter. Reported production results use 1,000,000 common draws, global screening, multiple local refinements, and separately seeded reevaluation of the retained estimates. Bounds impose $\kappa_A, \kappa_N \geq 1$, $\tau \in [0, 1)$, and finite mean skills.

Table 17 reports every empirical target and fitted value, rather than comparing objective values alone. The objective values are not comparable across the two panels as tests of the control sets because their empirical targets differ.

Table 17: Data and Fitted Values for All Eight SMM Targets

Moment	Data	Independent Fréchet	Bivariate lognormal	Fréchet–Frank
Panel A: Baseline (age and race)				
$\Pr(A \mid A\text{-born})$	0.5292	0.5286	0.5238	0.5313
$\Pr(A \mid N\text{-born})$	0.0551	0.0558	0.0560	0.0550
Mean ratio, agriculture	0.5344	0.5345	0.5361	0.5347
Mean ratio, nonagriculture	1.2643	1.2634	1.2748	1.2647
Variance, agriculture/A-born	0.7671	0.4920	0.9105	0.7729
Variance, agriculture/N-born	0.7488	0.4919	0.5568	0.7393
Variance, nonagriculture/A-born	0.3959	0.4923	0.3412	0.3511
Variance, nonagriculture/N-born	0.2593	0.4947	0.3412	0.2939
Panel B: Schooling and location				
$\Pr(A \mid A\text{-born})$	0.4693	0.4691	0.4645	0.4710
$\Pr(A \mid N\text{-born})$	0.0898	0.0894	0.0916	0.0896
Mean ratio, agriculture	0.6381	0.6382	0.6376	0.6380
Mean ratio, nonagriculture	1.1213	1.1214	1.1151	1.1218
Variance, agriculture/A-born	0.6908	0.4334	0.7985	0.6819
Variance, agriculture/N-born	0.6516	0.4338	0.5333	0.6609
Variance, nonagriculture/A-born	0.3367	0.4338	0.2887	0.3008
Variance, nonagriculture/N-born	0.2346	0.4358	0.2886	0.2629

Note: Mean ratios are ratios of adjusted arithmetic mean hourly earnings. Variances use adjusted log hourly earnings in the full-time/full-year sample. The four variances are targeted moments, not out-of-sample validation checks.

I next compare features of the adjusted earnings distributions that are not targeted directly. These include the pooled residual variance, cell-specific interquartile ranges, cell-specific P90–P50 spreads, and root-mean-squared deviations across the 1st through 99th percentiles of each centered cell distribution.

Table 18: Targeted and Untargeted Earnings-Distribution Fit

Diagnostic	Independent Fréchet	Bivariate lognormal	Fréchet–Frank
Panel A: Baseline (age and race)			
Pooled variance (level)	0.4939	0.4311	0.3883
Cell-variance MAE (targeted)	0.2159	0.1180	0.0237
Cell-IQR MAE	0.1892	0.2160	0.1245
Cell P90–P50 MAE	0.3203	0.2545	0.3636
Centered-quantile RMSE	0.2986	0.1965	0.2823
Panel B: Schooling and location			
Pooled variance (level)	0.4351	0.3693	0.3480
Cell-variance MAE (targeted)	0.1934	0.0820	0.0206
Cell-IQR MAE	0.1851	0.1805	0.0916
Cell P90–P50 MAE	0.3181	0.2629	0.3591
Centered-quantile RMSE	0.2801	0.1875	0.2642

Note: The cell-variance mean absolute error summarizes a targeted feature; all other fit errors are untargeted. Centered-quantile RMSE pools the four destination-by-parental-sector cells and percentiles 1 through 99. Empirical diagnostics use residuals from the same control specification and sample used to construct the four target variances.

The fit comparison separates targeted dispersion from broader distributional shape. Fréchet–Frank has the smallest cell-variance error and also fits the cell interquartile ranges best in both control specifications. The bivariate lognormal, however, has the smallest P90–P50 error and centered-quantile RMSE in both panels. I therefore use Fréchet–Frank for the headline exercise because it best fits the eight moments chosen to discipline choice, selection, and cell-specific dispersion, while retaining the lognormal results as a central warning that this targeted fit does not imply superior fit to the entire earnings distribution.

A.2.5. End-to-End Robustness across Skill Distributions

The distributional comparison also changes the mapping from relative payoffs to sector choices and effective labor throughout the macroeconomic exercise. Table 19 therefore carries each baseline estimate through its own historical inversion, 1880 regime reanchoring, forward counterfactual equilibria, and interval-level Shapley decomposition.

Table 19: Immediate and Dynamic Effects under Alternative Skill Distributions

Counterfactual	Immediate effect	Dynamic effect	Cumulative effect
	Percentage points		
Panel A: Independent Fréchet			
No inherited skills	−2.68	−4.33	−7.01
No inherited preferences	−3.31	2.64	−0.67
No intergenerational persistence	−3.73	−2.69	−6.42
Panel B: Bivariate lognormal			
No inherited skills	0.15	−5.27	−5.12
No inherited preferences	−5.23	4.81	−0.42
No intergenerational persistence	−3.16	−1.35	−4.51
Panel C: Fréchet–Frank (preferred)			
No inherited skills	−0.66	−8.65	−9.32
No inherited preferences	−1.35	1.21	−0.14
No intergenerational persistence	−1.10	−7.77	−8.87

Note: Each panel uses the age-and-race adjusted eight-moment estimates for the indicated joint skill distribution throughout the inversion and counterfactual solutions. Every counterfactual is separately anchored to the observed 1880 targets and subsequently faces the full-persistence growth in fundamentals inferred within that panel. Negative effects indicate additional reallocation out of agriculture relative to the panel's full-persistence benchmark.

Table 20: Relative-Technology Growth under Alternative Skill Distributions

Counterfactual	Benchmark growth	Counterfactual growth	Difference from benchmark
	1880–1940 log points		
Panel A: Independent Fréchet			
No inherited skills	55.53	92.02	36.49
No inherited preferences	55.53	58.23	2.71
No intergenerational persistence	55.53	93.10	37.57
Panel B: Bivariate lognormal			
No inherited skills	42.40	65.39	22.98
No inherited preferences	42.40	45.33	2.92
No intergenerational persistence	42.40	69.51	27.10
Panel C: Fréchet–Frank (preferred)			
No inherited skills	53.31	88.41	35.09
No inherited preferences	53.31	53.73	0.41
No intergenerational persistence	53.31	88.58	35.27

Note: Entries are cumulative 1880–1940 log changes in Z_N/Z_A. Each row re-inverts the same observed employment, relative-price, and real-output paths using the panel's estimated joint skill distribution.

The cumulative no-persistence effect remains negative under every distribution, but its magnitude ranges from 4.51 percentage points under bivariate lognormal to 6.42 under independent Fréchet and 8.87 under preferred Fréchet–Frank. Across all three models, removing inherited skills produces a much larger cumulative effect than removing inherited preferences because the preference channel's negative immediate contribution is substantially offset by positive intergenerational propagation. The inferred additional relative-technology growth required when persistence is omitted is likewise positive in every panel, ranging from 27.10 to 37.57 log points.

A.2.6. Allocation-Based Elasticity Illustration

As a complementary illustration of the model's distinction between aggregate and within-group labor-supply elasticities, I estimate the response of aggregate sectoral allocation to relative earnings using 1940–1950 commuting-zone-by-industry variation. For broad nonagricultural industries j, I estimate

$$\Delta \log \left(\frac{\phi_{j,r}}{\phi_{A,r}} \right) = \beta \Delta \log \left(\frac{y_{j,r}}{y_{A,r}} \right) + \alpha_{j,d(r)} + u_{j,r},$$

where r indexes commuting zones and $\alpha_{j,d(r)}$ denotes industry-by-Census-division fixed effects. The sample stacks seven broad nonagricultural industries, while agriculture is the common reference sector. I instrument relative earnings growth with a leave-one-commuting-zone-out shift-share measure,

$$Z_{j,r} = s_{j,r,1940} g_{j,-r} - s_{A,r,1940} g_{A,-r},$$

where $s_{j,r,1940}$ is industry j's initial CZ employment share and $g_{j,-r}$ is national 1940–1950 employment growth excluding CZ r. The instrument therefore varies across both commuting zones and industries rather than assigning every industry within a CZ the same shock.

Table 21 reports the resulting aggregate elasticities and applies the aggregation relationship in Proposition 1 using the observed 1940 and 1950 parental composition and subgroup agricultural shares. The population-weighted estimate is $\widehat{\beta} = 1.067$. The corresponding implied within-group elasticity is 1.533 using the 1940 shares and 1.422 using the 1950 shares, approximately one-third above the unadjusted aggregate elasticity. The unweighted estimates imply a smaller within-group elasticity near one. Because the regression stacks comparisons of each nonagricultural industry with agriculture, whereas the proposition is exact for the aggregate two-sector margin, I interpret this exercise as an illustration of the attenuation mechanism rather than as an alternative preferred estimator of γ.

Table 21: Aggregate Reallocation Elasticity Estimates

	(1) Unweighted	(2) Population weighted
Aggregate elasticity $\widehat{\beta}$	0.716	1.067
	(0.284)	(0.334)
First-stage F-statistic	24.21	30.17
Industry–CZ observations	2,423	2,423
CZ clusters	536	536
Industry $\times$ division FE	Yes	Yes

Note: The dependent variable is the 1940–1950 change in log employment share in nonagricultural industry j relative to agriculture. Relative mean hourly earnings growth is instrumented with the leave-one-CZ-out shift-share measure defined above. Employment shares and national employment growth use full-count records, while 1950 earnings moments apply the Census sample-line weights. Both columns include industry-by-Census-division fixed effects; column 2 weights observations by the CZ population of U.S.-born men ages 20–29 in 1940. Standard errors, clustered by commuting zone, are reported in parentheses.

A.2.7. Adjustment for Topcoded Earnings

This section returns to the independent common-shape analytical benchmark and checks the variance-based estimator's sensitivity to replacing each topcoded annual income with a multiple of the \$5,000 censoring threshold. In the final variance-estimation sample, 2,164 of 2,077,253 workers (0.104 percent) are topcoded, and their capped observations account for 0.501 percent of total wage and salary income. I calibrate the multiplier to 2.926 so that these workers instead account for 1.452 percent of total income, the 1940 income share of the top 0.1 percent reported by Atkinson, Piketty and Saez (2011). I then reconstruct hourly earnings and reestimate γ from the residual variance on the fixed main-text sample.

Table 22 shows that the estimate changes little over a broad range of multipliers. At the calibrated value, $\widehat{\gamma}$ declines from 2.195 to 2.180, while the quantile RMSE rises only from 0.322 to 0.325.

Table 22: Sensitivity to the Topcode Adjustment

Topcode multiplier	Topcoded income share (%)	$\widehat{\gamma}$	Observations	Quantile RMSE
1.00	0.501	2.195	2,077,253	0.322
1.25	0.625	2.193	2,077,253	0.323
1.50	0.749	2.190	2,077,253	0.323
1.75	0.873	2.188	2,077,253	0.324
2.00	0.997	2.186	2,077,253	0.324
2.93	1.452	2.180	2,077,253	0.325

Note: Each row replaces topcoded annual income with the stated multiple of \$5,000 before constructing hourly earnings and reestimating γ. The 2.93 multiplier matches the 1.452 percent target share; quantile RMSE is measured over the 1st–99th residual percentiles.

Table 23 applies the calibrated topcode adjustment to the estimates of inherited skills and preferences. The analytical-benchmark estimates are nearly unchanged, so the adjustment does not explain the difference between those estimates and the preferred SMM results.

Table 23: Inherited Skills and Preferences: Topcode Robustness

	Baseline	Location	Topcode Adjusted
Inherited Skills, $\widehat{\kappa}_A$	1.497	1.306	1.508
	(0.033)	(0.020)	(0.033)
Inherited Skills, $\widehat{\kappa}_N$	1.736	1.422	1.740
	(0.027)	(0.022)	(0.027)
Inherited Preferences, $\widehat{\tau}$	0.178	0.154	0.178
	(0.012)	(0.007)	(0.012)

Note: The first two columns reproduce the Baseline and Location analytical-benchmark estimates in Table 14. The Topcode Adjusted column replaces topcoded income with 2.926 times the censoring threshold when estimating γ, while holding the baseline earnings and sector-choice moments fixed. Commuting-zone cluster-bootstrap standard errors based on 499 resamples are reported in parentheses.

A.2.8. Visualization of Empirical vs Model Distribution

For a visual check, Figure 12 compares the residual earnings distribution with the log-Fréchet densities implied by the baseline and topcode-adjusted estimates.

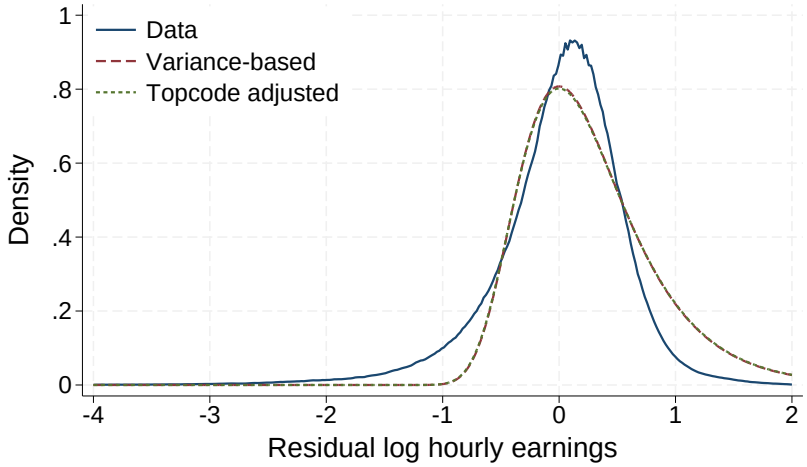


Figure 12: Residual Earnings Distribution in Data and Model

Notes: The solid line plots the kernel density of residual log hourly wage and salary income after removing worker-sector-by-parental-sector, age, and race fixed effects. The dashed red and short-dashed green lines use the baseline and topcode-adjusted estimates, $\widehat{\gamma} = 2.195$ and 2.180, respectively. The estimation sample has positive wage and salary income, at least 40 weeks worked, and at least 35 usual weekly hours, with no log-earnings restriction; the figure displays residuals between -4 and 2.

A.2.9. Alternative Earnings Measures

Hourly earnings are the closest empirical counterpart to the model's wage per efficiency unit, but they inherit measurement error from both weeks and hours. Annual and weekly earnings provide useful checks on whether this additional measurement affects the estimated dispersion of earnings. For comparability within the analytical benchmark, Table 24 estimates all three concepts on the common full-time/full-year sample used to construct the preferred specification's variance targets.

Table 24: Variance-Based Estimates of the Fréchet Shape Parameter by Income Measure

	Annual		Weekly		Hourly	
	Baseline	Location	Baseline	Location	Baseline	Location
$\widehat{\gamma}$	2.337	2.451	2.362	2.477	2.195	2.319
Residual SD of log wages	0.549	0.523	0.543	0.518	0.584	0.553
Observations	2,077,253	2,077,253	2,077,253	2,077,253	2,077,253	2,077,253
Topcoded observations	2,164	2,164	2,164	2,164	2,164	2,164
Topcoded share (%)	0.104	0.104	0.104	0.104	0.104	0.104
Earnings regression R^2	0.393	0.448	0.391	0.446	0.411	0.472

Note: Estimates use $\widehat{\gamma} = \pi/(\sqrt{6}\widehat{\sigma})$ after removing worker-sector-by-parental-sector, age, and race effects; Location columns additionally control for adult commuting zone. Weekly earnings divide annual wage and salary income by weeks worked, and hourly earnings additionally divide by usual weekly hours. All columns use the common unrestricted 2,077,253-observation sample described in the main text.

The estimates are similar across earnings concepts in both the baseline and location specifications. Figure 13 shows the corresponding distributional fits.

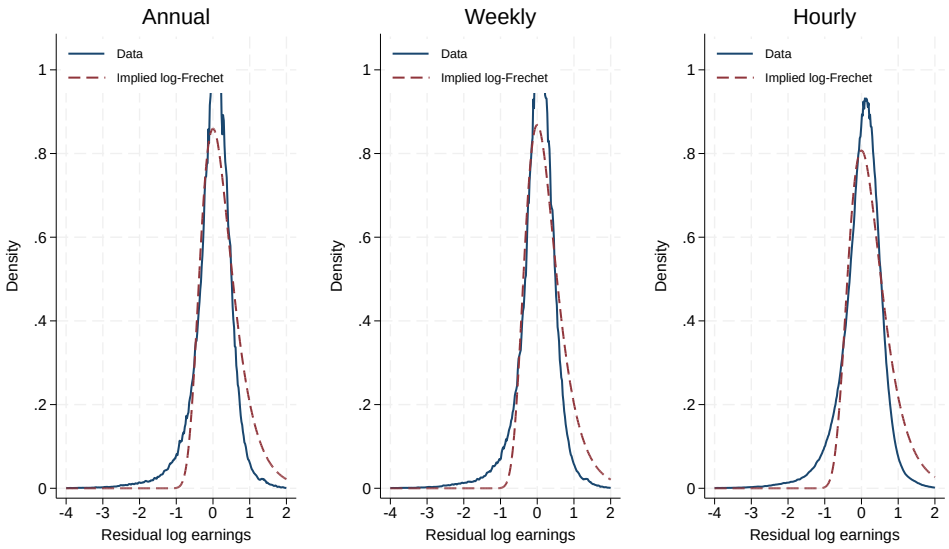


Figure 13: Residual Earnings Distributions and Implied Log-Fréchet Densities by Income Measure

Note: The solid lines plot kernel densities of residual log earnings after removing worker-sector-by-parental-sector, age, and race fixed effects. Dashed lines plot the log-Fréchet densities implied by the baseline estimates in Table 24. Panels use the common unrestricted estimation sample and display residuals between -4 and 2.

Finally, Table 25 carries each variance-based estimate of γ through the parameter calculations. The estimated inherited skills and preferences remain economically important across earnings concepts and control sets, although their magnitudes vary with the dispersion estimate and adjusted income gaps.

Table 25: Analytical-Benchmark Estimates by Income Measure

	Annual earnings		Weekly earnings		Hourly earnings	
	Baseline	Location	Baseline	Location	Baseline	Location
Panel A: Estimated Intergenerational Persistence						
Inherited Skills, $\widehat{\kappa}_A$	1.787	1.613	1.640	1.472	1.497	1.306
Inherited Skills, $\widehat{\kappa}_N$	1.754	1.467	1.694	1.416	1.736	1.422
Inherited Preferences, $\widehat{\tau}$	0.060	0.020	0.109	0.076	0.178	0.154
Panel B: Mean earnings						
Agriculture, ag-born: $\widehat{y}_A(A)$	330.46	347.96	8.72	9.19	0.19	0.20
Agriculture, nonag-born: $\widehat{y}_A(N)$	486.90	426.04	13.86	12.24	0.36	0.32
Nonagriculture, ag-born: $\widehat{y}_N(A)$	702.00	762.78	17.68	19.23	0.48	0.52
Nonagriculture, nonag-born: $\widehat{y}_N(N)$	913.76	896.53	22.30	21.86	0.60	0.59
Panel C: Intergenerational persistence						
Sector-choice odds ratio	19.270	9.122	19.270	9.122	19.270	9.122

Note: Each column uses the corresponding estimate of γ in Table 24. All columns adjust income and sector-choice moments for age and race; Location columns additionally adjust for adult commuting zone. Earnings moments use the relevant income sample; all sector-choice moments entering κ_j, τ, and Panel C use the full linked choice sample.

B. Appendix: Theoretical Results

B.1. Proofs and Supporting Derivations for the Model

This appendix provides the proofs of Propositions 1, 2, 3, and 4. The aggregate-elasticity result in Proposition 1 holds for general differentiable two-sector choice probabilities, while the subsequent comparative-static propositions use the independent common-shape Fréchet benchmark. The preferred quantitative specification evaluates its more flexible labor-supply block numerically. Throughout the general equilibrium derivations, I suppress time subscripts. All derivatives are evaluated at the common stationary employment share, $\bar{\Phi}_A$, before the technological improvement.

B.1.1. Aggregate Labor Supply Elasticity

Proof of Proposition 1. Let $x_t \equiv \log(w_{N,t}/w_{A,t})$. For any differentiable two-sector choice rule, $\phi_{N,t}(k) = 1 - \phi_{A,t}(k)$ implies

$$\frac{\partial \phi_{A,t}(k)}{\partial x_t} \;=\; -\phi_{A,t}(k)\phi_{N,t}(k)\mathcal{E}_t(k).$$

Holding parental composition fixed and aggregating across groups therefore gives

$$\frac{\partial \Phi_{A,t}}{\partial x_t} \;=\; -\sum_{k\in\{A,N\}} \Phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\mathcal{E}_t(k).$$

Since $\Phi_{N,t} = 1 - \Phi_{A,t}$, the aggregate relative sectoral labor supply elasticity satisfies

$$\mathcal{E}_t \;=\; \frac{\displaystyle\sum_{k\in\{A,N\}} \Phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\mathcal{E}_t(k)}{\Phi_{A,t}\Phi_{N,t}}.$$

Define $S_t \equiv \sum_{k\in\{A,N\}} \Phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)$. Under the conditions in the proposition, $S_t > 0$, and hence

$$\mathcal{E}_t \;=\; \overline{\mathcal{E}}_t \frac{S_t}{\Phi_{A,t}\Phi_{N,t}}.$$

The law of total variance for $D_{i,t} = \mathbb{1}(s_{i,t} = A)$ gives

$$\mathrm{Var}(D_{i,t}) \;=\; \mathbb{E}[\mathrm{Var}(D_{i,t} \mid k)] + \mathrm{Var}(\mathbb{E}[D_{i,t} \mid k]),$$

where

$$\mathrm{Var}(D_{i,t}) = \Phi_{A,t}\Phi_{N,t}, \qquad \mathbb{E}[\mathrm{Var}(D_{i,t} \mid k)] = S_t.$$

Consequently,

$$\mathcal{E}_t \;=\; \overline{\mathcal{E}}_t \left[1 - \frac{\mathrm{Var}(\mathbb{E}[D_{i,t} \mid k])}{\mathrm{Var}(D_{i,t})}\right].$$

The between-group variance is nonnegative, implying $0 < \mathcal{E}_t \leq \overline{\mathcal{E}}_t$. With both parental groups present, equality holds if and only if $\phi_{A,t}(A) = \phi_{A,t}(N)$. For two parental groups,

$$\mathrm{Var}(\mathbb{E}[D_{i,t} \mid k]) \;=\; \Phi_{A,t-1}(1 - \Phi_{A,t-1})[\phi_{A,t}(A) - \phi_{A,t}(N)]^2.$$

Under stationarity, equation (3) implies

$$\mathcal{E}_t \;=\; \overline{\mathcal{E}}_t \left[1 - [\phi_{A,t}(A) - \phi_{A,t}(N)]^2\right].$$

Finally, independent common-shape Fréchet skills imply $\partial\phi_{A,t}(k)/\partial x_t = -\gamma\phi_{A,t}(k)\phi_{N,t}(k)$, so $\mathcal{E}_t(k) = \overline{\mathcal{E}}_t = \gamma$ and the original common-shape identity is a special case. The same aggregation identity holds with respect to the relative common nonpecuniary benefit because workers' sector choices depend on the product $(B_{N,t}/B_{A,t})(w_{N,t}/w_{A,t})$. □

B.1.2. Immediate General Equilibrium Response

Let $\bar{\Phi}_N = 1 - \bar{\Phi}_A$ denote the common stationary employment share in nonagriculture. To simplify the comparison across economies, let $g \equiv \phi_A(A) - \phi_A(N)$ denote the gap between the agricultural employment shares of ag-born and nonag-born workers. Stationarity and equation (3) imply

$$\phi_A(A) = \bar{\Phi}_A + \bar{\Phi}_N g, \qquad \phi_A(N) = \bar{\Phi}_A(1-g), \tag{16}$$

with $\phi_N(k) = 1 - \phi_A(k)$ for both parental groups. The no-IGP economy has $g = 0$, while the two IGP economies have the same positive g by Assumption 1.

The aggregate effective labor supplies satisfy

$$\begin{aligned} H_A &= \bar{\gamma}\left[\bar{\Phi}_A\kappa_A\phi_A(A)^{1-1/\gamma} + \bar{\Phi}_N\phi_A(N)^{1-1/\gamma}\right], \\ H_N &= \bar{\gamma}\left[\bar{\Phi}_A\phi_N(A)^{1-1/\gamma} + \bar{\Phi}_N\kappa_N\phi_N(N)^{1-1/\gamma}\right]. \end{aligned} \tag{17}$$

Holding parental composition fixed, the group-specific sector-choice equations imply

$$\frac{\partial\Phi_A}{\partial\log(w_N/w_A)} \;=\; -\gamma\bar{\Phi}_A\bar{\Phi}_N(1-g^2). \tag{18}$$

Differentiating aggregate effective labor supply with respect to the relative wage yields

$$\begin{aligned} &-\frac{1}{\gamma-1}\frac{\partial\log(H_A/H_N)}{\partial\log(w_N/w_A)} \\ &\qquad = 1 - g\left[\frac{\bar{\Phi}_A\phi_A(A)h_A(A)}{H_A} - \frac{\bar{\Phi}_A\phi_N(A)h_N(A)}{H_N}\right]. \end{aligned} \tag{19}$$

The two fractions inside the brackets are the shares of effective agricultural and nonagricultural labor supplied by ag-born workers, respectively. Thus, even when the two IGP economies have identical worker allocations, their wage feedbacks can differ because inherited skills change these effective-labor shares.

Under homothetic demand, goods-market clearing, sectoral production, and competitive wages imply

$$d\log\left(\frac{w_N}{w_A}\right) = \frac{1-\sigma}{\sigma}\, d\log Z_A + \frac{1}{\sigma}\, d\log\left(\frac{H_A}{H_N}\right), \tag{20}$$

where all fundamentals other than Z_A are fixed. Combining this condition with (18) gives the immediate employment response:

$$-\frac{d\Phi_A}{d\log Z_A} = \frac{\gamma\bar{\Phi}_A\bar{\Phi}_N(1-g^2)(1-\sigma)}{\sigma - \dfrac{\partial\log(H_A/H_N)}{\partial\log(w_N/w_A)}} > 0. \tag{21}$$

The denominator is positive because an increase in the relative wage of nonagriculture reduces relative agricultural effective labor supply.

Proof of Proposition 2. In the no-IGP economy, $g = 0$ and the derivative in (19) equals $-(\gamma - 1)$. Consequently,

$$-\left.\frac{d\Phi_A}{d\log Z_A}\right|_{\text{no-IGP}} = \frac{\gamma\bar{\Phi}_A\bar{\Phi}_N(1-\sigma)}{\sigma+\gamma-1}.$$

Consider next the two economies with intergenerational persistence. They have the same g and therefore the same numerator in (21). Relative to the τ-IGP economy, inherited skills increase the fraction of effective agricultural labor supplied by ag-born workers and decrease their fraction of effective nonagricultural labor. Equation (19) therefore implies

$$-\left.\frac{\partial\log(H_A/H_N)}{\partial\log(w_N/w_A)}\right|_{\kappa\text{-IGP}} < -\left.\frac{\partial\log(H_A/H_N)}{\partial\log(w_N/w_A)}\right|_{\tau\text{-IGP}}.$$

Thus, the τ-IGP economy has the stronger wage feedback and the smaller immediate reallocation response.

It remains to show that the κ-IGP economy reallocates less than the no-IGP economy. In the κ-IGP economy, symmetric inherited skills and the matched sector-choice gap imply

$$\kappa^{2\gamma} = \frac{\phi_A(A)\phi_N(N)}{\phi_A(N)\phi_N(A)}, \qquad \kappa \equiv \kappa_A = \kappa_N. \tag{22}$$

The following short calculation establishes the required bound on effective-labor shares. For this calculation alone, define the common odds multiplier

$$\rho \equiv \left[\frac{\phi_A(N)\phi_N(N)}{\phi_A(A)\phi_N(A)}\right]^{1/(2\gamma)}.$$

Using stationarity and (22), the two effective-labor shares in (19) can be written as

$$\frac{\bar{\Phi}_A\phi_A(A)h_A(A)}{H_A} = \frac{\rho\phi_A(A)}{1+(\rho-1)\phi_A(A)}, \qquad \frac{\bar{\Phi}_A\phi_N(A)h_N(A)}{H_N} = \frac{\rho\phi_A(N)}{1+(\rho-1)\phi_A(N)}.$$

Their difference is therefore

$$\frac{\rho g}{[1+(\rho-1)\phi_A(A)][1+(\rho-1)\phi_A(N)]}.$$

The stationary relations imply

$$\phi_A(A)+\phi_A(N)-1 = (2\bar{\Phi}_A-1)(1-g),$$
$$\phi_A(N)\phi_N(N)-\phi_A(A)\phi_N(A) = (2\bar{\Phi}_A-1)g(1-g).$$

Thus, $\rho-1$ and $\phi_A(A)+\phi_A(N)-1$ have the same sign. It follows that

$$\begin{aligned}&[1+(\rho-1)\phi_A(A)][1+(\rho-1)\phi_A(N)]-\rho\\ &\quad=(\rho-1)[\phi_A(A)+\phi_A(N)-1+(\rho-1)\phi_A(A)\phi_A(N)]\geq 0.\end{aligned}$$

Consequently, the difference between the two effective-labor shares is no greater than g. By (19), this implies

$$-\left.\frac{\partial\log(H_A/H_N)}{\partial\log(w_N/w_A)}\right|_{\kappa\text{-IGP}} \geq (\gamma-1)(1-g^2).$$

Since $g>0$ and $\sigma>0$,

$$\sigma+(\gamma-1)(1-g^2) > (1-g^2)(\sigma+\gamma-1).$$

Substitution into (21) establishes that the no-IGP economy reallocates strictly more workers than the κ-IGP economy. Combining the two comparisons establishes Proposition 2. □

B.1.3. Dynamic Transmission Across Generations

To examine intergenerational transmission, differentiate the next generation's employment law of motion with respect to the previous generation's agricultural employment share. Holding the next generation's wages fixed gives

$$\left.\frac{\partial \Phi_{A,t+1}}{\partial \Phi_{A,t}}\right|_{w_{N,t+1}/w_{A,t+1}} = \phi_{A,t+1}(A) - \phi_{A,t+1}(N) = g.$$

The two IGP economies have the same positive direct effect, whereas the no-IGP economy has no direct effect.

Changes in parental composition also change the relative effective labor supplied to agriculture, holding wages fixed:

$$\begin{aligned} &\left.\frac{\partial \log(H_A/H_N)}{\partial \Phi_{A,t-1}}\right|_{w_N/w_A} \\ &= \frac{\phi_A(A)h_A(A) - \phi_A(N)h_A(N)}{H_A} - \frac{\phi_N(A)h_N(A) - \phi_N(N)h_N(N)}{H_N}. \end{aligned} \tag{23}$$

Combining this expression with (20) and (18) gives the total dynamic response:

$$\frac{d\Phi_{A,t+1}}{d\Phi_{A,t}} = g - \frac{\gamma \bar{\Phi}_A \bar{\Phi}_N (1-g^2)}{\sigma - \dfrac{\partial \log(H_A/H_N)}{\partial \log(w_N/w_A)}} \left.\frac{\partial \log(H_A/H_N)}{\partial \Phi_{A,t-1}}\right|_{w_N/w_A}. \tag{24}$$

The first term captures the direct effect of changing parental composition, while the second captures the opposing effect of the induced relative-wage adjustment.

To establish the sign of the dynamic response in the inherited-preferences economy, I use the following concavity inequality.

Lemma 1. *Under Assumption 1, when $\kappa_A = \kappa_N = 1$ and $g > 0$,*

$$g\left[-\frac{\partial \log(H_A/H_N)}{\partial \log(w_N/w_A)}\right] > \gamma \bar{\Phi}_A \bar{\Phi}_N (1-g^2) \left.\frac{\partial \log(H_A/H_N)}{\partial \Phi_{A,t-1}}\right|_{w_N/w_A}.$$

Proof. Let $f(x) = x^{1-1/\gamma}$, which is strictly concave because $\gamma > 1$. The stationary relations

imply

$$\begin{aligned}\phi_A(A)\phi_N(N) - \phi_A(N)\phi_N(A) &= g,\\ \phi_A(A)\phi_N(A) &= \phi_A(N)\phi_N(A) + g\phi_N(A),\\ \phi_A(N)\phi_N(N) &= \phi_A(N)\phi_N(A) + g\phi_A(N).\end{aligned}$$

Thus, the final two products lie strictly between $\phi_A(N)\phi_N(A)$ and $\phi_A(A)\phi_N(N)$. For any $x < y < z$, strict concavity implies

$$\frac{f(y)-f(x)}{y-x} > \frac{f(z)-f(x)}{z-x} > \frac{f(z)-f(y)}{z-y}.$$

Applying these inequalities at the two intermediate products, using $xf'(x) = (1-1/\gamma)f(x)$, and collecting terms gives

$$\begin{aligned}&\left(1-\frac{1}{\gamma}\right) g\Big\{\bar{\Phi}_A^2 f[\phi_A(A)\phi_N(A)] + \bar{\Phi}_N^2 f[\phi_A(N)\phi_N(N)]\\ &\quad + \bar{\Phi}_A\bar{\Phi}_N\big[(1-g)f[\phi_A(A)\phi_N(N)] + (1+g)f[\phi_A(N)\phi_N(A)]\big]\Big\}\\ &\quad > \bar{\Phi}_A\bar{\Phi}_N(1-g^2)\big\{f[\phi_A(A)\phi_N(N)] - f[\phi_A(N)\phi_N(A)]\big\}.\end{aligned}$$

The expression inside braces on the left is equal to

$$\frac{H_A H_N}{\bar{\gamma}^2} - \bar{\Phi}_A\bar{\Phi}_N g\big\{f[\phi_A(A)\phi_N(N)] - f[\phi_A(N)\phi_N(A)]\big\}.$$

Rearranging therefore yields

$$\begin{aligned}&(\gamma-1)g\frac{H_A H_N}{\bar{\gamma}^2}\\ &\quad > \bar{\Phi}_A\bar{\Phi}_N(\gamma-g^2)\big\{f[\phi_A(A)\phi_N(N)] - f[\phi_A(N)\phi_N(A)]\big\}.\end{aligned}$$

Substituting the effective labor expressions into the derivatives in (19) and (23) shows that this last inequality is equivalent to the claim of the lemma. □

Proof of Proposition 3. In the no-IGP economy, $g = 0$ and changes in parental composition leave relative effective labor unchanged. Consequently, (24) gives $d\Phi_{A,t+1}/d\Phi_{A,t} = 0$.

In the τ-IGP economy, Lemma 1 implies

$$g\left[\sigma - \frac{\partial \log(H_A/H_N)}{\partial \log(w_N/w_A)}\right] > \gamma \bar{\Phi}_A \bar{\Phi}_N (1-g^2) \left.\frac{\partial \log(H_A/H_N)}{\partial \Phi_{A,t-1}}\right|_{w_N/w_A},$$

where the inequality is strict because $\sigma > 0$ and $g > 0$. Since the denominator in (24) is positive, the total dynamic response is strictly positive.

In the κ-IGP economy, the direct effect remains positive, but the wage adjustment can dominate it. The weak-persistence expansion in the next subsection establishes that the dynamic response is negative for sufficiently small $g > 0$. As g approaches one, the group-specific agricultural employment shares approach one and zero, respectively, and the symmetric inherited skill advantage diverges. In this limit, (19) implies that the wage feedback tends to zero, while (23) tends to $1/(\bar{\Phi}_A \bar{\Phi}_N)$. Equation (24) therefore implies that the dynamic response tends to one. Hence, the total dynamic effect in the κ-IGP economy can have either sign. □

B.1.4. Cumulative Effects with Weak Intergenerational Persistence

I now examine how the immediate and dynamic effects combine when intergenerational persistence is sufficiently weak. These results concern local responses to a permanent improvement in agricultural technology around the initial stationary equilibrium. For each stylized economy, let $\mathcal{R}_h$ denote the increase in the nonagricultural employment share h generations after the shock, per unit increase in $\log Z_A$. Let $\mathcal{R}_0 = -d\Phi_A/d\log Z_A > 0$ denote the immediate response and let $\lambda = d\Phi_{A,t+1}/d\Phi_{A,t}$ denote the dynamic transmission coefficient.

For a permanent technological improvement, linearization around the stationary equilibrium implies

$$\mathcal{R}_h = \mathcal{R}_0 \left(1 + \lambda + \cdots + \lambda^h\right), \qquad h = 0, 1, 2, \ldots. \tag{25}$$

Each generation experiences the same direct effect of the technological improvement, together with the employment response inherited from the previous generation. If $|\lambda| < 1$, the eventual response is

$$\mathcal{R}_\infty = \frac{\mathcal{R}_0}{1-\lambda}. \tag{26}$$

Because $\lambda = 0$ in the no-IGP economy, intergenerational persistence increases eventual

reallocation relative to this benchmark if and only if

$$\lambda \;>\; 1 - \frac{\mathcal{R}_0}{\mathcal{R}_0^{\text{no-IGP}}}. \tag{27}$$

Thus, dynamic propagation must be sufficiently strong to overcome the initial dampening of reallocation.

Proof of Proposition 4. Fix the common interior stationary agricultural employment share, and let $g > 0$ denote the parental-sector employment gap. I examine the limit as g approaches zero, holding the other model parameters fixed. At $g = 0$, the two persistence economies converge to the no-IGP economy. Stationarity implies

$$\phi_A(A) = \bar{\Phi}_A + \bar{\Phi}_N g,$$

$$\phi_A(N) = \bar{\Phi}_A(1-g),$$

$$\phi_N(A) = \bar{\Phi}_N(1-g),$$

$$\phi_N(N) = \bar{\Phi}_N + \bar{\Phi}_A g.$$

When persistence arises exclusively from inherited preferences, a first-order expansion of (23) gives

$$\left.\frac{\partial \log(H_A/H_N)}{\partial \Phi_{A,t-1}}\right|_{\tau\text{-IGP}} \;=\; \frac{\gamma-1}{\gamma}\cdot\frac{g}{\bar{\Phi}_A\bar{\Phi}_N} + O(g^2). \tag{28}$$

For symmetric inherited skills, the matched sector-choice gap implies

$$\log\kappa \;=\; \frac{1}{2\gamma}\log\left[\frac{\phi_A(A)\phi_N(N)}{\phi_A(N)\phi_N(A)}\right] \;=\; \frac{g}{2\gamma\bar{\Phi}_A\bar{\Phi}_N} + O(g^2).$$

Substituting this expression into the same effective labor derivative gives

$$\left.\frac{\partial \log(H_A/H_N)}{\partial \Phi_{A,t-1}}\right|_{\kappa\text{-IGP}} \;=\; \frac{g}{\bar{\Phi}_A\bar{\Phi}_N} + O(g^2). \tag{29}$$

In both economies, the relative effective labor response to wages satisfies

$$-\frac{\partial \log(H_A/H_N)}{\partial \log(w_N/w_A)} \;=\; (\gamma-1) + O(g^2).$$

Consequently, the immediate response in (21) satisfies

$$\frac{\mathcal{R}_0^r}{\mathcal{R}_0^{\text{no-IGP}}} \;=\; 1 + O(g^2), \qquad r \in \{\kappa\text{-IGP}, \tau\text{-IGP}\}. \tag{30}$$

Substituting the two parental-composition derivatives into (24) yields

$$\begin{aligned} \lambda^{\tau\text{-IGP}} &= \frac{\sigma}{\sigma + \gamma - 1}\, g + O(g^2), \\ \lambda^{\kappa\text{-IGP}} &= -\frac{1-\sigma}{\sigma + \gamma - 1}\, g + O(g^2). \end{aligned} \tag{31}$$

Thus, the immediate dampening from intergenerational persistence is second order in the sector-choice gap, while the dynamic transmission is first order. For any fixed $h \geq 1$, (25) therefore gives

$$\begin{aligned} \frac{\mathcal{R}_h^{\tau\text{-IGP}}}{\mathcal{R}_h^{\text{no-IGP}}} &= 1 + \frac{\sigma}{\sigma + \gamma - 1}\, g + O(g^2), \\ \frac{\mathcal{R}_h^{\kappa\text{-IGP}}}{\mathcal{R}_h^{\text{no-IGP}}} &= 1 - \frac{1-\sigma}{\sigma + \gamma - 1}\, g + O(g^2). \end{aligned} \tag{32}$$

Because $0 < \sigma < 1$ and $\gamma > 1$, the first ratio is greater than one and the second is less than one for all sufficiently small positive g. Moreover, $\lambda \to 0$ as $g \to 0$, so the stationary equilibrium is locally stable for sufficiently weak persistence. The same expansions therefore apply to the eventual response in (26). □

The result establishes analytically that inherited preferences can accelerate structural transformation, even though they dampen the immediate response to technological change. For stronger persistence, the immediate dampening can instead dominate, as illustrated by the numerical exercise in the main text.

B.1.5. Additional Numerical Illustrations

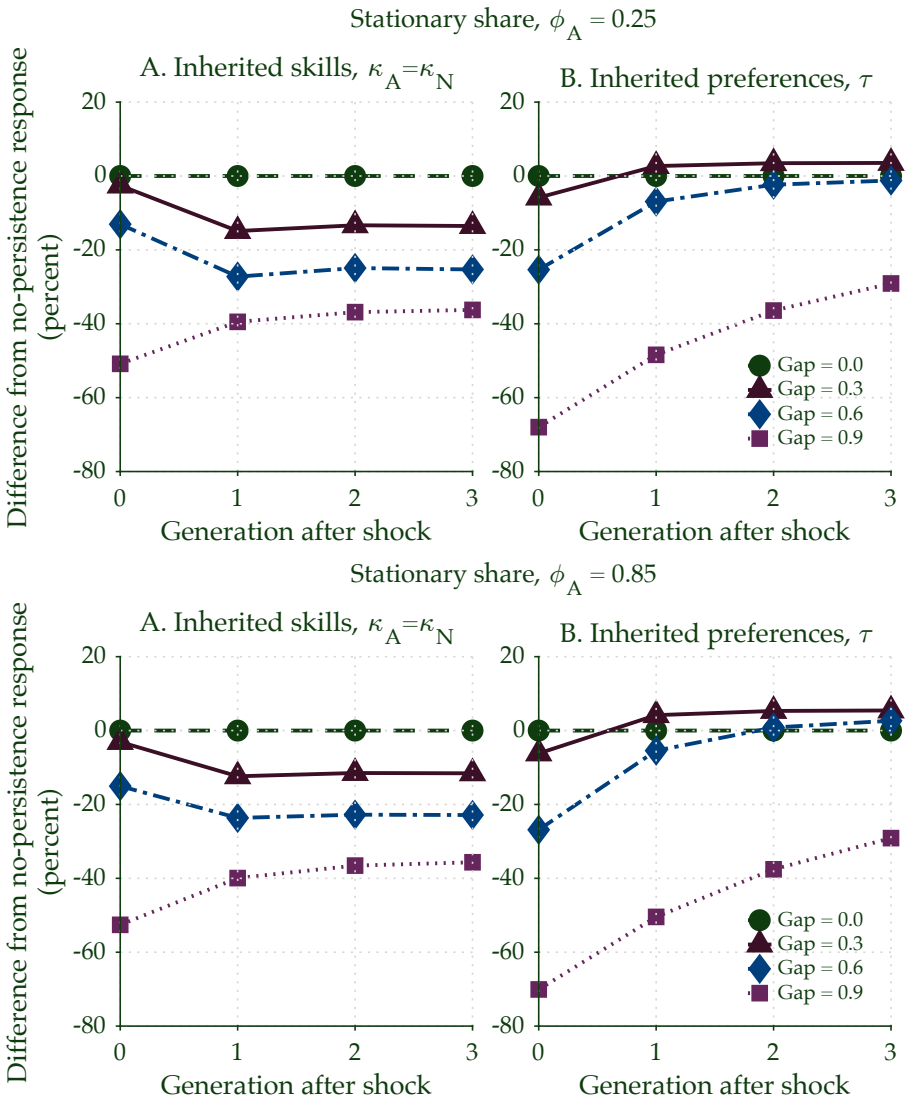


Figure 14: Numerical Responses at Alternative Stationary Agricultural Employment Shares

Note: The figure repeats the numerical exercise in Figure 3 at common stationary agricultural employment shares of $\bar{\Phi}_A = 0.25$ and 0.85, using the same permanent 0.01-log-unit agricultural-technology shock. Within each subfigure, every economy is separately re-anchored to the stated $\bar{\Phi}_A$; the legend reports the parent-sector employment gap, and Panels A and B generate that gap using symmetric inherited skills and inherited preferences, respectively. Responses equal $100 \times$ (IGP response/no-IGP response $- 1$), so positive values indicate faster reallocation than without IGP. All other parameters remain at the main-text benchmark: $\gamma = 2$, $\sigma = 0.3$, $\vartheta_A = \vartheta_N = 1$, and $\Omega_A = \Omega_N = 1$.

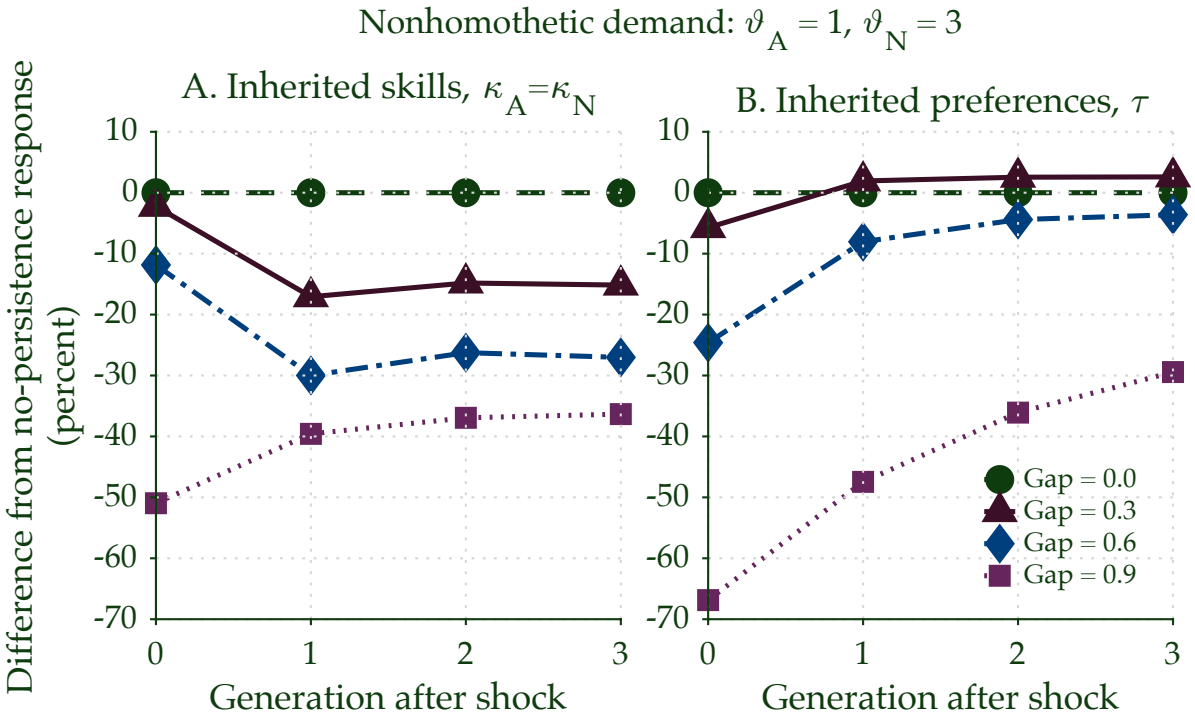


Figure 15: Numerical Responses with Nonhomothetic Demand

Note: The figure repeats the permanent 0.01-log-unit agricultural-technology shock with nonhomothetic demand, $(\vartheta_A, \vartheta_N) = (1, 3)$, rather than the homothetic benchmark in the main text. Every economy is separately re-anchored to $\bar{\Phi}_A = 0.6$; the legend reports the parent-sector employment gap, and Panels A and B generate that gap using symmetric inherited skills and inherited preferences, respectively. Responses equal $100 \times$ (IGP response/no-IGP response $- 1$). The remaining parameters are $\gamma = 2$, $\sigma = 0.3$, and $\Omega_A = \Omega_N = 1$.

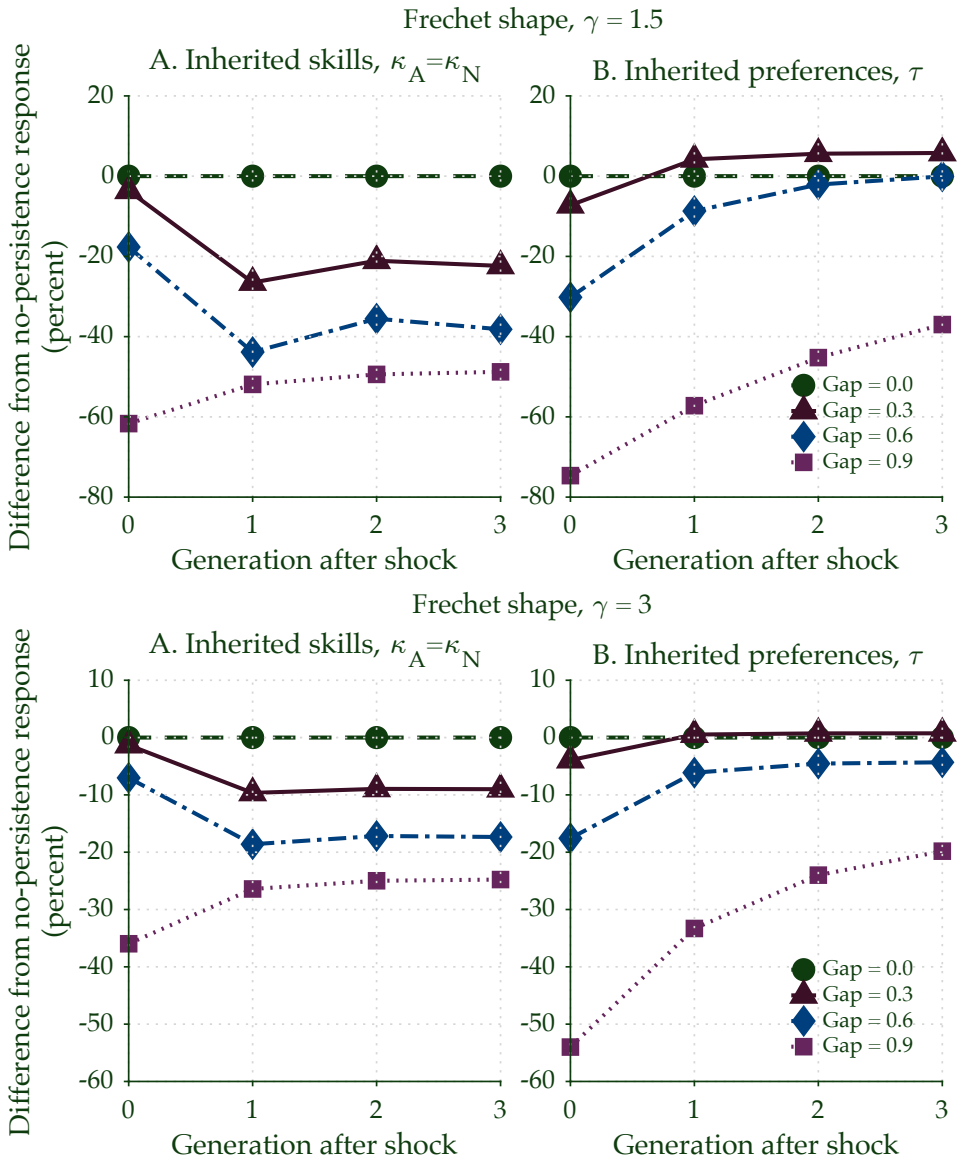


Figure 16: Numerical Responses at Alternative Fréchet Shape Parameters

Note: The figure repeats the permanent 0.01-log-unit agricultural-technology shock for $\gamma \in \{1.5, 3, 5\}$. For each γ, the inherited-skill and inherited-preference parameters are chosen separately to generate the common parent-sector employment gaps shown in the legend, and every economy is re-anchored to $\bar{\Phi}_A = 0.6$. Panels A and B vary symmetric inherited skills and inherited preferences, respectively; responses are percentage differences from the corresponding no-IGP response. All other parameters remain at the benchmark: $\sigma = 0.3$, $\vartheta_A = \vartheta_N = 1$, and $\Omega_A = \Omega_N = 1$.

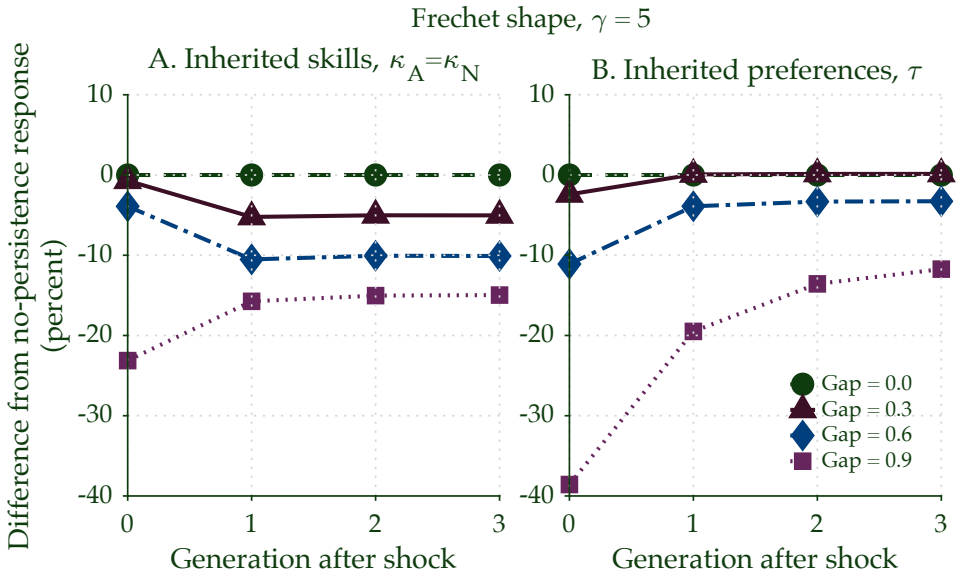


Figure 16: Numerical Responses at Alternative Fréchet Shape Parameters (continued)

Note: This panel completes Figure 16 with $\gamma = 5$; all other parameters and construction follow the preceding note.

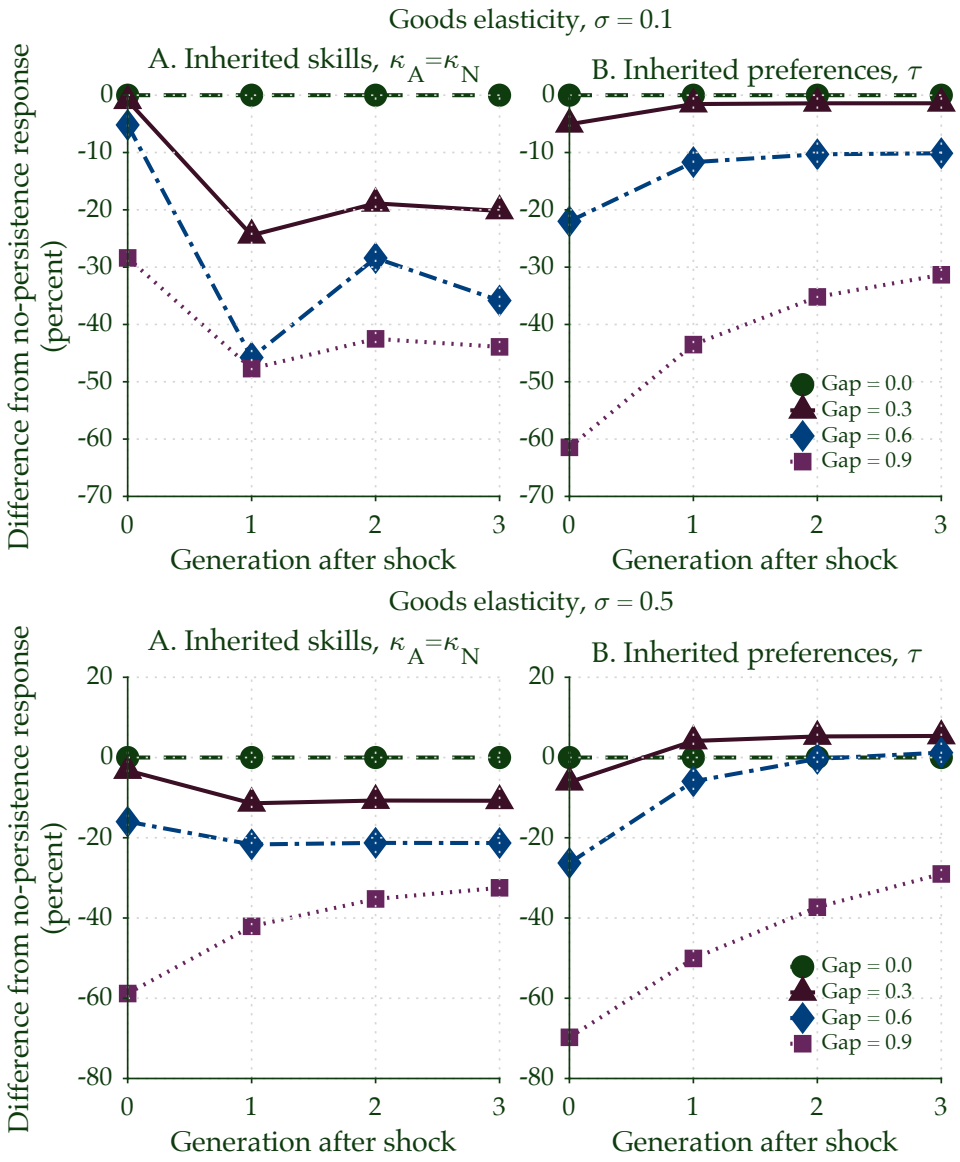


Figure 17: Numerical Responses at Alternative Elasticities of Substitution Across Goods

Note: The figure repeats the permanent 0.01-log-unit agricultural-technology shock for $\sigma \in \{0.1, 0.5, 0.9\}$. Within each subfigure, every economy is separately re-anchored to $\bar{\Phi}_A = 0.6$; the legend reports the parent-sector employment gap, and Panels A and B generate that gap using symmetric inherited skills and inherited preferences, respectively. Responses are percentage differences from the corresponding no-IGP response. All other parameters remain at the benchmark: $\gamma = 2$, $\vartheta_A = \vartheta_N = 1$, and $\Omega_A = \Omega_N = 1$.

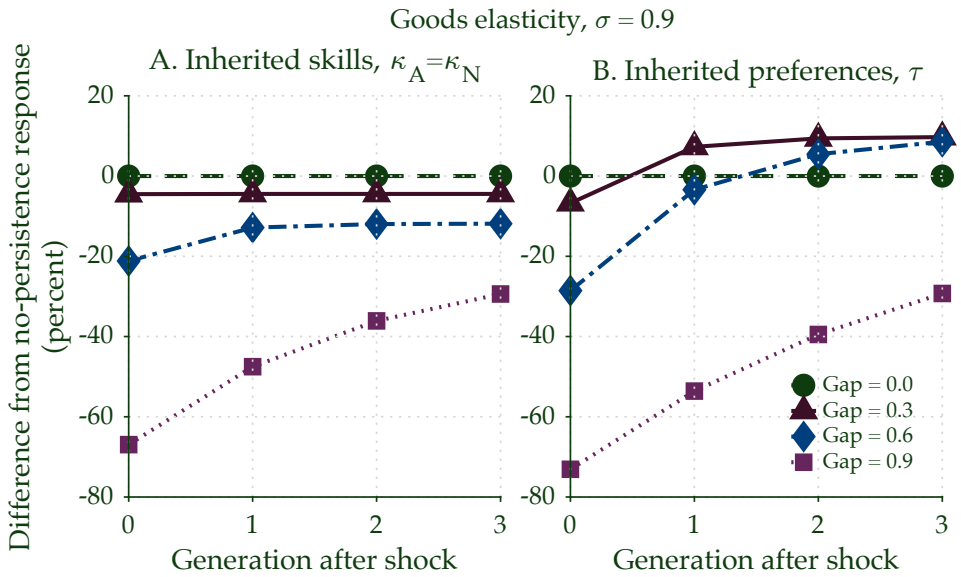


Figure 17: Numerical Responses at Alternative Elasticities of Substitution Across Goods (continued)

Note: This panel completes Figure 17 with $\sigma = 0.9$; all other parameters and construction follow the preceding note.

C. Appendix: Identification

C.1. Identification: Supporting Derivations

This appendix derives the identifying relationships for the labor supply parameters and the exogenous drivers of structural transformation established in Section 3. The results concern population moments implied by the model, rather than the estimation of those moments from the historical data. Throughout this appendix, I impose the identification restriction $\tau_A = \tau_N \equiv \tau$. The closed-form labor-supply derivations first specialize to the independent common-shape Fréchet benchmark used for the analytical results. The subsequent inversion proof permits the general continuous joint skill distribution used in the preferred quantitative specification.

C.1.1. Labor Supply Parameters

Conditional Log-Earnings Variance Fix a parental sector k and consider the indirect utilities of a type-k worker in each sector. Because the sector-specific skill draws are independent and satisfy $\epsilon_{i,j,t}^{-\gamma} \sim \text{Exp}(1)$, the workers' indirect utilities imply

$$V_{i,A,t}(k)^{-\gamma} \sim \text{Exp}\left([B_{A,t} w_{A,t} \kappa_A^{\mathbb{1}(k=A)} (1-\tau)^{\mathbb{1}(k=N)}]^{\gamma}\right),$$

$$V_{i,N,t}(k)^{-\gamma} \sim \text{Exp}\left([B_{N,t} w_{N,t} \kappa_N^{\mathbb{1}(k=N)} (1-\tau)^{\mathbb{1}(k=A)}]^{\gamma}\right).$$

The exponential variables are independent because the skill draws are independent. Maximizing indirect utility is equivalent to minimizing its negative γ-th power. For independent exponential random variables, the minimum is independent of which variable attains it and is exponentially distributed with a rate equal to the sum of the individual rates. Therefore, the maximum indirect utility, $V_{i,t}^{\max}(k) \equiv \max_{j \in \{A,N\}} V_{i,j,t}(k)$, is Fréchet distributed with shape parameter γ and is independent of the sector chosen by the worker. Since the variance of a standard Gumbel random variable is $\pi^2/6$, it follows that

$$\text{Var}\left[\log V_{i,t}^{\max}(k) \mid s_{i,t} = j,\ k\right] \quad = \quad \frac{\pi^2}{6\gamma^2}.$$

For a worker who chooses sector j, earnings differ from maximum indirect utility only by the nonpecuniary benefits associated with that sector:

$$\log y_{i,j,t}(k) \quad = \quad \log V_{i,t}^{\max}(k) - \log B_{i,j,t}(k) \ = \ \log V_{i,t}^{\max}(k) - \log B_{j,t} - \mathbb{1}(j \neq k) \log(1-\tau).$$

Conditional on destination and parental sector, the final two terms are constants and do not affect the variance of log earnings. Consequently,

$$\text{Var}\left[\log y_{i,j,t}(k) \mid s_{i,t} = j,\ k\right] \quad = \quad \frac{\pi^2}{6\gamma^2},$$

which identifies γ according to equation (ID-3).

Inherited Skills and Preferences The same independence between maximum indirect utility and sector choice delivers the conditional average effective labor supplied by each parental group. The conditional mean of maximum indirect utility equals $\bar{\gamma}$ times its Fréchet scale, while each group-specific choice probability equals the corresponding sectoral scale raised to γ divided by the sum across sectors. In particular, the expected idiosyncratic skill of a type-k worker who selects sector j is $\bar{\gamma}\phi_{j,t}(k)^{-1/\gamma}$, where $\bar{\gamma} = \Gamma(1 - 1/\gamma)$. Multiplying this conditional mean by the inherited skill advantage and the sectoral wage gives

$$y_{j,t}(k) \quad = \quad w_{j,t} \cdot \bar{\gamma} \cdot \kappa_k^{\mathbb{1}(k=j)} \cdot \phi_{j,t}(k)^{-1/\gamma}.$$

Under the maintained assumption that both parental groups face the same wage per efficiency unit within each sector, their earnings ratios therefore satisfy

$$\frac{y_{A,t}(A)}{y_{A,t}(N)} \quad = \quad \kappa_A \left(\frac{\phi_{A,t}(A)}{\phi_{A,t}(N)}\right)^{-1/\gamma}, \qquad \frac{y_{N,t}(N)}{y_{N,t}(A)} \quad = \quad \kappa_N \left(\frac{\phi_{N,t}(N)}{\phi_{N,t}(A)}\right)^{-1/\gamma}.$$

Rearranging these expressions gives the identifying equations (ID-4) and (ID-5) for inherited skills.

To identify inherited preferences, equation (1) implies that the odds of agricultural employment within each parental group satisfy

$$\frac{\phi_{A,t}(A)}{\phi_{N,t}(A)} \quad = \quad \left(\frac{B_{A,t} w_{A,t} \kappa_A}{B_{N,t} w_{N,t} (1-\tau)}\right)^{\gamma}, \qquad \frac{\phi_{A,t}(N)}{\phi_{N,t}(N)} \quad = \quad \left(\frac{B_{A,t} w_{A,t} (1-\tau)}{B_{N,t} w_{N,t} \kappa_N}\right)^{\gamma}.$$

Taking the ratio eliminates both relative wages and common nonpecuniary benefits:

$$\mathcal{O} \quad = \quad \frac{\phi_{A,t}(A)/\phi_{N,t}(A)}{\phi_{A,t}(N)/\phi_{N,t}(N)} \quad = \quad \left(\frac{\kappa_A \kappa_N}{(1-\tau)^2}\right)^{\gamma}.$$

Solving for τ yields equation (ID-6). Thus, the variance of log earnings, the two within-sector earnings ratios, and the intergenerational sector-choice odds ratio identify the four

baseline labor supply parameters from population moments under the maintained model assumptions.

C.1.2. Exogenous Drivers of Structural Transformation

I now derive the period-by-period identification of $Z_{A,t}$, $Z_{N,t}$, and $B_{N,t}/B_{A,t}$ from $\Phi_{A,t}$, p_t, and Q_t without imposing independence or a common Fréchet shape. The joint distribution $G(\epsilon_A, \epsilon_N)$, the inheritance parameters, and the previous generation's agricultural employment share, $\Phi_{A,t-1}$, are taken as given. Assume that G is continuous, has positive density and full support on $\mathbb{R}^2_{++}$, and has finite first moments; observed employment shares are interior and prices are strictly positive.

Let

$$q_t \equiv \frac{B_{N,t} w_{N,t}}{B_{A,t} w_{A,t}}, \qquad a_A \equiv \frac{1-\tau}{\kappa_A}, \qquad a_N \equiv \frac{\kappa_N}{1-\tau}.$$

A type-k worker chooses agriculture when $\epsilon_A/\epsilon_N \geq q_t a_k$. Hence define

$$\psi_k(q) \equiv \Pr_G\left(\frac{\epsilon_A}{\epsilon_N} \geq q a_k\right), \qquad k \in \{A, N\}.$$

Positive density and full support imply that each $\psi_k(q)$ is continuous and strictly decreasing in q, with limits one and zero as q approaches zero and infinity, respectively. It follows that

$$\Psi_t(q) \equiv \Phi_{A,t-1}\psi_A(q) + (1 - \Phi_{A,t-1})\psi_N(q) = \Phi_{A,t} \tag{33}$$

has exactly one solution $q_t \in \mathbb{R}_{++}$ for every $\Phi_{A,t} \in (0,1)$. This solution uniquely recovers both parental-group choice probabilities.

The same cutoff then uniquely determines aggregate effective labor through truncated moments of the known joint distribution:

$$H_{A,t} = \Phi_{A,t-1}\kappa_A \mathbb{E}_G[\epsilon_A \mathbb{1}\{\epsilon_A/\epsilon_N \geq q_t a_A\}] + (1 - \Phi_{A,t-1})\mathbb{E}_G[\epsilon_A \mathbb{1}\{\epsilon_A/\epsilon_N \geq q_t a_N\}],$$
$$H_{N,t} = \Phi_{A,t-1}\mathbb{E}_G[\epsilon_N \mathbb{1}\{\epsilon_A/\epsilon_N < q_t a_A\}] + (1 - \Phi_{A,t-1})\kappa_N \mathbb{E}_G[\epsilon_N \mathbb{1}\{\epsilon_A/\epsilon_N < q_t a_N\}].$$

Full support makes both quantities strictly positive, and finite marginal means make them finite.

Second, goods-market clearing and sectoral production imply $X_{j,t} = Z_{j,t} H_{j,t}$. Since

$\omega_{A,t}/\omega_{N,t} = X_{A,t}/(p_t X_{N,t})$, substituting the production functions into equation (9) gives

$$\frac{Z_{A,t}H_{A,t}}{Z_{N,t}H_{N,t}} = \frac{\Omega_A}{\Omega_N} \cdot p_t^{\sigma} \cdot Q_t^{(\vartheta_A - \vartheta_N)(1-\sigma)}.$$

Rearranging identifies $Z_{N,t}/Z_{A,t}$ according to equation (ID-7).

Third, the relative expenditure-share equation also implies the following relative quantity demand:

$$\frac{X_{A,t}}{X_{N,t}} = \frac{\Omega_A}{\Omega_N} \cdot p_t^{\sigma} \cdot Q_t^{(\vartheta_A - \vartheta_N)(1-\sigma)}.$$

Substituting this relationship into the nonhomothetic CES aggregator (8) and rearranging gives agricultural consumption as

$$X_{A,t} = \Omega_A Q_t^{\vartheta_A} \left[\Omega_A + \Omega_N \left(p_t Q_t^{\vartheta_N - \vartheta_A}\right)^{1-\sigma}\right]^{\frac{\sigma}{1-\sigma}}.$$

Because $X_{A,t} = Z_{A,t}H_{A,t}$ and $H_{A,t} > 0$, dividing by agricultural effective labor identifies the level of agricultural technology in equation (ID-8). The level of nonagricultural technology is then uniquely determined from the identified relative technology, $Z_{N,t} = (Z_{N,t}/Z_{A,t})Z_{A,t}$.

Finally, the competitive wage condition (7) implies $w_{N,t}/w_{A,t} = p_t Z_{N,t}/Z_{A,t}$. Because the employment inversion has already identified $q_t = (B_{N,t}/B_{A,t})(w_{N,t}/w_{A,t})$, the relative common nonpecuniary benefit is uniquely recovered as

$$\frac{B_{N,t}}{B_{A,t}} = \frac{q_t}{p_t Z_{N,t}/Z_{A,t}}.$$

Each step is unique and returns strictly positive sectoral technologies and relative nonpecuniary benefits. The recovered fundamentals satisfy the sector-choice equations, sectoral production, consumer demand, and goods-market clearing by construction. Therefore, the observed aggregate employment share, relative goods price, and real consumption uniquely identify the three exogenous fundamentals conditional on the model parameters and inherited worker composition.

D. Appendix: Quantitative Application

D.1. Quantitative Implementation and Additional Results

This appendix documents the data, consumer-demand calibration, model inversion, counterfactual construction, and channel decomposition used in the quantitative analysis.

Data and timing. The quantitative exercise covers 1880, 1900, 1920, and 1940. The period-t cohort consists of ag-born and nonag-born workers with shares $\Phi_{A,t-1}$ and $1 - \Phi_{A,t-1}$, respectively. Thus the agricultural employment share in 1860, $\Phi_{A,0} = 0.665393$, determines cohort composition in 1880; thereafter, the observed aggregate share in one census determines the composition used for the next census. Table 26 describes the inputs and their sources. Aggregate and subgroup employment shares are constructed from the linked censuses as described in Section 1. Agricultural output shares, relative prices, and real GDP per capita are from Dennis and İşcan (2009), Alvarez-Cuadrado and Poschke (2011), and Maddison (1995), respectively. I normalize the latter two series to one in 1880.

Table 26: Data Used in the Quantitative Analysis

Variable	Description and use	Source or construction
$\Phi_{A,0}$	Initial share of ag-born workers entering in 1880; determines initial cohort composition.	1860 agricultural employment share in the linked census sample.
$\Phi_{A,t}$	Aggregate agricultural employment share; targeted by the inversion and used as the next period's ag-born share.	Linked census sample, 1880–1940.
$\phi_{A,t}(A)$, $\phi_{A,t}(N)$	Agricultural employment shares of ag-born and nonag-born workers; untargeted validation moments.	Linked census sample, 1880–1940.
$\omega_{A,t}$	Agricultural output share; calibrates and evaluates the consumer-demand parameters.	Dennis and İşcan (2009).
$p_t = p_{N,t}/p_{A,t}$	Price of nonagricultural relative to agricultural goods; targeted by the inversion.	Alvarez-Cuadrado and Poschke (2011); indexed to one in 1880.
$Q_t = Y_t/P_t$	Real GDP per capita; targeted by the inversion.	Maddison (1995); indexed to one in 1880.

Note: Employment measures use the sample and sector definitions described in Section 1.

Consumer-demand calibration. I set $\sigma = 0.3$, normalize $\Omega_N = 1$ and $\vartheta_A = 1$, and choose

$$\Omega_A = \frac{\omega_{A,1880}}{1 - \omega_{A,1880}} = 0.449275$$

so that the model exactly matches the 1880 agricultural output share. For any $\vartheta_N > 1$, the implied share is

$$\widehat{\omega}_{A,t}(\vartheta_N) = \frac{R_t(\vartheta_N)}{1 + R_t(\vartheta_N)}, \qquad R_t(\vartheta_N) = \Omega_A p_t^{\sigma-1} Q_t^{(1-\vartheta_N)(1-\sigma)},$$

where $p_t = p_{N,t}/p_{A,t}$ and $Q_t = Y_t/P_t$. I estimate

$$\widehat{\vartheta}_N = \arg\min_{\vartheta_N > 1} \sum_{t \in \{1900,1920,1940\}} \left[\omega_{A,t} - \widehat{\omega}_{A,t}(\vartheta_N)\right]^2,$$

obtaining $\widehat{\vartheta}_N = 4.630$. The optimization uses the transformation $\vartheta_N = 1 + \exp(u)$ and multiple starting values; the reported fit is the solution with the smallest sum of squared residuals. For comparison, I also report the fit under the external two-sector mapping of Comin, Lashkari and Mestieri (2021), which sets $(\sigma, \vartheta_A, \vartheta_N) = (0.33, 1, 1/0.33)$.

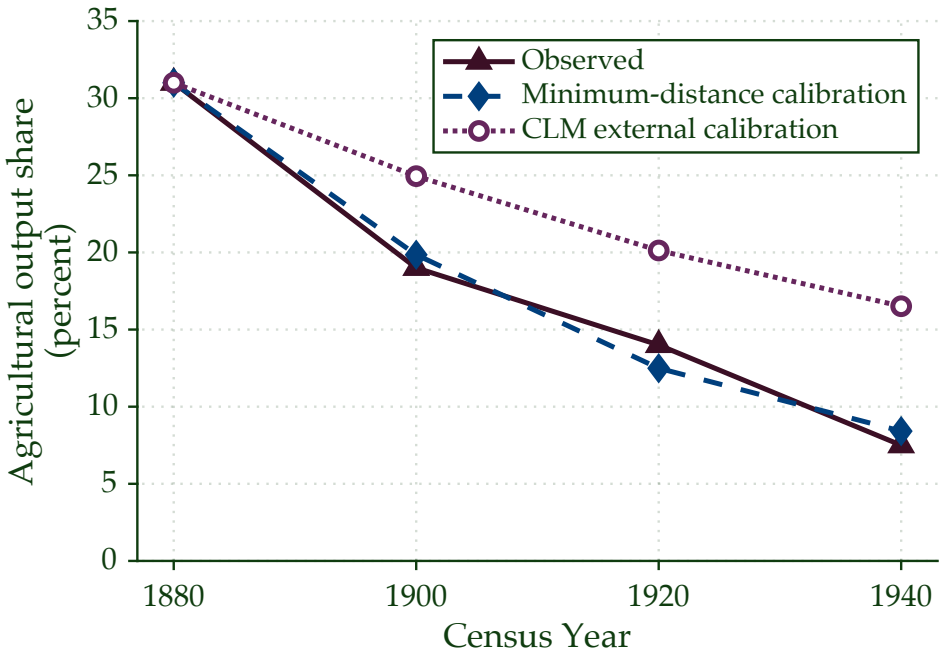


Figure 18: Consumer Demand Calibration of the Agricultural Output Share

Note: The figure compares the observed agricultural output share with the baseline minimum-distance calibration and the external parameterization of consumer demand. The baseline fixes $\sigma = 0.3$ and $\vartheta_A = 1$, estimates $\vartheta_N = 4.630$ from the 1900–1940 shares, normalizes $\Omega_N = 1$, and chooses Ω_A to match the 1880 share. The external specification uses $\sigma = 0.33$ and $\vartheta_N = 1/0.33 = 3.030$, following the two-sector mapping of Comin, Lashkari and Mestieri (2021); ϑ_A, Ω_N, and the 1880 normalization are unchanged. Output shares, relative prices, and real GDP per capita are from Dennis and İşcan (2009), Alvarez-Cuadrado and Poschke (2011), and Maddison (1995), respectively.

Period-by-period model inversion. Appendix C.1.2 provides the formal derivation and proof of the unique mapping from $(\Phi_{A,t}, p_t, Q_t)$ to $(Z_{A,t}, Z_{N,t}, B_{N,t}/B_{A,t})$ established in Proposition 5. The quantitative implementation evaluates those identifying equations period by period under the calibrated parameters and timing described above.

Recursive implementation and model fit. Starting from the observed or imposed initial condition $\Phi_{A,0}$, I apply the inversion sequentially because the recovered allocation in period t supplies the inherited composition $\Phi_{A,t}$ used in period $t+1$. In the historical inversion, I use the observed $\Phi_{A,t}$, rather than its fitted value, as the next period's inherited composition. At every date, the numerical implementation reconstructs the three targeted outcomes and the CES aggregator with absolute errors below 10^{-8}. The agricultural output share is not an additional identifying moment in this inversion. Once p_t and Q_t are imposed, equation (9) determines the agricultural output share from the calibrated consumer-demand parameters. The discrepancy from the observed output share there-

fore remains the consumer-demand calibration residual rather than a fourth equation in the period-by-period inversion.

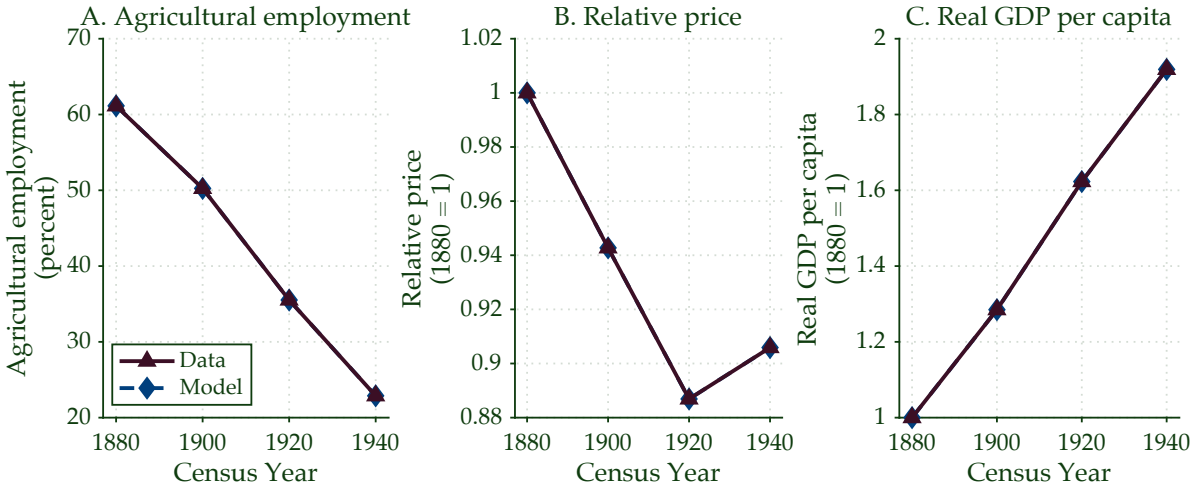


Figure 19: Targeted Moments for Model Inversion

Note: The figure compares the data with the outcomes reproduced by the period-by-period model inversion. Agricultural employment is reported in percent; the relative price, p_N/p_A, and real GDP per capita are indexed to one in 1880. The inversion targets all three series exactly in 1880, 1900, 1920, and 1940, recursively using the previous period's agricultural employment share as the current share of ag-born workers. Employment shares are calculated from the linked censuses; relative prices and real GDP per capita are from Alvarez-Cuadrado and Poschke (2011) and Maddison (1995), respectively.

Untargeted Subgroup Shares. The inversion also implies $\phi_{A,t}(A)$ and $\phi_{A,t}(N)$, but does not target their observed counterparts. Conditional on $\Phi_{A,t-1}$ and the time-invariant labor-supply parameters, the single aggregate target $\Phi_{A,t}$ identifies the subgroup choices through equation (33), which in turn fixes both subgroup shares. Their comparison with the data in Figure 20 is therefore an overidentifying check for both adjusted specifications. The inherited-skill and inherited-preference parameters are estimated from 1940 income and sector-choice data, so the model does not reproduce the substantially greater subgroup persistence observed in 1880. Matching these subgroup paths would require time-varying persistence parameters or an additional model mechanism.

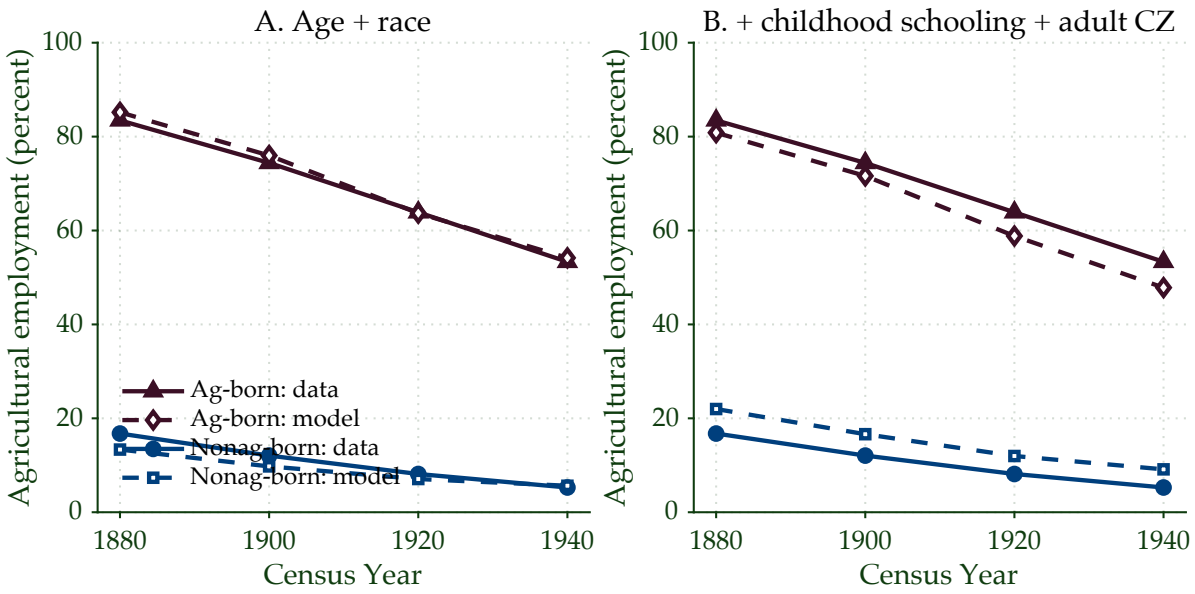


Figure 20: Untargeted Subgroup Employment Shares

Note: The figure compares observed and model-implied agricultural employment shares for ag-born and nonag-born workers in the linked censuses under the baseline and schooling-and-location specifications. These subgroup shares are not targeted: each inversion matches only the aggregate employment share, relative price, and real GDP per capita in each year, using that specification's time-invariant labor-supply parameters estimated from 1940.

Constructing the counterfactuals. Let superscript b denote the benchmark inversion and define the benchmark growth factors

$$g_t^A = \frac{Z_{A,t}^b}{Z_{A,1880}^b}, \qquad g_t^N = \frac{Z_{N,t}^b}{Z_{N,1880}^b}, \qquad g_t^B = \frac{(B_N/B_A)_t^b}{(B_N/B_A)_{1880}^b}.$$

I consider the benchmark and three counterfactuals: no inherited skills ($\kappa_A = \kappa_N = 1$), no inherited preferences ($\tau = 0$), and no intergenerational persistence ($\kappa_A = \kappa_N = 1, \tau = 0$). For each specification r, I first apply the inversion under the corresponding parameters to the common 1880 targets. This produces specification-specific anchors $Z_{A,1880}^r$, $Z_{N,1880}^r$, and $(B_N/B_A)_{1880}^r$. The subsequent fundamentals are

$$Z_{A,t}^r = Z_{A,1880}^r g_t^A, \qquad Z_{N,t}^r = Z_{N,1880}^r g_t^N, \qquad (B_N/B_A)_t^r = (B_N/B_A)_{1880}^r g_t^B.$$

I then solve the equilibrium forward, using the simulated $\phi_{A,t}^r$ as the parental composition of the next cohort. This construction makes all four specifications match the 1880 agricultural employment share, relative price, and real GDP per capita, while exposing them to the same proportional growth in fundamentals thereafter. Worker reallocation is

the signed change $100(\phi^r_{A,1940} - \phi^r_{A,1880})$ percentage points, so movement out of agriculture is negative.

Labor-supply elasticity benchmark. Figure 21 compares the preferred Fréchet–Frank results with the independent, common-shape Fréchet benchmark. In the latter, the No IGP elasticity and the Full IGP weighted within-group elasticity both equal γ at every date, so their visually redundant paths coincide and the entire difference from the Full IGP aggregate elasticity is aggregation attenuation. Under Fréchet–Frank, within-group elasticities instead vary with the prevailing choice thresholds, and this equivalence need not hold.

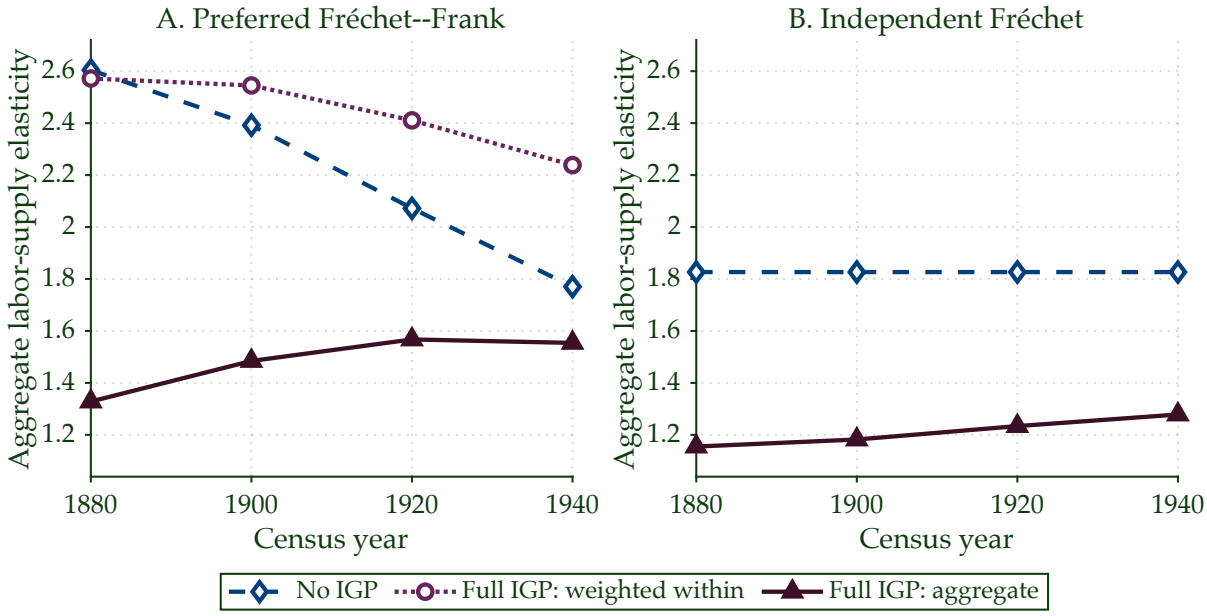


Figure 21: Aggregate Labor Supply Elasticities Across Skill Distributions

Note: Panel A uses the preferred age-and-race Fréchet–Frank estimates; Panel B uses the independently estimated common-shape Fréchet benchmark. All elasticities are partial-equilibrium responses to relative sectoral wages at the prevailing equilibrium, holding parental composition, common nonpecuniary benefits, inherited parameters, and the skill distribution fixed. In Panel B, the Full IGP weighted within-group series coincides with the No IGP series and is therefore not plotted separately.

Inferred Drivers Without Inherited Skills or Inherited Preferences. Table 27 reports the full-distribution inversion for each inherited-mechanism restriction and each adjusted Fréchet–Frank specification. Every row retains the specification's estimated joint skill distribution and re-inverts the same observed aggregate paths.

Table 27: Relative-Technology Growth by Control Specification and Persistence Restriction

Counterfactual	Benchmark growth	Counterfactual growth	Difference from benchmark
		1880–1940 log points	
Panel A: Baseline (age and race)			
No inherited skills	53.31	88.41	35.09
No inherited preferences	53.31	53.73	0.41
No intergenerational persistence	53.31	88.58	35.27
Panel B: Schooling and location			
No inherited skills	58.34	83.57	25.23
No inherited preferences	58.34	58.95	0.61
No intergenerational persistence	58.34	83.78	25.45

Note: Benchmark and counterfactual growth are cumulative 1880–1940 log changes in Z_N/Z_A. Each counterfactual retains the panel's estimated joint Fréchet–Frank skill distribution, re-infers the historical fundamentals, and changes only the indicated inherited mechanism.

The no-persistence restriction raises inferred relative-technology growth from 53.31 to 88.58 log points in the baseline and from 58.34 to 83.78 log points with schooling and location controls. In both specifications, inherited skills account for nearly all of this difference.

Interval-level Shapley decomposition. For specification r and an interval ending in t, let F_0^r and F_1^r denote the beginning- and end-of-interval fundamentals, and let M_0^r and M_1^r denote the corresponding inherited parental compositions. Write

$$y_{ab}^r = \Phi_A(F_a^r, M_b^r), \qquad a, b \in \{0, 1\}.$$

The two-player Shapley contributions to the signed change in the agricultural employment share are

$$C_t^r = \frac{1}{2}\left[(y_{10}^r - y_{00}^r) + (y_{11}^r - y_{01}^r)\right],$$
$$P_t^r = \frac{1}{2}\left[(y_{01}^r - y_{00}^r) + (y_{11}^r - y_{10}^r)\right],$$

where C_t^r is the immediate effect of technologies and the relative payoff shifter and P_t^r is the dynamic effect transmitted through parental composition. The averages span the two

possible orders in which the inputs change. They satisfy

$$C_t^r + P_t^r = y_{11}^r - y_{00}^r$$

in every interval. I sum each channel over 1880–1900, 1900–1920, and 1920–1940, then report its counterfactual-minus-benchmark difference in percentage points:

$$100 \sum_t (C_t^r - C_t^b), \qquad 100 \sum_t (P_t^r - P_t^b).$$

Their sum equals the difference between counterfactual and benchmark worker reallocation reported in Table 4.

D.1.1. Sensitivity to Schooling and Location Controls

The schooling-and-location specification is a complete alternative quantitative exercise rather than a partial substitution of wedges. It uses its own adjusted eight target moments, fitted joint skill distribution, historical inversion, initial-condition anchors, counterfactual equilibria, and interval-level Shapley calculations.

Table 28: Immediate, Dynamic, and Cumulative Effects by Control Specification

Counterfactual	Immediate effect	Dynamic effect	Cumulative effect
	Percentage points		
Panel A: Baseline (age and race)			
No inherited skills	−0.66	−8.65	−9.32
No inherited preferences	−1.35	1.21	−0.14
No intergenerational persistence	−1.10	−7.77	−8.87
Panel B: Schooling and location			
No inherited skills	0.02	−6.59	−6.57
No inherited preferences	−1.23	1.13	−0.10
No intergenerational persistence	−0.38	−5.74	−6.11

Note: Effects are counterfactual minus benchmark changes in the agricultural employment share. Each panel uses its own adjusted eight-moment estimates and full-distribution macro chain; immediate and dynamic effects add to the cumulative effect within numerical tolerance.

Table 28 shows that the channel results are qualitatively stable across the two control specifications. In the baseline, removing inherited skills lowers $\Delta\Phi_A$ through both chan-

nels; with schooling and location controls, its immediate contribution is essentially zero while its dynamic contribution remains large and negative. Removing inherited preferences produces a negative immediate effect that is substantially offset by a positive dynamic effect in both specifications.

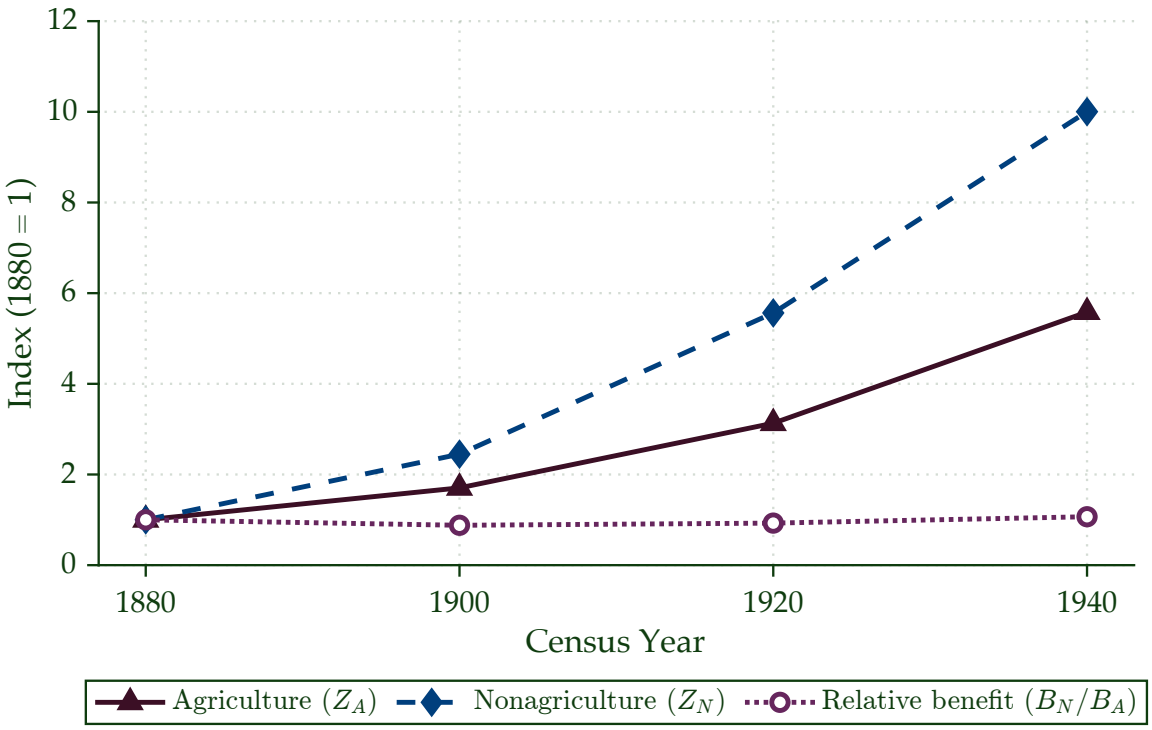


Figure 22: Inferred Fundamentals with Schooling and Location Controls

Note: The figure repeats the full-persistence historical inversion using the schooling-and-location adjusted Fréchet–Frank estimate. Each series is indexed to one in 1880.

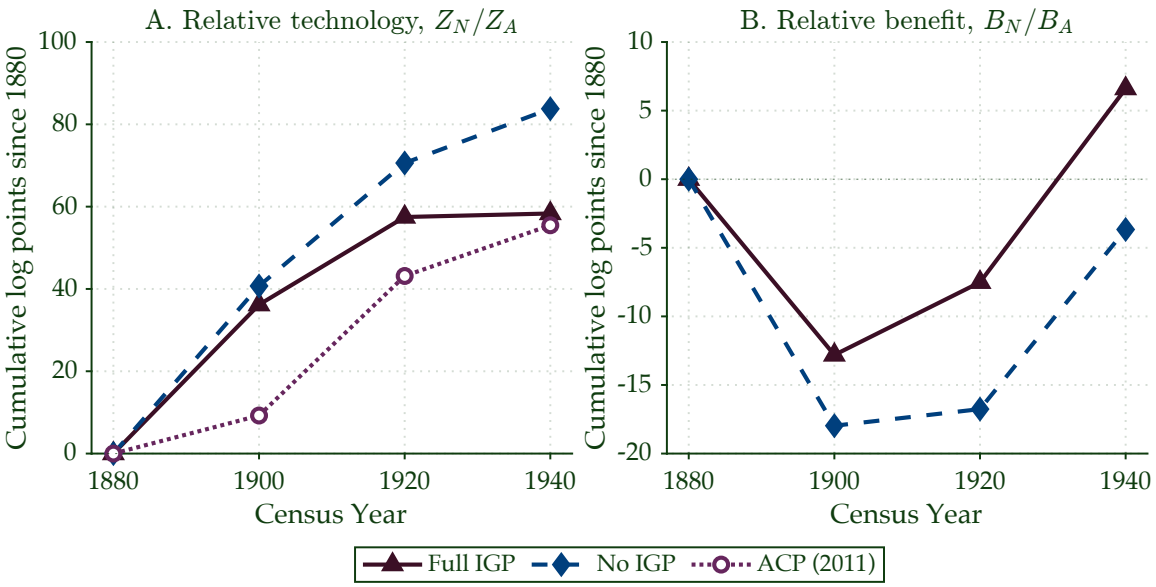


Figure 23: Relative Drivers with Schooling and Location Controls

Note: The figure repeats the full-IGP and no-IGP inversions using the schooling-and-location adjusted Fréchet–Frank estimate. Panel A additionally reports the nonfarm-to-farm relative TFP series from Alvarez-Cuadrado and Poschke (2011), with its 1880 level interpolated as in the original series construction.

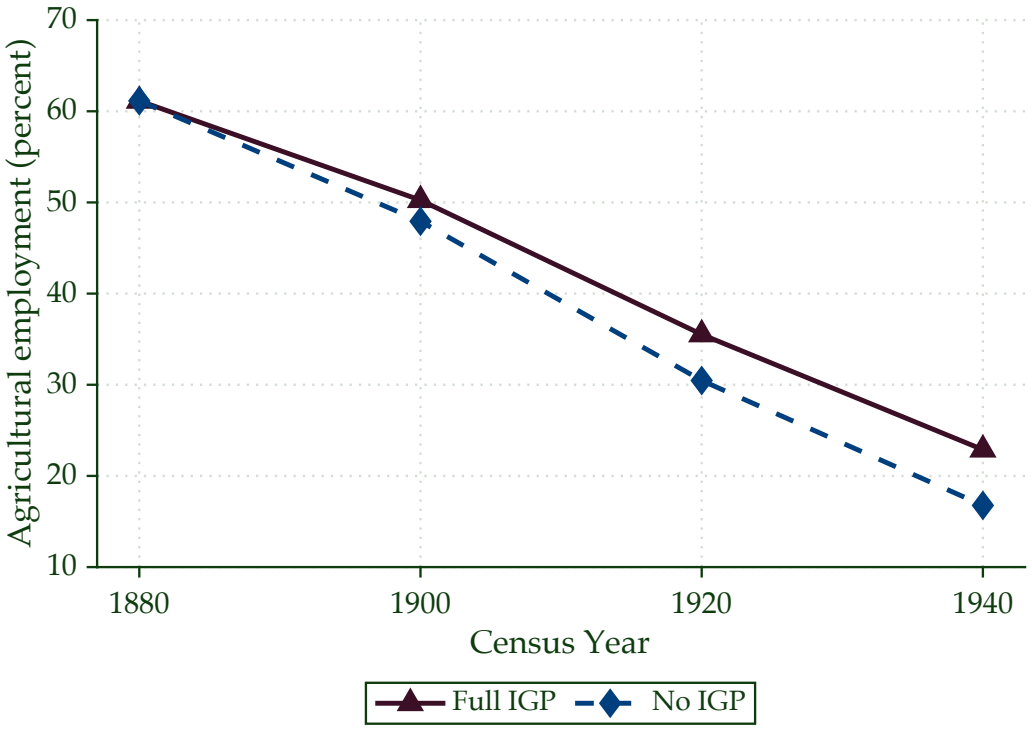


Figure 24: Counterfactual Employment Paths with Schooling and Location Controls

Note: The Full IGP and No IGP paths match the observed 1880 agricultural employment share and subsequently face the Full IGP growth in fundamentals inferred under the schooling-and-location specification. The No IGP path sets $\kappa_A = \kappa_N = 1$ and $\tau = 0$; parental composition evolves recursively along both paths.